Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY D/B/A NATIONAL GRID NY

AND

KEYSPAN GAS EAST CORPORATION D/B/A NATIONAL GRID

Direct Testimony

of

Joshua C. Nowak

April 28, 2023

Testimony of Joshua C. Nowak

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I. INTRODUCTION AND QUALIFICATIONS

Q. Please state your name, affiliation, and business address.

A. My name is Joshua C. Nowak. I am employed by Concentric Energy Advisors, Inc. ("Concentric") as Vice President. Concentric is a management consulting and economic advisory firm, focused on the North American energy and water industries. Based in Marlborough, Massachusetts and Washington, D.C., Concentric specializes in regulatory and litigation support, financial advisory services, energy market strategies, market assessments, energy commodity contracting and procurement, economic feasibility studies, and capital market analyses. My business address is 293 Boston Post Road West, Suite 500, Marlborough, Massachusetts 01752.

Q. On whose behalf are you submitting this Direct Testimony?

A. I am submitting this Direct Testimony on behalf of The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI"), collectively referred to as "the Companies," wholly-owned subsidiaries of National Grid USA ("National Grid"). My Direct Testimony is part of the Companies' rate case filings before the New York State Public Service Commission ("Commission").

Q. Please describe your experience in the energy and utility industries and your educational and professional qualifications.

A. I hold a Bachelor's degree in Economics from Boston College, and have more than 15 years of experience in providing economic, financial, and strategic advisory services. As a consultant, I primarily advise clients in regulated utility industries and have provided testimony regarding financial matters before multiple regulatory agencies. I have advised numerous energy and utility clients on a wide range of financial and economic issues with primary concentrations in valuation and utility rate matters. Many of these assignments have included the determination of the cost of capital for valuation and ratemaking purposes. I have provided testimony before the Federal Energy Regulatory Commission ("FERC") as well as state and provincial jurisdictions in the U.S. and Canada. A summary of my professional and educational background is presented as Attachment A.

Q. Have you previously submitted testimony before the Commission?

A. Yes. I have testified in on behalf of Niagara Mohawk Power Corporation d/b/a National Grid before the Commission in Cases 17-E-0238 and17-G-0239 and Cases 20-E-0380 and 20-G-0381.

II. PURPOSE AND OVERVIEW OF TESTIMONY

Q. What is the purpose of your Direct Testimony?

A. The purpose of my Direct Testimony is to present evidence and provide a recommended range for the Companies' cost of equity (sometimes referred

to as the Return on Equity or "ROE" for rate-setting purposes) and capital structure for the Companies' gas distribution utility operations. My analysis and recommendations are supported by the data presented in Exhibits __ (JCN-1) through __ (JCN-8).

Q. Please provide a brief overview of the analyses that led to your ROE recommendation.

A. As discussed in more detail in the remainder of my Direct Testimony, it is important to consider the results of several analytical approaches in determining a reasonable recommendation for the Companies' ROE. To establish my ROE recommendation, I developed a proxy group that consists of utilities that face investment risk generally comparable to that faced by the Companies. I developed a multi-stage Discounted Cash Flow ("DCF") model and two forms of the Capital Asset Pricing Model ("CAPM"). I have presented the DCF and CAPM results weighted equally, and with the Commission's conventional 2/3 weighting of the DCF and 1/3 weighting of the CAPM.

The use of a multi-stage DCF model and two forms of the CAPM is consistent with the approach employed by the Commission in prior cases. While my equal weighting of the DCF and CAPM results does not conform to the weighting typically employed in proceedings before the Commission in the past, I explain in my Direct Testimony why placing less emphasis on

the DCF model at this time is reasonable and consistent with the goals of the Recommended Decision ("Generic Finance RD") issued in the Generic Finance Proceeding ("GFP"), Case 91-M-0509, which is the docket that has been cited by the Commission to establish the ROE formula.

In addition, I have considered how current market conditions have affected the results of the ROE estimation models. The current economic environment is characterized by increased inflationary pressures. As a result of elevated inflation, the Federal Reserve's responsive tightening of monetary policy has caused an increase in interest rates. Further, I evaluated the Companies' risk profiles, including capital spending plans, the regulatory environment, and the emerging risks due to decarbonization policy.

Q. Please summarize the results of the ROE estimation models that you considered in your analyses.

A. As noted above, I considered the results of the multi-stage form of the DCF model and two versions of the CAPM. The results of my analyses are summarized in Figure 1 (below).

Figure 1: Summary of Analytical Results

	Low	Mean	High
DCF	9.13%	9.68%	10.31%
Mean CAPM	11.13%	11.14%	11.15%
67%/33% DCF/CAPM	9.79%	10.17%	10.59%
50%/50% DCF/CAPM	10.13%	10.41%	10.73%

Q. What are your conclusions regarding the appropriate cost of equity for the Companies?

A. Based on the quantitative and qualitative analyses and the weighting of the DCF and CAPM results presented in Figure 1, as well as my assessment of the business and financial risk environment of KEDLI and KEDNY relative to the proxy groups, I conclude that the appropriate ROE for the Companies is within the range of 9.80 percent and 10.70 percent. The Companies are requesting an ROE of 9.80 percent, which is at the low end of the range of reasonableness and is a conservative estimate of the investor-required ROE.

Q. Please summarize your analysis of the appropriate ratemaking capital structure for the Companies.

A. The analysis presented in Section VIII demonstrates that the Companies' requested equity ratio of 48 percent is below the mean equity ratio of 54.13 percent for the operating utility companies in the proxy group over the last four years. Therefore, I conclude that the Companies' requested equity ratio is reasonable, and in fact, conservative.

Q. How is the remainder of your Direct Testimony organized?

A. The remainder of my Direct Testimony is organized as follows:

Section III – Discusses the regulatory guidelines and financial considerations pertinent to the development of the cost of capital;

Section IV – Briefly discusses the current capital market conditions and the effect of those conditions on the Companies' cost of equity;

Section V – Explains my selection of the proxy group used to develop my analytical results;

Section VI – Explains my analyses and the analytical bases for the ROE recommendation;

Section VII – Summarizes the specific regulatory and business risks that have a direct bearing on the Companies' cost of equity;

Section VIII – Provides an assessment of the Companies' proposed capital structure;

Section IX – Provides an assessment of the effect of a Multi-Year Rate Plan on the ROE; and

Section X – Summarizes my conclusions and recommendations.

III. REGULATORY GUIDELINES AND FINANCIAL CONSIDERATIONS

Q. Please describe the guiding principles to be used in establishing the cost of capital for a regulated utility.

A. The foundations of public utility regulation require that utilities receive a fair rate of return sufficient to attract needed capital to maintain service to customers at reasonable rates. The basic tenets of this regulatory doctrine

originate from two decisions by the United States Supreme Court, notably *Bluefield Waterworks and Improvement Company v. Public Service Commission of West Virginia*, 262 U.S. 679 (1923) (*Bluefield*), and *Federal Power Commission v. Hope Natural Gas Company,* 320 U.S. 591 (1944) (*Hope*). In *Bluefield*, the Court stated:

> A public utility is entitled to such rates as will permit it to earn a return on the value of the property which it employs for the convenience of the public equal to that generally being made at the same time and in the same general part of the country on investments in other business undertakings which are attended by corresponding risks and uncertainties…
>
> The return should be reasonably sufficient to assure investor confidence in the financial soundness of the utility and should be adequate, under efficient and economical management, to maintain and support its credit and enable it to raise the money necessary for the proper discharge of its public duties.

Later, in *Hope*, the Court established a standard for the ROE that remains the guiding principle for ratemaking regulatory proceedings to this day:

> [T]he return to the equity owner should be commensurate with returns on investments in other enterprises having corresponding risks. That return, moreover, should be sufficient to assure confidence in the financial integrity of the enterprise, so as to maintain its credit and to attract capital.

Q. Please explain how these principles apply in the context of the regulated rate of return.

A. Regulated utilities rely primarily on common stock and long-term debt to finance permanent property, plant, and equipment. The allowed rate of return for a regulated utility is based on its weighted average cost of capital,

where the costs of the individual sources of capital (*i.e.*, debt and equity) are weighted by their respective book values. The ROE represents the cost of raising and retaining equity capital and is estimated with one or more analytical techniques that use market data to quantify investor requirements for equity returns. However, the ROE cannot be derived mechanically through quantitative metrics and models alone. To properly estimate the ROE, the financial, regulatory, and economic context must also be considered.

Based on the standards established in *Hope* and *Bluefield*, the authorized ROE in this proceeding should provide the Companies with the opportunity to earn a fair and reasonable return that is:

- Adequate to allow the Companies to attract the capital that is necessary to provide safe and reliable service (the "capital attraction standard");
- Sufficient to ensure the Companies' ability to maintain its financial integrity (the "financial integrity standard"); and
- At a level that is comparable to returns required on investments of similar risk (the "comparability standard").

Importantly, a fair return must satisfy all three of these standards. The allowed ROE should enable the Companies to finance capital expenditures on reasonable terms and provide the Companies with the ability to raise capital under a full range of capital market circumstances to serve its

customers. The DCF and CAPM approaches, while fundamental to the ROE determination, are still only models. The results of these models cannot be mechanically applied without also using informed judgment to consider economic and capital market conditions and the relative risk of KEDNY and KEDLI as compared to the proxy group companies.

Q. Did the Commission conduct a proceeding to review the standard for estimating the cost of capital for a regulated utility?

A. Yes. On August 21, 1991, the Commission established the GFP to review the Commission's then-current methodology for estimating the cost of equity and to examine various alternatives.[1]

Q. Please provide a brief summary of the purpose of the GFP.

A. The Commission's goal in opening the GFP was to eliminate controversy around ROE calculations and attempt to find common ground on contentious issues by developing a consensus approach for setting utility equity returns. Among other things, the Commission questioned whether there should be greater consistency in rate of return determinations from company to company, such that differences in returns could be directly attributed to differences in risk between companies, and whether the

[1] Case 91-M-0509, Proceeding on Motion of the Commission to Consider Financial and Regulatory Policies for New York State Utilities, Recommended Decision, (issued July 19, 1994) Generic Finance RD at 2.

Commission's historical primary reliance on DCF-based ROE determinations continued to provide fair returns.[2]

Q. Please describe the conclusions outlined in the Generic Finance RD.

A. The Generic Finance RD concluded that the Commission should implement a generic process for setting returns, based on proxy groups (not company-specific data), and that reliance on the DCF method should be replaced with a combination of the DCF and CAPM methodologies. The RD proposed to use as a preferred convention a methodology that gives a respective 2/3rd to 1/3rd weighting to the results of the DCF and CAPM analyses. The RD recognized that the CAPM "should figure prominently in the analysis" because this methodology provides fundamental information on interest rates and the returns required by stocks as a result of changes in interest rates. At that time, the CAPM was not accorded the same level of prominence as the DCF analysis, given that the former had previously only been used as a check.[3] However, while the Generic Finance RD recognized a benefit to establishing an "operating norm" with respect to setting the ROE, it also recognized that there may be good reason to adjust either the weightings of the DCF and CAPM models or to rely on different ROE estimation models. Specifically, the Generic Finance RD provides the following guidance:

> In either an annual proceeding to determine a rate of return or in individual proceedings, the 2/3 DCF and 1/3

[2] *Id.* at 13-14.
[3] *Id.* at 27.

> CAPM convention should be the presumption, but as Multiple Intervenors suggests, parties would not be barred from introducing new methods or different weightings. Such parties, however, would have the burden of convincing other parties and the Commission of the relevance or superiority of their proposals. [4]

To establish the "operating norm," the Generic Finance RD recommended specific forms of the ROE estimation models – a two-stage DCF approach and a Traditional and Zero Beta CAPM. In the DCF model, the first-stage growth was determined by the implied growth rate in Value Line dividend forecasts for four- to six-years in the future. The second growth rate began with the end of the four- to six-year period of the first stage and extended infinitely. The second stage included what is termed an "SV"[5] adjustment for external growth through additional equity issuances.[6] The CAPM result was proposed to be based on the average of the Traditional and Zero-Beta forms of the model.[7] Dividend yields in the DCF analysis and risk-free bond yields in the CAPM analyses were based on six months of yield data.[8] Equity ratios were capped at the upper end of the levels necessary to maintain an "A" bond rating.[9] Although the Generic Finance RD was never formally adopted by the Commission, it has served as a touchstone for the Commission's ROE determinations for more than 25 years.

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4 *Id*.
5 "SV" is the product of new shares issued as a percentage of current shares ("S") and the value of the market-to-book ratio accruing to shareholders ("V") as determined by: 1 - (book value / market value).
6 Generic Finance RD at 21.
7 *Id.* at 24.
8 *Id.* at 26.
9 *Id.* at 43.

Q. Does the analysis presented in the remainder of your Direct Testimony meet the intentions of the Generic Finance RD?

A. Yes, it does. As discussed in greater detail in Section VI, the methodologies that I have applied to estimate the return on equity are consistent with Commission precedent since the issuance of the Generic Finance RD in the GFP. Moreover, the models used in my analysis extend the principles advanced in the Generic Finance RD to best practices in financial analysis and current capital market conditions, as was contemplated in the Generic Finance RD.

Specifically, I rely on the weighted results of DCF and CAPM analyses. In developing these ROE estimation models, I rely on proxy groups of risk-comparable companies as discussed in Section V. I have used both the DCF and CAPM methodologies to estimate the return on equity. The multi-stage DCF model that I relied on is consistent with the methodology that the Commission has relied on in that it allows growth rates to vary over time. Consistent with the fundamental principles upheld by the Commission, I have applied two versions of the CAPM: Traditional and Zero Beta. Finally, consistent with the principles of the GFP, to reduce the volatility associated with the reliance on any one model, I have considered the DCF and CAPM results weighted equally, and with the Generic Finance RD's proposed 2/3 weighting on the DCF and 1/3 weighting of the CAPM.

Q. Do the principles and intentions of the Generic Finance RD in the GFP require adherence to a static formula?

A. No. The GFP and the Generic Finance RD did not require strict adherence to a static formula, rather, they promoted some basic principles and afforded parties the flexibility to investigate approaches to address changing financial market conditions. When faced with the task of estimating the cost of equity, analysts are inclined to gather and evaluate as much relevant data (both quantitative and qualitative) as can be reasonably analyzed. Analysts and academics understand that ROE models are tools to be used in the ROE estimation process, and that strict adherence to any single approach, or the specific results of any single approach, can lead to flawed conclusions. No model can exactly pinpoint the correct return on equity; rather, each model brings its own perspective and set of inputs that inform the estimate of ROE. That position is consistent with the *Hope* finding that "[u]nder the statutory standard of 'just and reasonable,' it is the result reached, not the method employed, which is controlling."[10]

Although each model brings a different perspective and adds depth to the analysis, each model also has its own set of inherent weaknesses and should not be relied upon individually without corroboration from other approaches. Changes to inputs as a result of changes in economic conditions could have widely different effects on the results of the various

[10] *Hope*, 320 U.S. at 602.

analyses. Regardless of which analyses are performed to estimate the investors' required return on equity, the analyst must apply judgment to assess the reasonableness of results and to determine the best weighting to apply to results under prevailing capital market conditions. No one model can reliably and consistently estimate the cost of capital that meets the fairness standard of *Hope* and *Bluefield* in all market conditions.

Q. What is your conclusion with regard to regulatory guidelines and the application of the models used to estimate the cost of equity?

A. The Generic Finance RD recognized that the CAPM provides fundamental information on interest rates and the returns on stocks required by investors as a result of changes in interest rates. Further, it also recognized that there may be good reason to adjust the weightings of the DCF and CAPM models in the future. As discussed in Section IV, interest rates have increased substantially in recent months and are a particular concern to investors in the current capital market environment. Given the recent changes in the financial markets, and the fact that the CAPM directly accounts for changes in interest rates (*i.e.*, risk-free rate) and the relative performance of utilities relative to the broader equity market (*i.e.*, Beta), it is necessary to evaluate the appropriate weight to give the CAPM in determining the ROE for the Companies.

IV. CAPITAL MARKET CONDITIONS

Q. Why is it important to analyze capital market conditions?

A. The ROE estimation models rely on market data that are either specific to the proxy group, in the case of the DCF model, or to the expectations of market risk, in the case of the CAPM. The results of the ROE estimation models can be affected by prevailing market conditions at the time the analysis is performed. While the ROE that is established in a rate proceeding is intended to be forward-looking, the analyst uses current and projected market data, specifically stock prices, dividends, growth rates and interest rates in the ROE estimation models to estimate the required return for the subject company.

As is discussed in the remainder of this section, analysts and regulatory commissions have concluded that current market conditions have affected the results of the ROE estimation models. As a result, it is important to consider the effect of these conditions on the ROE estimation models when determining the appropriate range and recommended ROE for a future period. If investors do not expect historical or current market conditions to be sustained in the future, it is possible that the ROE estimation models will not provide an accurate estimate of investors' required return during that rate period. Therefore, it is important to consider projected market data to estimate the return for that forward-looking period.

Q. Please describe the circumstances that have affected capital market conditions.

A. Inflation and rising interest rates are the predominant factors in the current economic environment. However, it is important to put inflation and the current monetary policy in historical context. Following a period of significant inflation and multiple recessions in the 1970s and early 1980s, the past 30 years have been marked by low inflation and stable economic growth as central banks have adopted inflation-targeting policies. From the mid-1980s until 2008, an extended period of low inflation was characterized by tighter monetary policy as compared with the prior period. In 2008, the global financial crisis created a severe recession that was met with comparatively looser monetary policy (*i.e.,* lower target interest rates) and significant fiscal policy stimulus. However, inflationary pressures were offset by the significant demand shock generated by the underlying credit crisis. As a consequence, the economy avoided significant inflation in the period following the global financial crisis despite lower interest rates. In 2018 and 2019, in the period immediately preceding the worldwide pandemic of Coronavirus Disease 2019 ("COVID-19"), inflationary pressures remained stable. However, current capital market conditions have been significantly affected by the economic repercussions of the COVID-19 pandemic and the subsequent fiscal and monetary policy reactions.

Q. What steps did the Federal Reserve take to stabilize financial markets and support the economy in response to COVID-19?

A. In response to the economic effects of COVID-19, the Federal Reserve decreased the federal funds rate twice in March 2020, resulting in a target range of 0.00 percent to 0.25 percent (which remained in effect until March 2022), and increased its holdings of both Treasury and mortgage-backed securities, and supported increased credit to both businesses and individual borrowers. This increase in the money supply kept short-term interest rates low and increased the ability of banks to lend to consumers and businesses.

Q. Please summarize the fiscal policy response from the U.S. Congress.

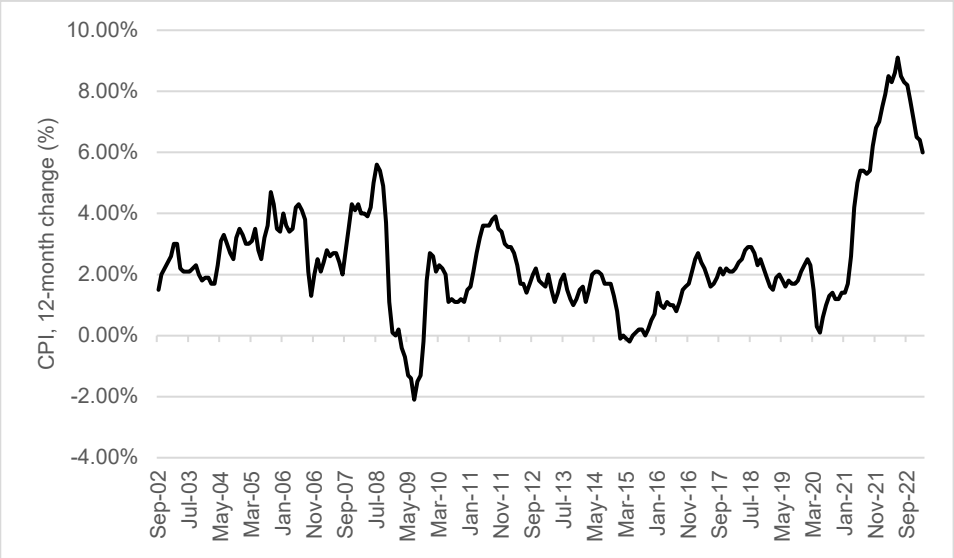
A. In addition to the Federal Reserve's response, the U.S. Congress passed approximately $4.5 trillion in fiscal stimulus programs. On March 27, 2020, the Coronavirus Aid, Relief, and Economic Security Act was signed into law, providing a large fiscal stimulus package aimed at mitigating the economic effects of the coronavirus. Further, in March 2021, the U.S. Congress approved additional fiscal stimulus of $1.9 trillion in response to the ongoing economic effects of COVID-19. The extraordinary measures taken by the Federal Reserve and U.S. Congress to support the economy and stabilize financial markets impacted bond markets (deliberately driving government and corporate yields lower) and equity markets (creating upward pressure on valuations and downward pressure on yields for dividend paying companies such as utilities). However, these fiscal

stimulus programs have increased inflationary pressures, as evidenced in the sharp run-up in inflation and corresponding upward spike in bond yields since late November 2021.

Q. Please describe the effect of these policies on inflation.

A. Inflation has been a significant cause for concern among both policymakers and investors. As shown in Figure 2 below, inflation levels have moved within a relatively narrow band over the past twenty years (other than during the Great Recession of 2007/08). Starting with the initial stages of the COVID-19 pandemic in 2020, inflation levels have been driven higher, reaching 9.1 percent in June of 2022. Inflation is being driven by strong consumer demand and supply constraints, some of which have been exacerbated by COVID-19 related government stimulus programs in the U.S. and international lockdowns. More recently, the war in Ukraine, which began in February 2022, is constraining the world's supply of food and fuel, pushing overall inflation higher and feeding into the cost of other products and services.

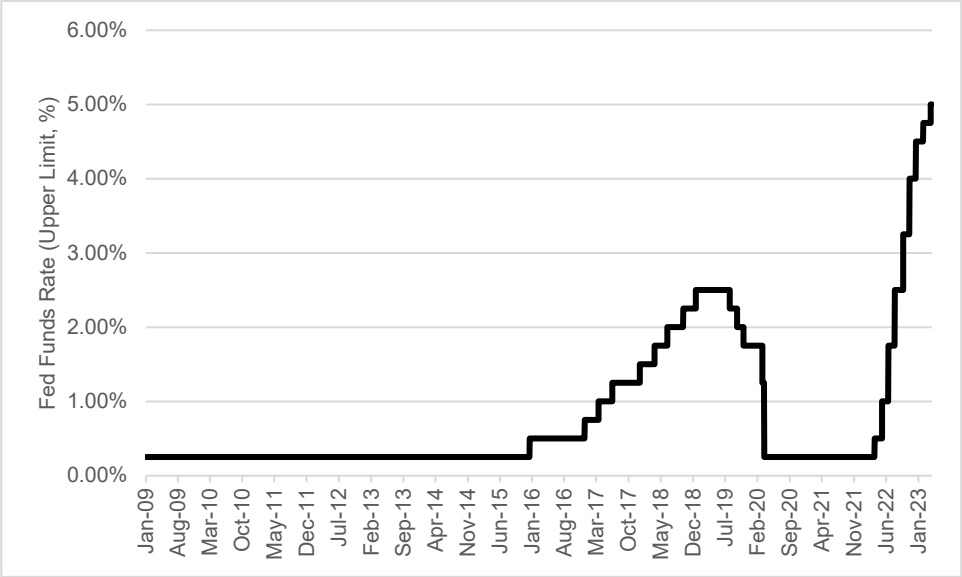
Figure 2: Consumer Price Index, 12-month Percentage Change[11]



As a result of these substantially higher inflation rates, the Federal Reserve has been left little choice but to pull back on its COVID-related monetary policies and apply tighter monetary policy with higher interest rates. In 2022, the Federal Reserve increased the target rate seven times, and in 2023 another two times, as illustrated in Figure 3.

[11] Source: Bureau of Labor Statistics, https://www.bls.gov/charts/consumer-price-index/consumer-price-index-by-category-line-chart.htm.

Figure 3: Federal Funds Rate, Target Range, Upper Limit[12]



This demonstrates the level of Federal Reserve action necessary to reel in inflation. The Federal Reserve is willing to risk substantially higher interest rates and a slowdown in the economy, and it is clear that the era of record low interest rates and moderate inflation has closed. In addition, the Federal Reserve is executing its plan to reduce the size of its balance sheet by selling $95 billion in bonds each month, which will further tighten financial markets. In its most recent policy statement, the Fed confirmed its commitment to a 2 percent inflation target.

[12] Source: Federal Reserve Bank of St. Louis, FRED Economic Data, https://fred.stlouisfed.org/series/DFEDTARU .

> The Committee will closely monitor incoming information and assess the implications for monetary policy. The Committee anticipates that some additional policy firming may be appropriate in order to attain a stance of monetary policy that is sufficiently restrictive to return inflation to 2 percent over time. In determining the extent of future increases in the target range, the Committee will take into account the cumulative tightening of monetary policy, the lags with which monetary policy affects economic activity and inflation, and economic and financial developments. In addition, the Committee will continue reducing its holdings of Treasury securities and agency debt and agency mortgage-backed securities, as described in its previously announced plans. The Committee is strongly committed to returning inflation to its 2 percent objective.[13]

This is significant because the costs of capital are affected by the Federal Reserve's actions, even though it only sets the short-term interest rates. Further, the Inflation Reduction Act of 2022 ("IRA") was introduced intending to curb inflation by targeting drug costs, enabling investments in clean energy, raise tax revenues, and decrease the deficit.

Q. Are conditions expected to stabilize in the near term?

A. No, although difficult to predict in the long term, capital market conditions remain highly uncertain. The economy and capital markets remain in a tenuous state with ongoing global supply constraints and expectations of weaker consumer demand in the face of persistent inflation and higher interest rates. Interest rates may not fully reflect the effects of inflation

[13] Federal Reserve Press Release, March 22, 2023.

because no one knows exactly how high interest rates must go before inflation returns to more acceptable levels. The Federal Reserve has been targeting inflation of around 2.0 percent for many years, as compared with current annualized inflation of 6.0 percent at the consumer level for the 12-month period ending in February 2023.

Looking forward, the inflation risks in the market are twofold: (1) either inflation becomes embedded in the economy if the Federal Reserve does not move aggressively enough in tightening monetary policy, or (2) the Federal Reserve responds to inflationary pressure by raising short-term interest rates to a level that causes a slowdown in economic growth or a recession. Consequently, there is elevated market volatility and investors are concerned the Federal Reserve's push for tighter monetary policy in reaction to inflation may lead to a "hard landing." Recent developments in the banking sector suggest that we may already be seeing evidence of distress in equity markets.

As it relates to utilities, Moody's revised its outlook for the US regulated utilities industry from "stable" to "negative," citing concerns related to inflation and rising interest rates.[14] While the IRA contains credit positive provisions, they "are not sufficient to offset the combined adverse effects of

[14] Moody's Investors Service, "Regulated Electric and Gas Utilities – US 2023 outlook negative due to higher natural gas prices, inflation and rising interest rates," November 10, 2022, at 1.

high natural gas prices, inflation and rising interest rates."[15] In addition, Moody's notes that "[f]inancial metrics already under pressure with little cushion entering 2023," with funds from operations ("FFO")-to-debt ratios expected to decline further with anticipated delays in cost recovery.[16]

Q. Have you factored these circumstances into your cost of equity estimates?

A. Yes. I have relied on the most recent market data and forecasts available to me in my analysis. Long-term interest rates have increased substantially since the historical lows of 2020 and are expected to continue to increase as the Federal Reserve focuses on inflation. This supports the use of both current and forecast bond yields in the CAPM. These circumstances also reinforce the importance of considering the results of multiple models, as I have with the DCF and CAPM approaches.

Q. What is your conclusion with regard to the effect of capital market conditions on the cost of equity?

A. The current capital market conditions are heavily influenced by monetary policy aimed at mitigating inflationary pressures. This has caused both short-term and long-term interest rates to increase substantially since the historical lows of 2020 suggesting a higher cost of capital. Therefore, it is

[15] *Id.*, at 5.
[16] *Id.*, at 1.

important that these factors are accounted for in the cost of equity models and the appropriate weight is given to the approaches that best account for the current dynamics of the capital market environment.

V. PROXY GROUP SELECTION

Q. Please explain why you have used a group of proxy companies to determine the cost of equity for the Companies.

A. In these proceedings, we are focused on estimating the cost of equity for the Companies' rate-regulated, natural gas distribution utility operations in New York. Because ROE is a market-based concept and the Companies are not publicly traded, it is necessary to establish a group of companies that are both publicly traded and comparable to the Companies. Even if the Companies were publicly traded entities, it is possible that transitory events could bias the Companies' market value in one way or another in a given period of time. A significant benefit of using a proxy group is the ability to mitigate the effects of short-term events that may be associated with any specific company. The proxy companies used in my ROE analyses possess a set of business and operating characteristics similar to the Companies' utility operations, and thus provide a reasonable basis for estimating the Companies' ROE.

Q. Please provide a summary profile of the Companies.

A. KEDNY's business consists of its regulated natural gas transportation and distribution operations in New York State. KEDNY distributes natural gas to retail customers in the New York City boroughs of Brooklyn, Staten Island, and Queens.[17] KEDNY's long-term issuer ratings are Baa2 (Moody's, Outlook: Stable), BBB+ (S&P, Outlook: Stable) and BBB (Fitch, Outlook: Stable).[18]

KEDLI's business consists of its regulated natural gas transportation and distribution operations in New York State. KEDLI distributes natural gas to retail customers in Long Island, New York and Queens, New York.[19] KEDLI's long-term issuer ratings are Baa1 (Moody's, Outlook: Stable), BBB+ (S&P, Outlook: Stable) and BBB+ (Fitch, Outlook: Stable).[20] The Capital Structure Panel provides a more detailed discussion of the Companies' credit ratings.

Q. How did you select the companies included in your proxy groups?

A. The Commission has historically relied on proxy groups generally composed of electric and combination utilities even for the purposes of establishing the ROE for a natural gas distribution utility. In recognition of that practice, I began with the companies that Value Line classifies as

[17] KEDNY, Consolidated Financial Statements, at 11.
[18] Source: SNL Financial, accessed February 13, 2023.
[19] KEDLI, Consolidated Financial Statements, at 10.
[20] Source: SNL Financial, accessed February 13, 2023.

"Electric Utilities." However, it is important to recognize that the Companies' operations are entirely natural gas distribution operations. Therefore, I also included companies that Value Line classifies as and "Natural Gas Distribution Companies." That combined group includes 47 domestic U.S. utilities. I applied the following screening criteria to establish a risk-comparable proxy group that includes electric utility companies and natural gas distribution companies:

- To ensure that information regarding the proxy group companies is consensus-based, I eliminated the companies that are not covered by at least two utility industry equity analysts;
- I eliminated companies that do not have investment grade corporate credit ratings and/or senior unsecured bond ratings according to S&P and Moody's because such companies do not have a similar financial risk profile to that of the Companies;
- I eliminated companies that have not paid regular dividends or do not have positive earnings growth projections from at least one source because such characteristics are incompatible with the DCF model;
- To ensure that the proxy group consists of companies that are primarily transmission and distribution regulated utilities, I eliminated companies that have owned generation comprising greater than 60 percent of the company's MWh sales to ultimate customers;

- To ensure that the proxy group consists of companies that are primarily regulated utilities, I eliminated companies with less than 70 percent of total operating income derived from regulated utility operations; and
- I eliminated companies known to be party to a merger, acquisition, or other transformational transaction as such activities may have a temporary effect on such companies' stock prices and projections unrelated to the overall cost of capital.

Q. What is the composition of your proxy group?

A. My proxy group consists of the 19 companies presented in Figure 4.

Figure 4: Proxy Group Composition

Company	Ticker
ALLETE, Inc.	ALE
American Electric Power Company, Inc.	AEP
Atmos Energy Corporation	ATO
Avista Corporation	AVA
Black Hills Corporation	BKH
CMS Energy Corporation	CMS
Edison International	EIX
Eversource Energy	ES
Exelon Corporation	EXC
Hawaiian Electric Industries, Inc.	HE
NiSource Inc.	NI
Northwest Natural Gas Company	NWN
NorthWestern Corporation	NWE
OGE Energy Corporation	OGE
ONE Gas, Inc.	OGS
PPL Corporation	PPL
Public Service Enterprise Group Inc.	PEG
Spire, Inc.	SR
Xcel Energy Inc.	XEL

Q. Has the Commission typically relied on similar screening criteria when estimating the ROE?

A. Yes. The Commission has typically relied on screening criteria that are similar to those that I have used to develop my proxy groups. The proxy group that is typically relied on by the Commission is composed of a large group of dividend-paying companies with investment grade bond ratings and regulated revenues of at least 70 percent that are not subject to merger-related or corporate restructuring activities.[21] However, in establishing my proxy group, I relied on the percentage of net operating income derived

[21] See, e.g., Case 13-E-0030, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Electric Service, Testimony of Craig E. Henry, at 14-16.

from regulated operations instead of the percentage of total revenue derived from regulated operations because net operating income is more representative of the contribution of that business segment to earnings and the corporation's overall financial position. Specifically, a significant portion of gas and electric utility company revenue is derived from the costs of purchased gas, purchased fuel, and purchased power, which, in most cases, are recoverable through tracking mechanisms and do not, therefore, contribute to earnings. Furthermore, this portion of total revenue can fluctuate considerably based on the cost of gas and other inputs. Therefore, relying exclusively on a revenue screen does not provide a clear or necessarily consistent indicator of the contribution of the regulated utility operations to a company's earnings. Net operating income excludes the cost of purchased commodity and therefore more closely represents the contribution of the business segment to earnings.

VI. COST OF EQUITY ESTIMATION

Q. How is the required ROE determined?

A. While the cost of debt can be directly observed, the cost of equity and the required ROE are market-based and, therefore, must be estimated based on observable market information. The required ROE is determined by using one or more analytical techniques that rely on market data to quantify investor expectations regarding the range of required equity returns. Informed judgment is applied, based on the results of those analyses, to

determine where within the range of results the cost of equity for a company falls. As a general proposition, the key consideration in determining the cost of equity is to ensure that the methodologies employed reasonably reflect investors' views of the financial markets, the proxy group companies, and the subject company's risk profile.

Q. What methods did you use to determine the Companies' cost of equity?

A. Consistent with Commission precedent, I used the DCF model and CAPM approaches. In establishing my recommended ROE, I relied on a multi-stage form of the DCF model, and, consistent with the Commission's stated preference, I used both the traditional form of the CAPM as well as the Zero-Beta form of that model. In both forms of the CAPM, I incorporated a forward-looking measure of the Market Risk Premium.

Q. Why do you believe it is important to use more than one analytical approach?

A. Because the cost of equity is not directly observable, it must be estimated based on both quantitative and qualitative information. When faced with the task of estimating the cost of equity, analysts and investors are inclined to gather and evaluate as much relevant data as reasonably can be analyzed. As a result, a number of models have been developed to estimate the cost of equity. For that reason, I use multiple approaches to estimate the cost of equity. However, all of the models available for estimating the cost

of equity are subject to limiting assumptions or other methodological constraints.

Q. How are current market conditions affecting the results of the DCF and CAPM models?

A. As discussed in Section IV, current capital market conditions are heavily influenced by monetary policy aimed at mitigating inflationary pressures. While long-term interest rates have increased substantially since the historical lows of 2020, near-term conditions may not reflect investors long-term expectations. Importantly, the risk-free rate in the CAPM analysis should also consider projected yields on Treasury bonds, and the market risk premium should be based on a forward-looking computation of the expected return on the total market less the risk-free rate. Market risk premiums based on long-term historical averages are unresponsive to movements in interest rates and would likely understate the market risk premium and, accordingly, the cost of equity.

Fluctuations in utility valuations impact the results of the DCF model. The dividend yield is calculated using historical average stock prices, which may not fully reflect forward market expectations, especially as utility valuations may not reflect the effect of recent increases in interest rates. Long-term interest rates have increased substantially since the historical lows of 2020 as the Federal Reserve continues to mitigate strong inflationary pressures. Importantly, the CAPM directly accounts for

changes in interest rates. These circumstances collectively reinforce the importance of using multiple models, as I have with the DCF and CAPM approaches.

A. DISCOUNTED CASH FLOW MODEL

Q. Please describe the DCF approach.

A. The DCF approach is based on the theory that a stock's current market price represents the present value of all expected future cash flows. In its most general form, the DCF model is expressed as follows:

$$P_0 = \frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \cdots + \frac{D_n}{(1+r)^n} \qquad [1]$$

Where P_0 represents the current market stock price, D_1, D_2, … D_n are all expected future dividends, and *r* is the discount rate, or required ROE. As discussed below, I have not included the constant growth form of the DCF model, but instead have focused on a multi-stage form of the DCF model.

Q. Please generally describe the DCF model you relied on.

A. The multi-stage DCF model is an extension of the constant growth form that enables the analyst to specify growth rates over multiple stages. As with the constant growth form of the DCF model, the multi-stage form defines the cost of equity as the discount rate that sets the current price equal to the discounted value of future cash flows. A multi-stage DCF model addresses the possibility that mean five-year growth rates may not be reasonable in perpetuity and that payout ratios could vary over time.

Q. Please describe the structure of the multi-stage DCF model.

A. The multi-stage DCF model that I have used sets the proxy company's current stock price equal to the present value of future cash flows received over three time periods. In all three periods, cash flows are equal to the annual dividend payments that stockholders receive. The first period is a short-term growth period that consists of the first five years; the second period is a transition period from the short-term growth rate to the long-term growth rate that occurs over five years (*i.e.*, years 6 through 10); and the third period is a long-term growth period that begins in year 11 and continues in perpetuity. The ROE is then calculated as the rate of return that results from the initial stock investment and the dividend payments over the analytical period.

Q. Has the Commission relied on a multi-stage DCF model in prior cases?

A. Yes, the Commission has relied on a two-stage form of the DCF model in prior cases.[22] The two-stage model that the Commission has relied on and the multi-stage model that I rely on both define the cost of equity as the discount rate that sets the current stock price equal to the discounted value of future cash flows that are expressed as projected dividends. Both models project dividends using growth rates over multiple periods.

[22] See Case 10-E-0362, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Orange and Rockland Utilities, Inc. for Electric Service, Order Establishing Rates for Electric Service, (issued June 17, 2011) ("2011 O&R Rate Order"), at 68-69.

Q. Is the multi-stage form of the DCF model consistent with the intent of the two-stage model relied upon by the Commission?

A. Yes. Both the construction of the multi-stage model and the underlying assumptions are consistent with the two-stage model relied upon by the Commission. The constant growth DCF model assumes the expected growth rate will be constant in perpetuity. The multi-stage forms of the DCF model, including both the two-stage model that the Commission has relied upon and the multi-stage form of the model that is relied on in my analysis, recognize short and long-term growth prospects.

Q. Does the multi-stage form of the DCF model offer improvements over the two-stage model traditionally relied upon by the Commission?

A. Yes. The general form of the two-stage model relied upon by the Commission involves a near-term growth stage based on projected dividends and a long-term growth stage employing an estimated long-term growth rate in dividends.[23] The Commission's application of a two-stage DCF assumes that a company's growth abruptly shifts to a long-run growth state after the initial five-year period. In contrast, the multi-stage model relies on growth rates over three periods, as described above. The multi-stage form of the DCF model provides for a gradual transition to a

[23] See, e.g., Case 10-E-0362, Case 06-E-1433, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Orange and Rockland Utilities, Inc., for Electric Service, Case 08-E-0539, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Electric Service.

company's expected long-term growth, whereas the two-stage DCF model assumes the transition from short to long-term growth occurs in one year.

Q. What market data did you use to calculate the current stock price in your DCF model?

A. The stock prices that I relied on in my DCF model are based on the average market closing prices for the proxy companies over the three months ended March 31, 2023.

Q. What growth rates did you rely on in the multi-stage DCF model?

A. As shown in Exhibit __ (JCN-1), I began with the current annualized dividend as of March 31, 2023 for each proxy group company. In the first stage of the model, the current annualized dividend is escalated based on the average of the three-to five-year earnings growth estimates reported by First Call, Zacks, and Value Line. For the third stage of the model, I relied on long-term projected growth in Gross Domestic Product ("GDP"). The second stage growth rate is a transition from the first stage growth rate to the long-term growth rate on a geometric average basis.

Q. Why do you believe that earnings growth rates are the appropriate growth rates in the DCF model?

A. Earnings are the fundamental driver of a company's ability to pay dividends; therefore, earnings growth is the appropriate measure of a

company's long-term growth. In contrast, changes in a company's dividend payments are based on management decisions related to cash management and other factors. For example, a company may decide to retain certain earnings rather than include those earnings in a dividend issuance. Therefore, dividend growth rates are less likely than earnings growth rates to reflect investor perceptions of a company's growth prospects.

Q. Is there support for the use of analysts' earnings growth estimates in the DCF model?

A. Yes, there is significant academic support for the use of analysts' earnings growth rates. In addition, the majority of the data that are publicly available to investors sets forth analysts' projections of earnings growth rates. Value Line is the only publication I am aware of that provides projected dividend growth rates.

Q. Please summarize the academic research on growth rates and stock valuation.

A. The relationship between various growth rates and stock valuation metrics has been the subject of much academic research. Many published articles specifically support the use of analysts' earnings growth projections in the DCF model in general, as well as for a method of calculating the expected market risk premium. While this article is focused on the calculation of the CAPM, Dr. Robert Harris demonstrates that financial analysts rely on

earnings forecasts (referred to in the article as "FAF") and the use of a constant growth DCF formula to estimate the expected market risk premium.[24] Dr. Harris made the following observations:

> […] a growing body of knowledge shows that analysts' earnings forecasts are indeed reflected in stock prices. Such studies typically employ a consensus measure of FAF calculated as a simple average of forecasts by individual analysts.[25]
>
> *****
>
> Given the demonstrated relationship of FAF to equity prices and the direct theoretical appeal of expectational data, it is no surprise that FAF have been used in conjunction with DCF models to estimate equity return requirements.[26]

Dr. Harris's work demonstrates that analysts rely on earnings as the appropriate measure of growth in the DCF model. Professors Carleton and Vander Weide also performed a study to determine whether projected earnings growth rates are superior to historical measures of growth in the implementation of the DCF model.[27] Although the purpose of that study was to "investigate what growth expectation is embodied in the firm's current stock price,"[28] the authors clearly indicate the importance of earnings projections in the context of the DCF model., concluding that:

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[24] Robert S. Harris, Using Analysts' Growth Forecasts to Estimate Shareholder Required Rates of Return, Financial Management, Spring 1986, at 66.
[25] *Id*. at 59.
[26] *Id*. at 60.
[27] James H. Vander Weide, Willard T. Carleton, *Investor growth expectations: Analysts vs. history,* The Journal of Portfolio Management, Spring 1988.
[28] *Id*. at 78.

> […] our studies affirm the superiority of analysts' forecasts over simple historical growth extrapolations in the stock price formation process. Indirectly, this finding lends support to the use of valuation models whose input includes expected growth rates.[29]

Similarly, Harris and Marston presented "estimates of shareholder required rates of return and risk premia which are derived using forward-looking analysts' growth forecasts."[30] In addition to other findings, Harris and Marston reported that,

> […] in addition to fitting the theoretical requirement of being forward-looking, the utilization of analysts' forecasts in estimating return requirements provides reasonable empirical results that can be useful in practical applications.[31]

The Carleton and Vander Weide study was updated to determine whether the finding that analysts' earnings growth forecasts are relevant in the stock valuation process still holds. The results of that updated study continued to demonstrate the importance of analysts' earnings forecasts, including the application of those forecasts to utility companies.[32] Similarly, Brigham, Shome and Vinson noted that "evidence in the current literature indicates that (1) analysts' forecasts are superior to forecasts based solely on time series data; and (2) investors do rely on analysts' forecasts."[33]

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[29] *Id*. at 82.
[30] Robert S. Harris, Felicia C. Marston, *Estimating Shareholder Risk Premia Using Analysts' Growth Forecasts*, Financial Management, Summer 1992.
[31] *Id*. at 63.
[32] Advanced Research Center, *Investor Growth Expectations*, Summer, 2004.
[33] The Risk Premium Approach to Measuring a Utility's Cost of Equity, Financial Management, Spring 1985.

Q. Do you believe the Commission should adjust its historical reliance on dividend per share growth rates during the initial five-year term of its Two-Stage DCF?

A. Yes. Sole reliance on Value Line projections of dividend per share growth is not appropriate for several reasons. First, the use of only dividend growth rates ignores the substantial body of academic research demonstrating that earnings growth rates are the most relevant factor in stock price valuation.[34] Second, projections of dividend growth, which would not include growth in retained earnings, only measure a portion of a company's growth. Therefore, earnings growth projections are more complete estimates of total company growth than projected dividend growth rates. Finally, Value Line's 4-6 year projections are not consensus estimates, but reflect the viewpoint of a single analyst. Therefore, the Commission's models, which have historically relied only on projected dividend per share growth rates from Value Line, reflect the growth expectations of a single analyst in the first stage of the model. In contrast, there are several consensus estimates of projected earnings per share growth rates that are publicly available and widely used by investors, including Zacks Investment Research and First Call. Each of these consensus forecasts considers the growth expectations for each company based on the expectations of multiple analysts. It is not reasonable to exclude these timely and widely available sources of

[34] The Generic Finance RD indicates that the Telecommunications Group, which included Commission Staff, supported the use of earnings per share growth in the DCF models employed to estimate the ROE (RD at 9).

information from the analysis when these real-time sources have become the more common data points relied on by investors.

Q. How did you calculate the long-term GDP growth rate?

A. As shown in Exhibit __ (JCN-2), the long-term growth rate of 5.46 percent is based on the real GDP growth rate of 3.16 percent from 1929 through 2022,[35] and a projected long-term inflation rate of 2.23 percent.[36] The projected rate of inflation is based on three measures beyond the 2023-2028, near-term growth period: (1) the average long-term projected growth rate in the Consumer Price Index ("CPI") of 2.10 percent, as reported by Blue Chip Financial Forecasts;[37] (2) the compound annual growth rate of the CPI for all urban consumers for 2031-2050 of 2.35 percent as projected by the Energy Information Administration ("EIA") in the Annual Energy Outlook 2022; and (3) the compound annual growth rate of the GDP chain-type price index for 2031-2050 of 2.25 percent, also reported by the EIA in the Annual Energy Outlook 2022.[38]

[35] U.S. Department of Commerce, Bureau of Economic Analysis, National Income and Product Accounts Tables, Table 1.1.6, February 28, 2023.
[36] (1 + 0.0316) x (1 + 0.0223) -1 = 0.0546 or 5.46%.
[37] Blue Chip Financial Forecasts, Vol. 41, No. 12, December 1, 2022, at 14.
[38] U.S. Energy Information Administration, Annual Energy Outlook 2022, Table 20.

Q. Why is the long-term GDP growth rate a reasonable estimate of long-term growth in the multi-stage DCF model?

A. Long-term estimates of GDP growth are commonly used in regulatory proceedings as a proxy for the long-term growth rate in the multi-stage DCF analysis. That application is based on the common theoretical assumption that, over the long-run, all companies in the economy will tend to grow at the same constant rate. That assumption is designed to address the uncertainty associated with estimating individual company growth rates over very long time horizons and is not meant to suggest that company growth rates in the economy will indeed converge in practice over any given period.

Q. Is your calculation of GDP growth consistent with the way in which other analysts' compute estimates of long-term GDP growth?

A. Yes. Investors understand that the U.S. economy goes through cycles of growth and contraction. Therefore, it is appropriate to consider the longest period possible to measure historical real growth in GDP. This view is consistent with Morningstar's explanation about measuring GDP growth:

> Growth in real GDP (with only a few exceptions) has been reasonably stable over time; therefore, its historical performance is a good estimate of expected long-term future performance. By combining the inflation estimate with the real growth rate estimate, a long-term estimate of nominal growth is formed.[39]

[39] Ibbotson and Associates, Stocks, Bonds, Bills and Inflation, 1926-2012, 2013 Valuation Yearbook, at 52.

Furthermore, Morningstar supports the use of long-term historical data:

> The 87-year period starting with 1926 is representative of what can happen: it includes high and low returns, volatile and quiet markets, war and peace, inflation and deflation, and prosperity and depression. Restricting attention to a shorter historical period underestimates the amount of change that could occur in a long future period. Finally, because historical event-types (not specific events) tend to repeat themselves, long-run capital market return studies can reveal a great deal about the future. Investors probably expect "unusual" events to occur from time to time, and their return expectations reflect this.[40]

Q. How does your estimate of long-term GDP growth compare with investor expectations of long-term utility industry growth rates?

A. The Commission has traditionally relied on Bank of America Merrill Lynch's ("BAML") market return calculations in estimating a company's ROE using the CAPM. Exhibit __ (JCN-3) includes the relevant pages from the BAML *Quantitative Profiles* reports for December 2022 through February 2023. BAML derives the Implied Return using a multi-stage Dividend Discount Model ("DDM"). As shown in Exhibit __ (JCN-3), the Implied Returns for the utility industry were 9.70 percent to 9.80 percent each month.[41] For those same months, the average dividend yield for the utility industry was 3.00 percent to 3.10 percent.[42] Because the total return consists of capital appreciation (*i.e.*, growth) and dividend yield, that data suggest an expected utility growth rate of approximately 6.60 percent to

[40] *Id*., at 59.
[41] Bank of America Merrill Lynch, *Quantitative Profiles,* December 26, 2022, at 36, January 16, 2023 at 29, and February 10, 2023 at 29.
[42] *Id*.

6.80 percent, which is considerably higher than the long-term growth estimate of 5.46 percent used in my multi-stage DCF analysis.

Q. How does your estimate of long-term growth differ from the estimate the Commission has traditionally relied on?

A. The final stage of both the two-stage DCF model that the Commission has relied on and my multi-stage DCF model extends into the future indefinitely. My long-term growth estimate reflects investors' long-term growth expectations for the period from 2032 through 2050. Therefore, the third stage of my multi-stage DCF model reflects investor growth expectations beginning in the first year of the third stage of the model. In contrast, the growth estimate for the two-stage model that the Commission has typically relied on is based on short-term growth rate forecasts. The use of the sustainable growth rate, calculated using Value Line's published projections, provides an estimate of growth four- to six-years in the future. Relying on the sustainable growth rate in perpetuity in the second stage of a two-stage DCF model does not provide a long-run estimate of growth. Rather, the use of the sustainable growth rate assumes that the short-term estimate for the four- to six-year period from the Value Line report date is sustained in perpetuity.

In contrast, the long-term growth rate in my DCF analyses reflects both economic forecasts and market-derived projections of inflation over the longest available time period (30 or more years). Those estimates of long-

term inflation expectations are combined with the long-term average historical real GDP growth rate to calculate an expected nominal GDP growth rate. Consequently, the long-term growth estimate in my multi-stage DCF model represents investors and economists' views of nominal long-term GDP growth well beyond the time horizon reflected in the four- to six-year Value Line sustainable growth estimate relied on by the Commission in prior cases.

Q. Are there other problems with the use of the sustainable growth rate as an estimate of long-term growth?

A. Yes. Using the sustainable growth rate to estimate the long-term growth of a company uses a very narrowly-defined set of short-term projections based on Value Line data. Specifically, it relies on the following assumptions: (1) projected dividends for year 2; (2) projected dividends for years 4-6; (3) projected earnings for years 4-6; (4) projected book value for year 2; (5) projected book value for years 4-6; (6) current estimate of actual outstanding shares of stock; (7) projected shares of outstanding stock for years 4-6; and (8) current three-month stock price. Each of these assumptions is estimated at most for six years into the future. As defined using these assumptions, the sustainable growth rate, which is applied over the long-term in the Commission's two-stage model, does not consider any actual long-term forecasts for a specific company or the economy.

Q. What is your conclusion regarding the methodology typically relied on by the Commission to estimate the sustainable growth rate in the two-stage DCF model?

A. There are several reasons why the Commission's current view of the sustainable growth rate should not be continued in the two-stage DCF model. First, the sustainable growth rate is not a long-term measure of growth and as such should not be applied in perpetuity in the second stage of the model. Second, the exclusive use of Value Line data, which is a single analyst's viewpoint, to establish the sustainable growth rate assumes that investors do not consider any of the other financial information that is widely available when establishing future dividend expectations. In addition, the sustainable growth rate calculation includes Value Line's ROE projections as an input, implicitly accepting them as reasonable. However, Value Line's ROE projections are often significantly different from the ROE estimates produced by the two-stage DCF model. Finally, the Commission's sustainable growth rate methodology implicitly assumes that investors establish long-term growth expectations based entirely on short-term, company-specific projections. It is unreasonable to conclude that investors would ignore the expectations of long-term macroeconomic growth in establishing the long-term growth estimates for an electric or natural gas distribution utility or any other company.

Q. What are the results of your DCF analyses?

A. As shown in Exhibit __ (JCN-1), the multi-stage DCF analysis based on a three-month average stock price and a range of near-term growth rate assumptions produces a mean ROE of 9.68 percent.

Q. What are your conclusions about the results of the DCF model?

A. The conditions that warranted the Commission's GFP inquiry and the subsequent Generic Finance RD in the early 1990s exist again today with DCF results considerably lower than those from other models, such as the CAPM, as well as returns authorized in other jurisdictions. Therefore, it is important to recognize that the results of the DCF model are likely understated.

B. CAPITAL ASSET PRICING MODEL

Q. Please briefly describe the Capital Asset Pricing Model.

A. The CAPM is a risk premium approach that estimates the cost of equity for a given security as a function of a risk-free return plus a risk premium (to compensate investors for the non-diversifiable or "systematic" risk of that security).[43] As shown in Equation [3], the CAPM is defined by four components, each of which must theoretically be a forward-looking estimate:

[43] Systematic risks are fundamental market risks that reflect aggregate economic measures and therefore cannot be mitigated through diversification. Unsystematic risks reflect company-specific risks that can be mitigated and ultimately eliminated through investments in a portfolio of companies and/or market sectors.

$$k_e = r_f + \beta(r_m - r_f) \qquad [2]$$

where:

k_e = the required market ROE

β = Beta coefficient of an individual security

r_f = the risk-free rate of return

r_m = the required return on the market as a whole

The term (r_m – r_f) represents the Market Risk Premium ("MRP"). According to the theory underlying the CAPM, unsystematic risk can be diversified away, therefore investors should be concerned only with systematic or non-diversifiable risk. Non-diversifiable risk is measured by Beta, which is defined as:

$$\beta = \frac{Covariance(r_e, r_m)}{Variance(r_m)} \qquad [3]$$

where:

r_e = the rate of return for the individual security or portfolio.

The variance of the market return, noted in Equation [4], is a measure of the uncertainty of the general market, and the covariance between the return on a specific security and the market reflects the extent to which the return on that security will respond to a given change in the market return. Thus, Beta represents the risk that the selected security will not be effective in diversifying systematic market risks.

Q. What risk-free rate did you use in your CAPM model?

A. I used three estimates of the yield on Treasury bonds: (1) the current three-month average yield on 30-year Treasury bonds (3.75 percent);[44] (2) the projected 30-year Treasury yield for 2023-2024 (3.88 percent);[45] and (3) the projected 30-year Treasury yield for the period 2024-2028 (3.90 percent).[46] In determining the security most relevant to the application of the CAPM, it is important to select the term (or maturity) that best matches the life of the underlying investment. As noted by Morningstar:

> The traditional thinking regarding the time horizon of the chosen Treasury security is that it should match the time horizon of whatever is being valued… Note that the horizon is a function of the investment, not the investor. If an investor plans to hold stock in a company for only five years, the yield on a five-year Treasury note would not be appropriate since the company will continue to exist beyond those five years.[47]

Because utility companies represent long-duration investments, it is appropriate to use yields on long-term Treasury bonds as the risk-free rate component of the CAPM. In my view, the 30-year Treasury bond is the appropriate security for that purpose. Because the cost of capital is intended to be forward-looking, it is appropriate to consider projected measures of the market risk premium and interest rates.

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44 Bloomberg Professional.
45 Aspen Publishers, Blue Chip Financial Forecasts, Vol. 42, No. 3 March 1, 2023, at 2.
46 Aspen Publishers, Blue Chip Financial Forecasts, Vol. 41, No. 12 December 1, 2023, 14.
47 Morningstar Inc., Ibbotson SBBI 2013 Valuation Yearbook, at 44.

Q. Please describe your estimate of the market risk premium used in your CAPM.

A. The estimated market risk premium is based on the expected return on the S&P 500 Index, less the 30-year Treasury bond yield. The expected return on the S&P 500 Index is calculated using a DCF model for all companies in the index based on market capitalization-weighted growth rates and dividend yields. The market risk premium implied by each of the three Treasury yields discussed above is used in the CAPM analysis.

The Commission previously has relied upon the calculation of a projected market risk premium, based on the difference between the estimated forward-looking required market return for the S&P 500, as provided by BAML, and the risk-free rate.[48] As a practical matter, that approach is consistent with the Market DCF-derived forward-looking market risk premium estimate discussed above (*see also* Exhibit __ (JCN-5). Nonetheless, I have relied on the average of: (1) forward-looking required market return for the S&P 500 as calculated using a DCF model for all companies in the index; and (2) the estimated forward-looking required market return for the S&P 500, as provided by BAML.

[48] *See e.g.,* 2011 O&R Rate Order, at 77.

Q. What Beta coefficient did you use in your CAPM model?

A. I considered the average Beta coefficients for the proxy group companies as reported by Bloomberg and Value Line (*see* Exhibit __ (JCN-4)). The Beta coefficients reported by Bloomberg were calculated using ten years of weekly returns relative to the S&P 500 Index. Value Line's calculation is based on five years of weekly returns relative to the New York Stock Exchange Composite Index.

Q. Did you consider another form of the CAPM in your analysis?

A. Yes. In prior proceedings, the Commission has relied upon the Zero-Beta CAPM (the form of which is sometimes referred to as the "Empirical CAPM"[49]) in estimating the cost of equity. The Zero-Beta CAPM calculates the product of the adjusted Beta coefficient and the market risk premium and applies a weight of 75.00 percent to that result. The model then applies a 25.00 percent weight to the market risk premium, without any effect from the Beta coefficient. The results of the two calculations are summed, along with the risk-free rate, to produce the Zero-Beta CAPM result, as noted in Equation [4] below:

$$k_e = r_f + 0.75\beta(r_m - r_f) + 0.25(r_m - r_f) \qquad [4]$$

where:

k_e = the required market ROE

β = Adjusted Beta coefficient of an individual security

[49] *See e.g.,* Roger A. Morin, New Regulatory Finance, Public Utilities Reports, Inc., 2006, at 189.

r_f = the risk-free rate of return

r_m = the required return on the market as a whole

In essence, the Zero-Beta form of the CAPM addresses the tendency of the "traditional" CAPM to underestimate the cost of equity for companies with low Beta coefficients such as regulated utilities. In that regard, the Zero-Beta CAPM is not redundant to the use of adjusted Betas; rather, it recognizes the results of academic research indicating that the risk-return relationship is different (in essence, flatter) than estimated by the CAPM, and that the CAPM underestimates the "alpha," or the constant return term.[50]

As with the CAPM, my application of the Zero-Beta CAPM uses the forward-looking market risk premium estimates, the three yields on 30-year Treasury securities noted earlier as the risk-free rate, and the Bloomberg and Value Line Beta coefficients. Exhibit __ (JCN-4) show the results of the CAPM models. The traditional CAPM model results range from 10.96 percent to 10.99 percent. The Zero-Beta CAPM model results range from 11.29 percent to 11.31 percent. The range established by the traditional CAPM model and the Zero-Beta CAPM model results is 10.96 percent to 11.31 percent with a mean of 11.14 percent.

[50] *Id*. at 191.

C. Weighted Average Results

Q. Please summarize the results of your analysis and your recommended ROE.

A. As shown in Figure 5 (below), I have presented the DCF and CAPM results weighted equally, and with the Generic Finance RD's proposed 2/3 weighting of the DCF and 1/3 weighting of the CAPM.

Figure 5: Weighted Average Analytical Results

	Low	Mean	High
DCF	9.13%	9.68%	10.31%
Mean CAPM	11.13%	11.14%	11.15%
67%/33% DCF/CAPM	9.79%	10.17%	10.59%
50%/50% DCF/CAPM	10.13%	10.41%	10.73%

Q. What was the reasoning for developing its weighting of the DCF and CAPM methodologies in the Generic Finance RD?

A. At the time of the Generic Finance RD, the Commission did not have a significant amount of experience with the CAPM. The Generic Finance RD noted that the Commission had historically used the CAPM as a check on its DCF results, and was somewhat undecided as to "how far the Commission should go in elevating the status of CAPM."[51] The Generic Finance RD opted for a gradual transition towards the CAPM, ultimately settling on a 1/3 weighting, indicating that "proposals have simply not shown that the CAPM should be raised all at once to parity with the DCF

[51] RD, at 27.

analysis in the setting of returns on equity."[52] To the extent that this was a consideration in the RD's weighting determination, the Commission's 25 years of experience with the CAPM since that time now provides a sound basis for altering the weighting of the two ROE methodologies and raising the CAPM "to parity with the DCF analysis...."

Several regulators, including the FERC, have recognized the challenges associated with relying solely on the DCF to establish the authorized ROE for regulated utilities in recent years. The FERC considered what it termed "anomalous" market conditions as a basis for abandoning its practice of sole reliance on the DCF model to set ROEs for electric transmission companies which began in the 1980s. FERC reasoned:

> The fact that utility stock prices appear to have performed in a manner inconsistent with the theory underlying the DCF methodology during the periods at issue in these four complaint proceedings is an example of what NETOs have described as "model risk" — the risk that in some circumstances a model will produce results that do not reflect real world experience. It appears that, for whatever the reason, investors during this period have seen greater value in utility stocks than the DCF methodology would predict. This suggests that the ROE estimated by that methodology may be correspondingly inaccurate.[53]

[52] *Id*.
[53] 165 FERC ¶ 61,030, December 16, 2018, at 46.

The importance of relying on multiple methodologies, while deemphasizing the DCF, has been consistently recognized by a number of state jurisdictions.[54]

Q. Please summarize your conclusion regarding the relative weighting of the CAPM and DCF results.

A. While the Generic Finance RD proposed the 2/3 weighting on the DCF, the weightings and methodologies used to estimate the ROE were left open for additional consideration in future rate proceedings. Since then, the Commission has employed the CAPM as one component of the formula used to develop ROE estimates. There does not appear to be any reason to infer that the Commission has less confidence in the results of the CAPM than those of the DCF. The conditions that warranted the Commission's GFP inquiry and the subsequent Generic Finance RD in the early 1990s exist again today with DCF results considerably lower than those from other models, such as the CAPM, as well as returns authorized in other jurisdictions. Current capital market conditions are heavily influenced by monetary policy aimed at mitigating inflationary pressures and

[54] See, e.g., Pennsylvania Public Utility Commission, PPL Electric Utilities, R-2012-2290597, meeting held December 5, 2012, at 80-81; State of Illinois Commerce Commission, Docket No. 16-0093, Illinois-American Water Company Initial Brief, August 31, 2016, at 10, Illinois Staff's analysis and recommendation in that proceeding were based on its application of the multi-stage DCF model and the CAPM to a proxy group of water utilities, State of Illinois Commerce Commission Decision, Docket No. 16-0093, Illinois-American Water Company, 2016 WL 7325212 (2016), at 55; and File No. GR-2017-0215 and File No. GR-2017-0216, Missouri Public Service Commission, Report and Order, Issue Date February 21, 2018, at 34.

consequently long-term interest rates have increased substantially since the historical lows of 2020. Since the CAPM directly accounts for changes in interest rates, it is appropriate to give the CAPM more weight as compared to recent Commission decisions in the current capital market environment. Therefore, it is reasonable to apply equal weighting to the DCF and CAPM methods when determining the ROE for the Companies.

Q. Are the assumptions used in the CAPM less reliable than the assumptions used in the DCF model?

A. No. As discussed previously, the CAPM relies on a risk-free rate, Beta and the MRP. The risk-free rate is readily observable and can be projected for the forward-looking period. Beta is estimated using the historical relationship between the risk of the stock and the overall market. Finally, the market risk premium, while not observable, can be estimated for the forward-looking period. My testimony discusses how the current economic environment is directly affected by Federal Reserve policy and may not be representative of forward-looking market conditions. Therefore, while the CAPM is sometimes criticized as relying on unobservable assumptions, the ability to rely on forward-looking assumptions allows for a prospective analysis of investors' long-term expectations.

VII. REGULATORY AND BUSINESS RISKS

A. DECARBONIZATION POLICY

Q. Are there recent developments that have increased the business risk to gas distribution utilities generally in New York?

A. Yes. New York passed the Climate Leadership and Community Protection Act ("CLCPA"), which envisions a net zero carbon economy by 2050. As it relates to the Companies, the New York Climate Action Council's Final Scoping Plan indicates that "a substantial reduction of fossil natural gas use and strategic downsizing and decarbonization of the gas system"[55] may be required to achieve emissions limits. This could result in significant financial and market risk for the Companies. While the implementation of this legislation continues to be a work-in-progress, the uncertainty surrounding it creates significant financial and market risk for utility companies, especially utilities with natural gas distribution operations.

Q. Have rating agencies commented on the effect of the decarbonization policies of New York on the risk profile of KEDNY and KEDLI?

A. Yes. For both Companies, Moody's identified that "New York State's commitment to significantly reduce greenhouse gas emissions poses long-term challenges."[56] In the most recent Moody's report for KEDLI, this point

[55] New York State Climate Action Council. 2022. "New York State Climate Action Council Scoping Plan," http://climate.ny.gov/ScopingPlan, Full report December 2022, at 350.

[56] Moody's Investors Service, Credit Opinion, "The Brooklyn Union Gas Company, Update following downgrade to Baa2," November 8, 2021, at 2; Moody's Investors

is emphasized due to "KEDLI's network operations being, unlike peers, all gas rather than primarily electricity."[57] In a report addressing the effect on all gas LDCs, Moody's elaborated on the issue describing decarbonization policies as a threat to the gas distribution business model:

> Emissions reduction initiatives call into question the long-term ability of LDCs to maintain their position as natural gas monopoly service providers. Efforts in some jurisdictions to reduce greenhouse gas emissions as they transition to a low-carbon future may threaten the ability of LDCs to remain competitive. Government incentive programs are likely to accelerate the adoption of electric alternatives to natural gas usage, while clean energy innovations will expose the sector to technology risk. While there are still significant barriers to replacing LDCs, continued policy and regulatory support will be necessary to facilitate the sector's energy transition and maintain credit quality over the long term. The characteristics of each individual utility are important considerations when assessing these long-term risks.[58]

Moody's expects a differentiation in financial models as LDCs reflect increasing risks relative to electric utilities. Of the 24 LDCs Moody's identified as operating in jurisdictions with aggressive emissions reduction targets, only three were rated "highly negative," from a credit risk perspective. Two of those three companies are KEDNY and KEDLI.[59] In this case, New York's expansive decarbonization policies place the

Service, Credit Opinion, "KeySpan Gas East Corporation, Update to credit analysis," February 20, 2023, at 2.

[57] Moody's Investors Service, Credit Opinion, "KeySpan Gas East Corporation, Update to credit analysis," February 20, 2023, at 2.

[58] Moody's Investors Service, Credit Opinion, "Local Gas Distribution Utilities – North America, Emissions reduction, electrification threaten long-term competitiveness," November 14, 2022, at 2.

[59] *Id*. at 8.

Companies at the high end of the risk spectrum in the industry. As it relates to the Companies' ROE, Moody's indicates "With the increase in credit risk that LDCs will face, we expect that they will seek increases in their allowed return on equity and equity capital."[60] Therefore, given the Companies' higher relative risk as it relates to clean energy policy, a credit supportive rate plan would be appropriate to mitigate the credit challenges to KEDNY's and KEDLI's financial profiles.

Q. Are you proposing a specific premium to compensate investors for the risk of potential stranded assets?

A. No, my recommendation assumes the Companies will be allowed full and fair recovery of their prudent investments.

B. RISK ASSESSMENT

Q. Have you performed an analysis of the level of regulatory support that the Companies receive in New York as compared to the proxy group companies?

A. Yes. I conducted an analysis of the regulatory protections that are in place for KEDNY and KEDLI compared with those for the operating utility companies held by the proxy group companies. The results of my analysis are presented in Exhibit ___ (JCN-6). Specifically, I examined the following factors that affect the business risk of KEDNY and KEDLI and

[60] *Id*. at 7-8.

the proxy group companies: (1) test year convention; (2) fuel cost recovery; (3) revenue decoupling; and (4) capital cost recovery.

As shown in Exhibit ___ (JCN-6), 48 percent of the operating companies (*i.e.,* 44 out of 91) in the proxy groups provide service in jurisdictions that allow the use of a fully or partially forecast test year. All of the operating companies held by the proxy groups are allowed to pass through fuel costs and purchased power costs directly to customers, so that the utility does not incur any risk associated with fuel or purchased power costs. It is important to recognize that fuel and purchased power costs typically account for 50 – 60 percent of the total operating costs for a regulated utility. Like KEDNY and KEDLI, 57 percent of the operating utilities held by the proxy group (*i.e*., 52 out of 91) have revenue decoupling mechanisms or weather normalization adjustment clauses that allow them to break the link between customer usage and revenues. Finally, approximately 62 percent of the operating utilities held by the proxy group (56 out of 91) have capital cost recovery mechanisms that allow them to recover capital investments that are placed into service between rate cases.

Q. Based on these analyses, what is your conclusion regarding the level of regulatory support for KEDNY and KEDLI relative to that of the proxy group companies?

A. My conclusion is that KEDNY and KEDLI have comparable regulatory protection to the proxy group companies. While the Commission has been

a leader in implementing mechanisms that reduce the business risk of regulated utilities in New York, many other jurisdictions have taken similar steps in more recent years, leading to no meaningful difference in business risk difference from the proxy group.

Q. Are there other risks to the Companies that are specific to New York utility regulation?

A. Yes. In addition to the low equity returns that are typically authorized by the Commission for New York's gas utilities (in 2020-2022 average authorized ROEs in New York were 65 basis points below the national average for gas utilities),[61] New York utilities are subject to strictly-enforced customer service quality, electric reliability, and gas safety measures where the utility is required to achieve predetermined performance benchmarks, or be subject to a negative revenue adjustment for any shortfall.

Q. Please describe the Companies' customer service quality and gas safety measures.

A. The Companies are subject to a number of customer service quality and gas safety performance metrics for which negative revenue adjustments are incurred for specific levels of non-performance. Figure 6 below

[61] Source: Regulatory Research Associates. 2020-2022 average authorized ROEs for states other than New York (excluding limed issue riders) were 9.58 percent compared to an average authorized ROE of 8.93 percent in New York.

summarizes the Companies' potential annual exposure to these negative revenue adjustments.

Figure 6: Summary of Service Quality and Gas Safety Performance Metrics[62]

Performance Metric	Maximum Annual Negative Revenue Adjustment KEDLI	Maximum Annual Negative Revenue Adjustment KEDNY
Customer Service Quality	$9.9 million	$11.7 million
Gas Safety	150 Pre-Tax Basis Points	150 Pre-Tax Basis Points

Q. Do the negative revenue adjustments associated with these performance metrics differentiate the Companies from the proxy group companies?

A. Yes, they do. The magnitude of the Companies' exposure places the Companies at greater risk than proxy companies on average.

C. REGULATORY ENVIRONMENT

Q. Please explain how the regulatory framework affects investors' risk assessments.

A. The ratemaking process is premised on the principle that, for investors and companies to commit the capital needed to provide safe and reliable utility services, the subject utility must have the opportunity to recover invested capital and the market-required return on such capital. Regulatory commissions recognize that because utility operations are capital intensive,

[62] Case 19-G-0309 and 19-G-0310, Joint Proposal dated May 14, 2021, at 101 and 106-107.

regulatory decisions should enable the utility to attract capital at reasonable terms, which balance the long-term interests of investors and customers. In that respect, the regulatory framework in which a utility operates is one of the most important factors considered in both debt and equity investors' risk assessments.

Q. Please explain how credit rating agencies consider the regulatory framework in establishing a company's credit rating.

A. S&P and Moody's both consider the overall regulatory framework in establishing credit ratings. As shown in Figure 7, Moody's establishes credit ratings based on four key factors:

Figure 7: Moody's Rating Factors

Factor	Weighting
Regulatory Framework	25%
Ability to Recover Costs and Earn Returns	25%
Diversification	10%
Financial Strength	40%
Total	100%

Two of these factors (*i.e.*, regulatory framework and the ability to recover costs and earn returns) are based on the regulatory environment such that half of Moody's overall assessment of business and financial risk for

regulated utilities is based upon the regulatory environment.[63] Moody's further subdivides the first two factors, regulatory framework and the ability to recover costs and earn authorized returns, into sub-factors to help "provide more granularity and transparency on the overall regulatory environment, which is the most important consideration for this sector."[64]

With respect to the regulatory framework, Moody's looks for transparency, predictability, and supportiveness of regulatory commissions.[65] For the second factor, ability to recover costs and earn returns, Moody's evaluates the regulatory elements that directly affect the ability of the utility to generate cash flow and service its debt over time.[66] Moody's views the ability to recover costs on a timely basis and to attract debt and equity capital as crucial credit considerations noting that "[t]he inability to recover costs…has been one of the greatest drivers of financial stress in this sector."[67] This is particularly true as utilities are often cash flow negative due to large capital expenditures, so any lack of timely recovery or sufficiency of rates can strain access to capital markets.

S&P has also identified the regulatory environment as an important factor, stating, "we believe the fundamental regulatory environment in the

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63 Moody's Investor Service, Rating Methodology, Regulated Electric and Gas Utilities, December 23, 2013, at 6.
64 *Id.* at 3.
65 *Id.* at 9-10.
66 *Id.* at 15.
67 Id.

jurisdictions in which a utility operates often influence credit quality the most."[68]

Q. How does the regulatory environment in which a utility operates affect its access to and cost of capital?

A. The proportion and cost of debt capital available to utility companies are influenced by the rating agencies' assessment of the regulatory environment. Moody's has highlighted the relevance of a stable and predictable regulatory environment to a utility's credit quality, stating that "[b]roadly speaking, the Regulatory Framework is the foundation for how all the decisions that affect utilities are made (including the setting of rates), as well as the predictability and consistency of decision-making provided by that foundation."[69]

Q. Have the Companies received any recent ratings actions?

A. Yes. The Companies were downgraded by S&P and Moody's in 2021. On March 3, 2021 S&P downgraded the Companies' issuer ratings from A- to BBB+. Subsequently, on June 4, 2021 Moody's downgraded the senior unsecured and issuer ratings of KEDNY from Baa1 to Baa2 and the senior unsecured and issuer ratings of KEDLI from A3 to Baa1. Moody's pointed to the combination of a further reduction in the Companies' ROEs, material

[68] Standard & Poor's, Assessing U.S. Utility Regulatory Environments, March 11, 2010, at 2.

[69] *Id*.

capital expenditure programs, and limits in rate increases as drivers that are weakening the Companies' credit profiles.[70] Fitch similarly points to below average ROEs, limits to base rate increases, and the CLCPA as concerns and key ratings drivers. [71]

D. CAPITAL EXPENDITURES

Q. Did you consider any other information regarding the Companies' risks relative to the proxy group companies?

A. Yes, I also considered the risk related to the Companies' future capital expenditures as compared with the proxy group's capital spending plans.

Q. Please summarize the projected capital expenditure requirements for KEDLI and KEDNY.

A. The combined capital expenditure projections for KEDLI and KEDNY are approximately $8.18 billion for the period from Fiscal Year 2024 through 2028. The Companies' respective construction programs include significant capital investment in infrastructure projects designed to enhance the safety and reliability of their systems.

[70] Moody's Investors Service, Credit Opinion, "The Brooklyn Union Gas Company, Update following downgrade to Baa2," November 8, 2021, at 5; Moody's Investors Service, Credit Opinion, "KeySpan Gas East Corporation, Update following downgrade to Baa1," November 8, 2021 at 4.

[71] FitchRatings, "Fitch Rates Brooklyn Union Gas Co.'s Senior Unsecured Notes 'BBB+'," August 2, 2022 at 1.

Q. Do credit rating agencies recognize the risks associated with significant capital expenditures?

A. Yes, they do. From a credit perspective, the additional pressure on cash flows associated with high levels of capital expenditures exerts corresponding pressure on credit metrics and, therefore, credit ratings. A 2016 S&P report noted:

> When applicable, a jurisdiction's willingness to support large capital projects with cash during construction is an important aspect of our analysis. This is especially true when the project represents a major addition to rate base and entails long lead times and technological risks that make it susceptible to construction delays. Broad support for all capital spending is the most credit-sustaining. Support for only specific types of capital spending, such as specific environmental projects or system integrity plans, is less so, but still favorable for creditors. Allowance of a cash return on construction work-in-progress or similar ratemaking methods historically were extraordinary measures for use in unusual circumstances, but when construction costs are rising, cash flow support could be crucial to maintain credit quality through the spending program. Even more favorable are those jurisdictions that present an opportunity for a higher return on capital projects as an incentive to investors.[72]

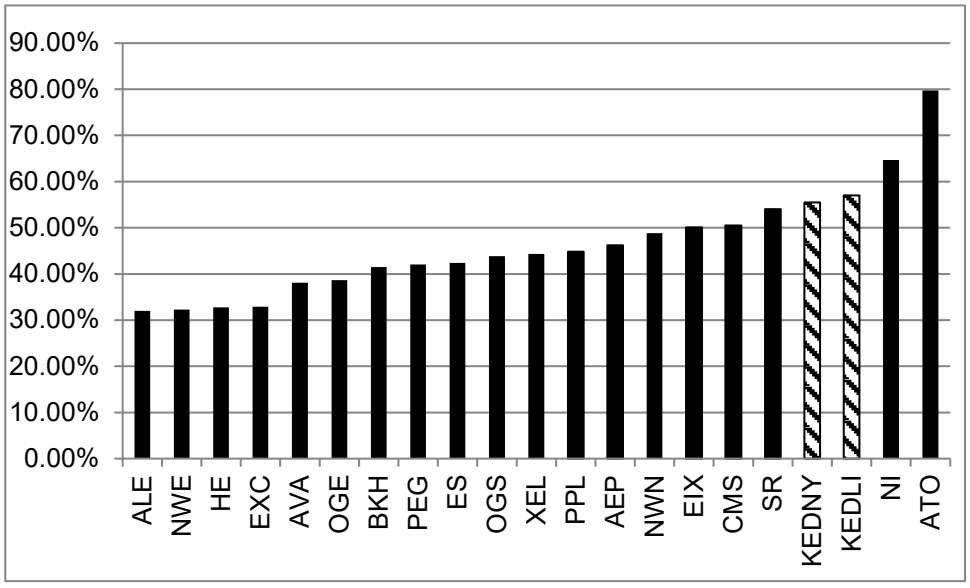
Q. Have you conducted any analysis of the Companies' projected capital expenditures relative to the proxy companies?

A. Yes. I compared the ratio of projected capital expenditures from 2024 through 2027 to net utility plant as of December 31, 2021, for KEDNY and KEDLI with each of the proxy group companies. Exhibit __ (JCN-7) shows

[72] S&P Global Ratings, Ratings Direct, "Assessing U.S. Investor-Owned Utility Regulatory Environments," August 10, 2016, at 7.

the ratio of three years of projected capital expenditures to net plant for the proxy groups based on data reported by Value Line. Figure 8 demonstrates that KEDLI's and KEDNY's ratio of projected capital expenditures to net plant are higher than 17 of the proxy group companies (there are 19 companies), respectively. Furthermore, as shown in Exhibit __ (JCN-7), KEDLI's planned investment ratio of 56.98 percent and KEDNY's planned investment ratio of 55.51 percent are more than the mean and median of the proxy groups, which suggests that the Companies face greater risk from their construction programs than the typical proxy group company.

Figure 8: Capital Expenditures/Net Plant[73]



[73] Source: Value Line, S&P Capital IQ Pro, and Company data.

Q. What are your conclusions regarding the effect of the projected capital expenditure plans on the risk profiles of KEDNY and KEDLI and the cost of equity?

A. It is clear that the Companies' capital expenditure requirements as a percentage of net utility plant are higher than the majority of the proxy group companies. This elevated level of capital expenditures relative to the proxy group increases the importance of setting a return for KEDNY and KEDLI that is within the range of reasonableness as established by the returns for that group.

VIII. CAPITAL STRUCTURE

Q. Please summarize the Companies' proposed capital structure.

A. As explained in Direct Testimony of the Capital Structure Panel, the Companies are requesting a 48 percent equity ratio for ratemaking purposes, which reflects a lower equity ratio than the Companies' actual stand-alone equity ratios.[74] The requested equity ratio is consistent with recent Commission precedent regarding the authorized capital structure for utilities.[75]

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[74] Direct Testimony of Capital Structure Panel.

[75] "However, as Staff notes the 48% equity ratio is consistent with the equity ratio of the proxy group of companies with similar risk characteristics as St. Lawrence and the level that we have adopted for other New York utilities." Case 15-G-0382, St. Lawrence Gas Company, Inc., Order Establishing Multi-Year Rate Plan, issued July 15, 2016, 2016 WL 3916059, at *15. See, e.g., Case 19-E-0378 and 19-G-0379, New York State Electric & Gas Corporation, Order Approving Electric and Gas Rate Plans in Accord with Joint Proposal, with Modifications, issued November 25, 2020; Cases

Q. Have you conducted any analysis of the Companies' proposed capital structure as compared with the proxy companies?

A. Yes. I have reviewed KEDNY and KEDLI's proposed capital structure as compared with the actual capital structures of the operating companies in the proxy group for the most recently reported four years. As shown on Exhibit __ (JCN-8), the mean annual equity ratio of the proxy companies over that period is 54.13 percent.

Q. What do you conclude from this analysis?

A. The requested 48 percent equity ratio is conservative considering the equity ratios of the proxy companies and the current business and financial risks of KEDNY and KEDLI, including significant capital investment programs, credit metrics pressures and credit rating agency pressures. This information indicates that the utility operating subsidiaries owned by holding companies with similar business characteristics to KEDNY and KEDLI have for the last four years maintained average common equity

19-E-0380 and 19-G-0381, Rochester Gas and Electric Corporation, Order Approving Electric and Gas Rate Plans in Accord with Joint Proposal, with Modifications, issued November 25, 2020; Cases 19-G-0309 and 19-G-0301, The Brooklyn Union Gas Company d/b/a National Grid NY, et al, Order Approving Joint Proposal, As Modified, and Imposing Additional Requirements, issued August 12, 2021);. Note, however, that in a Central Hudson case the Commission allowed an equity ratio of 50%, stating: "Although the recommended ROE is somewhat higher than others we have recently approved for other utilities, interest rates and equity return requirements have generally increased since those proceedings were completed. Further, decreasing the common equity ratio in Rate Years 2 and 3 will reduce the overall cost of capital in those rate years." Case 20-E-0428, Central Hudson Gas & Electric Corp., Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plan, issued November 18, 2021, 2021 WL 5496045, at *21.

ratios more than 6.00 percentage points above the 48.0 percent equity ratio that the Commission approved for KEDNY and KEDLI in the Companies' last rate proceeding. These higher proxy equity ratios reflect a level of financial risk that is lower than the financial risk implied by the proposed 48.0 percent equity ratio. In addition, equity ratios that are typically authorized by the Commission for New York's gas utilities (in 2020-2022 average authorized equity ratios in New York were 4.23 percentage points below the national average for gas utilities),[76] are considerably lower than equity ratios authorized by other jurisdictions. Therefore, I conclude that the requested equity ratio should be considered a lower bound on the equity ratio relative to the range of reasonableness established by industry practice. As such, it would be reasonable for the Commission to use higher equity ratios for KEDNY and KEDLI closer to those of the proxy group operating companies for ratemaking purposes.

IX. MULTI-YEAR RATE PLAN

Q. Would a multi-year rate plan impact your ROE recommendation?

A. Yes, it would. The Commission has in many cases approved three-year rate case settlements that often include stay-out premiums. It is my

[76] Source: Regulatory Research Associates. 2020-2022 average authorized equity ratios for states other than New York (excluding limed issue riders) were 52.43 percent compared to an average authorized equity ratio of 48.20 percent in New York. Rate cases in Arkansas, Florida, Indiana, and Michigan have been excluded from the analysis since the authorized capital structure approved in the cases includes deferred taxes and other credits as non-investor supplied capital.

understanding that the Companies will provide four years of forecast data in their rate filings. In keeping with Commission precedent, a stay-out premium would reflect the increased risk faced by the Companies under a multi-year rate plan.

Q. What are the relevant risk factors to equity investors inherent in a multi-year stay-out?

A. As described in Section IV, inflation and rising interest rates are the predominant factors in the current economic environment. As such, the Companies, are exposed to changes in underlying costs that deviate from expected levels of inflation. However, the stay-out premium associated with a multi-year rate plan should not only compensate investors for changes in the level of interest rates or inflation, but also for the potential risk of under-earning that is introduced by "staying out." By staying out of rate cases, the utility may not fully recover material amounts of capital expenditures and may be required to absorb losses due to differences between the cost of service established in the rate plan and actual levels of revenue and expense. The premium should compensate the utility and its investors for these additional risks over and above interest rate risk.

Q. What do you propose as the stay-out premium for a four-year rate plan?

A. The ROE proposed by the Companies of 9.80 percent will not provide the Companies a return commensurate with the return available on investments

of similar risk over the term of the multi-year rate plan without an adequate stay-out premium. Consistent with prior cases in which a stay-out premium was included in multi-year rate plans, I recommend that a stay-out premium be included in a multi-year rate plan. I believe 50 basis points is a reasonable, albeit conservative, reflection of the incremental risk to the Company under a multi-year stay-out provision.

X. CONCLUSION AND RECOMMENDATION

Q. What is your conclusion regarding a fair return on book equity for KEDNY and KEDLI?

A. My recommended return on book equity considers the results of the DCF and CAPM models, summarized in Figure 9 (below), and the specific risks to which the Companies are exposed. I conclude that the appropriate ROE for the Companies is within the range of 9.80 percent and 10.70 percent. The Companies are requesting an ROE of 9.80 percent, which is at the low end of this range of reasonableness and is a conservative estimate of the investor-required ROE.

Figure 9: Summary of Analytical Results

	Low	Mean	High
DCF	9.13%	9.68%	10.31%
Mean CAPM	11.13%	11.14%	11.15%
67%/33% DCF/CAPM	9.79%	10.17%	10.59%
50%/50% DCF/CAPM	10.13%	10.41%	10.73%

Furthermore, if the Commission approves a multi-year rate plan, a premium of 50 basis points should be added to the ROE; resulting in an ROE of 10.3 percent.

Q. Does this conclude your Direct Testimony?

A. Yes, it does.

JOSHUA C. NOWAK
Vice President

Mr. Nowak is a financial and economic consultant with more than fifteen years of experience in the energy industry. He has provided expert testimony on regulatory issues in several proceedings before the Federal Energy Regulatory Commission and regulatory commissions in Alaska, Connecticut, Kentucky, New Brunswick, New Hampshire, New York, Ohio, and Texas. Mr. Nowak specializes in providing rate case services on economic conditions and financial market matters related to the cost of capital. He is also experienced in providing strategic direction on financing activities including bond offerings, credit rating analysis, and investor relations. Previously, Josh was the Director of Regulatory Strategy & Integrated Analytics at National Grid where he was responsible for issues related to the cost of capital across its federal and state jurisdictional operating companies. He holds a Bachelor's Degree in Economics and History from Boston College.

REPRESENTATIVE EXPERIENCE

Expert Testimony and Litigation Support

Mr. Nowak's work includes regulatory project management, research, and analysis for expert witness testimony. His work has included:

- Expert testimony on cost of capital, financial markets, return on equity, capital structure, and debt financing issues
- Regulatory strategy in return on equity proceedings, including coordination across several utilities in joint-party proceedings
- Extensive support for expert testimony in cost of capital and return on equity proceedings through research, financial analysis, and testimony development
- Expert testimony, sponsoring lead-lag studies, in support of utility cash working capital requirements
- Project management of expert testimony assignments, including all phases of the regulatory schedule
- Performing analysis to support expert testimony regarding affiliate expenses and allocations

Policy Analysis

Mr. Nowak has contributed to projects related to policy review including:

- A review of natural gas capacity options and a cost-benefit analysis for state regulators seeking to reduce energy costs for ratepayers
- Analysis of the economic and environmental benefits of changes to natural gas ratemaking/expansion policy

Management and Operations Consulting

Mr. Nowak has taken a lead analytical role in developing benchmarking analyses and process reviews. Specifically, he has:

- Developed benchmarking analyses, in support of expert testimony, comparing electric and gas utilities' cost and operational efficiency, taking into account a situational assessment of exogenous factors
- Performed a process review of a gas utility's expansion projects, including an evaluation of policies, procedures, and financial models
- Supported analysis for a report of the reasonableness of a shared service company's administrative and general costs

Financial Analysis

Other financial analysis Mr. Nowak has conducted include:

- Extensive analysis on issues related to utilities' cost of capital
- Developing dispatch models to estimate revenues for merchant powerplants
- Estimating damages for breach of contract in fuel delivery commitment
- Researching strategic investment opportunities for merchant generators
- A report on the profitability of various generation technologies in a deregulated energy market
- Reviewing internal financial models used by utility clients
- Supporting utility asset appraisals, including research and analysis for income approach, cost approach, and sales comparison approach

Other Experience

In his previous work, Mr. Nowak contributed to the evaluation of regulatory policy for government clients. His experience included performing policy analysis, including economic impact assessments, for federal regulations.

PROFESSIONAL HISTORY

Concentric Energy Advisors, Inc. (2018 – Present)
Vice President
Assistant Vice President

National Grid USA (2017 – 2018)
Director, Regulatory Strategy & Integrated Analytics

ScottMadden, Inc. (formerly Sussex Economic Advisors, LLC) (2012 – 2016)
Director
Principal

Concentric Energy Advisors, Inc. (2007 – 2012)
Senior Consultant
Consultant
Assistant Consultant
Analyst

RTI International (2006 – 2007)
Economist

EDUCATION

Boston College
B.A., Economics and History, 2006

SPONSOR	DATE	CASE/APPLICANT	DOCKET	SUBJECT
Regulatory Commission of Alaska				
ENSTAR Natural Gas Company, a Division of Semco Energy, Inc.	06/16	ENSTAR Natural Gas Company, a Division of Semco Energy, Inc.	TA 285-4	Cash Working Capital
Connecticut Public Utilities Regulatory Authority				
Aquarion Water Company of Connecticut	08/22	Aquarion Water Company of Connecticut	Docket No. 22-07-01	Return on Equity
Aquarion Water Company of Connecticut	01/22	Aquarion Water Company of Connecticut	Docket No. 13-02-20RE06	Return on Equity and Cost of Debt
Federal Energy Regulatory Commission				
Central Hudson Gas & Electric Corporation, Consolidated Edison Company of New York, Inc., Niagara Mohawk Power Corporation d/b/a National Grid, New York State Electric & Gas Corporation, Orange and Rockland Utilities, Inc., and Rochester Gas and Electric Corporation	04/21	Central Hudson Gas & Electric Corporation, Consolidated Edison Company of New York, Inc., Niagara Mohawk Power Corporation d/b/a National Grid, New York State Electric & Gas Corporation, Orange and Rockland Utilities, Inc., and Rochester Gas and Electric Corporation	EL21-66-000, ER21-1647-000	Transmission Ownership Risk and Returns
Central Hudson Gas & Electric Corporation	12/19	Central Hudson Gas & Electric Corporation	ER20-715-000	Return on Equity
Kentucky Public Service Commission				
Duke Energy Kentucky, Inc.	12/22	Duke Energy Kentucky, Inc.	Case No. 2022-00372	Return on Equity
Public Utilities Commission of New Hampshire				
Liberty Utilities (Granite State Electric) Corp. d/b/a Liberty Utilities	04/16	Liberty Utilities (Granite State Electric) Corp. d/b/a Liberty Utilities	Docket No. DE 16-383	Cash Working Capital
New Brunswick Energy and Utilities Board				
New Brunswick Power Corporation (NB Power)	11/22	New Brunswick Power Corporation (NB Power)	Matter 541	Macroeconomic Environment and Capital Market Conditions

SPONSOR	DATE	CASE/APPLICANT	DOCKET	SUBJECT
New York Public Service Commission				
Niagara Mohawk Power Corporation d/b/a National Grid	07/20	Niagara Mohawk Power Corporation d/b/a National Grid	Case 20-E-0380/ Case 20-G- 0381	Return on Equity
Niagara Mohawk Power Corporation d/b/a National Grid	07/17	Niagara Mohawk Power Corporation d/b/a National Grid	Case 17-E-0238 / Case 17-G- 0239	Capital Structure and Overall Cost of Capital
Public Utilities Commission of Ohio				
Duke Energy Ohio, Inc.	01/23	Duke Energy Ohio, Inc.	Case No. 22-1153-EL-UNC	Return on Equity
Public Utility Commission of Texas				
Wind Energy Transmission Texas, LLC	05/15	Wind Energy Transmission Texas, LLC	Docket No. 44746	Cash Working Capital
Lone Star Transmission, LLC	05/14	Lone Star Transmission, LLC	Docket No. 42469	Cash Working Capital
Railroad Commission of Texas				
Texas Gas Service Company, a Division of One Gas, Inc.	06/16	Texas Gas Service Company, a Division of One Gas, Inc.	GUD No. 10526	Cash Working Capital
Texas Gas Service Company, a Division of One Gas, Inc.	03/16	Texas Gas Service Company, a Division of One Gas, Inc.	GUD No. 10506	Cash Working Capital
Texas Gas Service Company, a Division of One Gas, Inc.	12/15	Texas Gas Service Company, a Division of One Gas, Inc.	GUD No. 10488	Cash Working Capital
CenterPoint Energy Resources Corp., d/b/a CenterPoint Energy Entex and CenterPoint Energy Texas Gas	03/14	CenterPoint Energy Resources Corp., d/b/a CenterPoint Energy Entex and CenterPoint Energy Texas Gas	GUD No. 10432	Cash Working Capital

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3-Month Multi-Stage DCF

THREE-MONTH MULTI-STAGE DCF - PROXY GROUP ANALYSIS - MEAN GROWTH RATE

		[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]
								Second Stage Growth						
Company	Ticker	Stock Price	Annualized Dividend	Value Line EPS Growth	Yahoo! Finance EPS Growth	Zacks EPS Growth	First Stage Growth	Year 6	Year 7	Year 8	Year 9	Year 10	Third Stage Growth	ROE
ALLETE, Inc.	ALE	$62.52	$2.71	6.00%	8.70%	7.30%	7.33%	7.02%	6.71%	6.40%	6.09%	5.78%	5.46%	10.84%
American Electric Power Company, Inc.	AEP	$91.47	$3.32	6.00%	5.76%	6.10%	5.95%	5.87%	5.79%	5.71%	5.63%	5.55%	5.46%	9.59%
Atmos Energy Corporation	ATO	$113.73	$2.96	7.00%	7.80%	7.50%	7.43%	7.11%	6.78%	6.45%	6.12%	5.79%	5.46%	8.70%
Avista Corporation	AVA	$41.41	$1.84	3.50%	5.20%	5.20%	4.63%	4.77%	4.91%	5.05%	5.19%	5.33%	5.46%	10.13%
Black Hills Corporation	BKH	$65.90	$2.50	6.00%	5.40%	2.20%	4.53%	4.69%	4.84%	5.00%	5.15%	5.31%	5.46%	9.40%
CMS Energy Corporation	CMS	$61.47	$1.95	6.50%	8.00%	8.00%	7.50%	7.16%	6.82%	6.48%	6.14%	5.80%	5.46%	9.43%
Edison International	EIX	$67.54	$2.95	16.00%	7.00%	3.00%	8.67%	8.13%	7.60%	7.07%	6.53%	6.00%	5.46%	11.33%
Eversource Energy	ES	$78.99	$2.70	6.50%	6.70%	6.50%	6.57%	6.38%	6.20%	6.02%	5.83%	5.65%	5.46%	9.50%
Exelon Corporation	EXC	$41.68	$1.44	NMF	6.30%	6.60%	6.45%	6.29%	6.12%	5.96%	5.79%	5.63%	5.46%	9.51%
Hawaiian Electric Industries, Inc.	HE	$40.52	$1.44	4.50%	1.30%	3.10%	2.97%	3.38%	3.80%	4.22%	4.63%	5.05%	5.46%	8.78%
NiSource Inc.	NI	$27.36	$1.00	9.50%	n/a	6.80%	8.15%	7.70%	7.25%	6.81%	6.36%	5.91%	5.46%	10.22%
Northwest Natural Gas Company	NWN	$48.34	$1.94	6.50%	2.80%	4.30%	4.53%	4.69%	4.84%	5.00%	5.15%	5.31%	5.46%	9.64%
NorthWestern Corporation	NWE	$57.10	$2.56	3.50%	4.50%	1.70%	3.23%	3.61%	3.98%	4.35%	4.72%	5.09%	5.46%	9.76%
OGE Energy Corporation	OGE	$37.61	$1.66	6.50%	-12.34%	10.20%	1.45%	2.12%	2.79%	3.46%	4.13%	4.80%	5.46%	9.21%
ONE Gas, Inc.	OGS	$79.26	$2.60	6.00%	5.00%	5.00%	5.33%	5.36%	5.38%	5.40%	5.42%	5.44%	5.46%	9.03%
PPL Corporation	PPL	$28.32	$0.96	3.50%	17.21%	n/a	10.36%	9.54%	8.72%	7.91%	7.09%	6.28%	5.46%	10.51%
Public Service Enterprise Group Inc.	PEG	$60.57	$2.28	4.50%	2.40%	4.30%	3.73%	4.02%	4.31%	4.60%	4.89%	5.18%	5.46%	9.17%
Spire, Inc.	SR	$70.91	$2.88	8.00%	6.10%	4.20%	6.10%	5.99%	5.89%	5.78%	5.68%	5.57%	5.46%	10.13%
Xcel Energy Inc.	XEL	$67.48	$2.08	6.00%	6.40%	6.60%	6.33%	6.19%	6.04%	5.90%	5.75%	5.61%	5.46%	9.04%
Mean														9.68%
Median														9.51%

Notes
[1] Source: Bloomberg Professional, equals 3-month average as of 03/31/2023
[2] Source: Bloomberg Professional
[3] Source: Value Line
[4] Source: Yahoo! Finance
[5] Source: Zacks
[6] Equals Average ([3], [4], [5])
[7] Equals [6] + ([12] - [6]) / 6
[8] Equals [7] + ([12] - [6]) / 6
[9] Equals [8] + ([12] - [6]) / 6
[10] Equals [9] + ([12] - [6]) / 6
[11] Equals [10] + ([12] - [6]) / 6
[12] Source: Exhibit__(JCN-2)
[13] Equals internal rate of return of cash flows for Year 0 through Year 200

THREE-MONTH MULTI-STAGE DCF - PROXY GROUP ANALYSIS - LOW GROWTH RATE

		[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]
								Second Stage Growth						
Company	Ticker	Stock Price	Annualized Dividend	Value Line EPS Growth	Yahoo! Finance EPS Growth	Zacks EPS Growth	First Stage Growth	Year 6	Year 7	Year 8	Year 9	Year 10	Third Stage Growth	ROE
ALLETE, Inc.	ALE	$62.52	$2.71	6.00%	8.70%	7.30%	6.00%	5.91%	5.82%	5.73%	5.64%	5.55%	5.46%	10.42%
American Electric Power Company, Inc.	AEP	$91.47	$3.32	6.00%	5.76%	6.10%	5.76%	5.71%	5.66%	5.61%	5.56%	5.51%	5.46%	9.54%
Atmos Energy Corporation	ATO	$113.73	$2.96	7.00%	7.80%	7.50%	7.00%	6.74%	6.49%	6.23%	5.98%	5.72%	5.46%	8.61%
Avista Corporation	AVA	$41.41	$1.84	3.50%	5.20%	5.20%	3.50%	3.83%	4.15%	4.48%	4.81%	5.14%	5.46%	9.80%
Black Hills Corporation	BKH	$65.90	$2.50	6.00%	5.40%	2.20%	2.20%	2.74%	3.29%	3.83%	4.38%	4.92%	5.46%	8.83%
CMS Energy Corporation	CMS	$61.47	$1.95	6.50%	8.00%	8.00%	6.50%	6.33%	6.15%	5.98%	5.81%	5.64%	5.46%	9.19%
Edison International	EIX	$67.54	$2.95	16.00%	7.00%	3.00%	3.00%	3.41%	3.82%	4.23%	4.64%	5.05%	5.46%	9.58%
Eversource Energy	ES	$78.99	$2.70	6.50%	6.70%	6.50%	6.50%	6.33%	6.15%	5.98%	5.81%	5.64%	5.46%	9.48%
Exelon Corporation	EXC	$41.68	$1.44	NMF	6.30%	6.60%	6.30%	6.16%	6.02%	5.88%	5.74%	5.60%	5.46%	9.47%
Hawaiian Electric Industries, Inc.	HE	$40.52	$1.44	4.50%	1.30%	3.10%	1.30%	1.99%	2.69%	3.38%	4.08%	4.77%	5.46%	8.42%
NiSource Inc.	NI	$27.36	$1.00	9.50%	n/a	6.80%	6.80%	6.58%	6.35%	6.13%	5.91%	5.69%	5.46%	9.84%
Northwest Natural Gas Company	NWN	$48.34	$1.94	6.50%	2.80%	4.30%	2.80%	3.24%	3.69%	4.13%	4.58%	5.02%	5.46%	9.19%
NorthWestern Corporation	NWE	$57.10	$2.56	3.50%	4.50%	1.70%	1.70%	2.33%	2.95%	3.58%	4.21%	4.84%	5.46%	9.35%
OGE Energy Corporation	OGE	$37.61	$1.66	6.50%	-12.34%	10.20%	-12.34%	-9.37%	-6.41%	-3.44%	-0.47%	2.50%	5.46%	6.64%
ONE Gas, Inc.	OGS	$79.26	$2.60	6.00%	5.00%	5.00%	5.00%	5.08%	5.15%	5.23%	5.31%	5.39%	5.46%	8.96%
PPL Corporation	PPL	$28.32	$0.96	3.50%	17.21%	n/a	3.50%	3.83%	4.15%	4.48%	4.81%	5.14%	5.46%	8.73%
Public Service Enterprise Group Inc.	PEG	$60.57	$2.28	4.50%	2.40%	4.30%	2.40%	2.91%	3.42%	3.93%	4.44%	4.95%	5.46%	8.85%
Spire, Inc.	SR	$70.91	$2.88	8.00%	6.10%	4.20%	4.20%	4.41%	4.62%	4.83%	5.04%	5.25%	5.46%	9.60%
Xcel Energy Inc.	XEL	$67.48	$2.08	6.00%	6.40%	6.60%	6.00%	5.91%	5.82%	5.73%	5.64%	5.55%	5.46%	8.96%
Mean														9.13%
Median														9.19%

Notes
[1] Source: Bloomberg Professional, equals 3-month average as of 03/31/2023
[2] Source: Bloomberg Professional
[3] Source: Value Line
[4] Source: Yahoo! Finance
[5] Source: Zacks
[6] Equals Minimum ([3], [4], [5])
[7] Equals [6] + ([12] - [6]) / 6
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[11] Equals [10] + ([12] - [6]) / 6
[12] Source: Exhibit__(JCN-2)
[13] Equals internal rate of return of cash flows for Year 0 through Year 200

THREE-MONTH MULTI-STAGE DCF - PROXY GROUP ANALYSIS - HIGH GROWTH RATE

		[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]
								Second Stage Growth						
Company	Ticker	Stock Price	Annualized Dividend	Value Line EPS Growth	Yahoo! Finance EPS Growth	Zacks EPS Growth	First Stage Growth	Year 6	Year 7	Year 8	Year 9	Year 10	Third Stage Growth	ROE
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American Electric Power Company, Inc.	AEP	$91.47	$3.32	6.00%	5.76%	6.10%	6.10%	5.99%	5.89%	5.78%	5.68%	5.57%	5.46%	9.62%
Atmos Energy Corporation	ATO	$113.73	$2.96	7.00%	7.80%	7.50%	7.80%	7.41%	7.02%	6.63%	6.24%	5.85%	5.46%	8.77%
Avista Corporation	AVA	$41.41	$1.84	3.50%	5.20%	5.20%	5.20%	5.24%	5.29%	5.33%	5.38%	5.42%	5.46%	10.30%
Black Hills Corporation	BKH	$65.90	$2.50	6.00%	5.40%	2.20%	6.00%	5.91%	5.82%	5.73%	5.64%	5.55%	5.46%	9.79%
CMS Energy Corporation	CMS	$61.47	$1.95	6.50%	8.00%	8.00%	8.00%	7.58%	7.15%	6.73%	6.31%	5.89%	5.46%	9.56%
Edison International	EIX	$67.54	$2.95	16.00%	7.00%	3.00%	16.00%	14.24%	12.49%	10.73%	8.98%	7.22%	5.46%	14.22%
Eversource Energy	ES	$78.99	$2.70	6.50%	6.70%	6.50%	6.70%	6.49%	6.29%	6.08%	5.88%	5.67%	5.46%	9.53%
Exelon Corporation	EXC	$41.68	$1.44	NMF	6.30%	6.60%	6.60%	6.41%	6.22%	6.03%	5.84%	5.65%	5.46%	9.55%
Hawaiian Electric Industries, Inc.	HE	$40.52	$1.44	4.50%	1.30%	3.10%	4.50%	4.66%	4.82%	4.98%	5.14%	5.30%	5.46%	9.14%
NiSource Inc.	NI	$27.36	$1.00	9.50%	n/a	6.80%	9.50%	8.83%	8.15%	7.48%	6.81%	6.14%	5.46%	10.63%
Northwest Natural Gas Company	NWN	$48.34	$1.94	6.50%	2.80%	4.30%	6.50%	6.33%	6.15%	5.98%	5.81%	5.64%	5.46%	10.19%
NorthWestern Corporation	NWE	$57.10	$2.56	3.50%	4.50%	1.70%	4.50%	4.66%	4.82%	4.98%	5.14%	5.30%	5.46%	10.13%
OGE Energy Corporation	OGE	$37.61	$1.66	6.50%	-12.34%	10.20%	10.20%	9.41%	8.62%	7.83%	7.04%	6.25%	5.46%	11.93%
ONE Gas, Inc.	OGS	$79.26	$2.60	6.00%	5.00%	5.00%	6.00%	5.91%	5.82%	5.73%	5.64%	5.55%	5.46%	9.19%
PPL Corporation	PPL	$28.32	$0.96	3.50%	17.21%	n/a	17.21%	15.25%	13.29%	11.34%	9.38%	7.42%	5.46%	12.83%
Public Service Enterprise Group Inc.	PEG	$60.57	$2.28	4.50%	2.40%	4.30%	4.50%	4.66%	4.82%	4.98%	5.14%	5.30%	5.46%	9.36%
Spire, Inc.	SR	$70.91	$2.88	8.00%	6.10%	4.20%	8.00%	7.58%	7.15%	6.73%	6.31%	5.89%	5.46%	10.71%
Xcel Energy Inc.	XEL	$67.48	$2.08	6.00%	6.40%	6.60%	6.60%	6.41%	6.22%	6.03%	5.84%	5.65%	5.46%	9.10%
Mean														10.31%
Median														9.79%

Notes
[1] Source: Bloomberg Professional, equals 3-month average as of 03/31/2023
[2] Source: Bloomberg Professional
[3] Source: Value Line
[4] Source: Yahoo! Finance
[5] Source: Zacks
[6] Equals Maximum ([3], [4], [5])
[7] Equals [6] + ([12] - [6]) / 6
[8] Equals [7] + ([12] - [6]) / 6
[9] Equals [8] + ([12] - [6]) / 6
[10] Equals [9] + ([12] - [6]) / 6
[11] Equals [10] + ([12] - [6]) / 6
[12] Source: Exhibit__(JCN-2)
[13] Equals internal rate of return of cash flows for Year 0 through Year 200

Exhibit ___ (JCN-2)

Calculation of Long-Term GDP Growth Rate

CALCULATION OF LONG-TERM GDP GROWTH RATE

Step 1	
Real GDP ($ Billions) [1]	
1929	$ 1,110.2
2022	$ 20,015.4
Compound Annual Growth Rate	**3.16%**
Step 2	
Consumer Price Index (YoY % Change) [2]	
2029-2033	2.10%
Average	2.10%
Consumer Price Index (All-Urban) [3]	
2031	3.38
2050	5.26
Compound Annual Growth Rate	2.35%
GDP Chain-type Price Index (2009=1.000) [3]	
2031	1.49
2050	2.27
Compound Annual Growth Rate	2.25%
Average Inflation Forecast	**2.23%**
Long-Term GDP Growth Rate	**5.46%**

Notes:
[1] Bureau of Economic Analysis, February 28, 2023
[2] Blue Chip Financial Forecasts, Vol. 41, No. 12, December 2, 2022 at 14
[3] Energy Information Administration, Annual Energy Outlook 2022 at Table 20, March 2022

Exhibit ___ (JCN-3)

Bank of America Merrill Lynch: Quantitative Profiles

BofA GLOBAL RESEARCH

BofA SECURITIES

Quantitative Profiles

2023: Value > Growth; Cash Flow based Value > Deep Value

29 December 2022
Equity & Quant Strategy
United States

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Value won so far in 2022; stick with it…

While Value factors (+5.1% on avg.) trailed Growth factors (+6.0% on avg.) in Nov., Value remains ahead of Growth so far in 2022 (-3.7% vs -4.0%, respectively). The Russell 1000 Value Index leads the Russell 1000 Growth Index by 20ppt year-to-date (YTD). 2022 was marked by broad market leadership, as 73% of the almost 50 quantitative factors we follow outperformed the equal-weighted S&P 500 (a top-quintile year for market breadth in our data since 1987; Exhibit 49). All Value factors we follow except High EPS Yield (or Low P/E, -10.7% YTD) led the index. Historically, in prior years of strong market breadth (>73% of stocks outperforming, Value led Growth in the subsequent year in 67% of the cases and outperformed by a median of 1ppt. We see multiple reasons to stick with Value: 1) Growth remains stretched vs Value on most valuations metrics vs. history, 2) Value remains deeply underweight by active managers (Exhibit 39), 3) During high inflation periods Value led Growth 3-mth prior to Downturn (more below) and up to 12-mth after, 4) During high inflation periods Value led Growth for 12mos after the Fed stopped hiking, 5) Value fared well during periods of heightened rates volatility.

…but be selective

Our US Regime Indicator edged lower in November (to 0.2 from 0.3, Exhibit 11), still in Late Cycle but approaching Downturn. Performance of Value in Downturns has been mixed: while deep Value (such as Low Trailing P/E, Low Forward P/E, Low Price to Book) underperformed, cash producing companies (our High Free Cash Flow to Enterprise Value (FCF/EV) and Low Price / Free Cash Flow (P/FCF) factors) led the index. Our DDM Alpha factor also fared well. These factors have also tended to outperform in "Late Late Cycle" phases like we are in today (Exhibit 3). Other factor groups for Downturns include High Quality, Low Risk and Large Cap, which historically outperformed with 71% to 86% success rate and about 5ppt avg. alpha (Exhibit 15) – though we see differences this cycle and expect small caps to outperform large (see SMID Year Ahead).

Duration will likely remain under pressure

While inflation has declined from recent highs, the market prices in additional 50bp of Fed hikes, even if achieved at a slower pace. Our High Duration factor, which includes secular growth companies with eventual payoffs far in the future and is sensitive to rising discount rates and cost of capital, was the second worst performing factor overall YTD, underperforming the index by 13.7ppt (second biggest margin in our history since 1987). A continued rise in interest rates near-term (BofA Rates Strategists view) could continue to pressure High Duration stocks, especially given their elevated valuations.

Dividends should become a bigger share of returns

Dividend Yield was the third best factor overall YTD with a 17.1ppt lead over the index, its second best outperformance margin for the Jan – Nov period since 1987. Dividends' contribution to total returns lagged over the last decade, but returning to historical norms would imply that >50% of total returns will be from income vs. 14% today (see US Equity Strategy Year Ahead). Demand for dividend paying stocks should remain elevated, especially given low valuations and low institutional ownership. **See the Research Library for all data in Excel.**

Exhibit 1: Best and worst performing screens
As of 11/30/2022

High Foreign Exposure	11.9%
ROE (5-Yr Avg. Adj. by Debt)	8.9%
ROA	8.6%
High Duration	8.2%
Low EPS Tornado	8.2%
S&P 500 (Equal-weighted)	**6.7%**
Price Returns (0 Month)	0.4%
Price Returns (11 mo since 1 mo ago)	1.4%
Rel. Strength (Price/200 Day MA)	1.5%
Price Returns (12 Month)	1.9%
Price Returns (12 mo plus 1 mo)	1.9%
S&P 500 (Equal-weighted)	**6.7%**

Source: FactSet, BofA US Equity & Quant Strategy

Disclaimer: The valuations and screens contained herein are useful in assessing comparative valuations and comparative earnings prospects and are not intended to recommend transactions relating to any specific security. These indicators should be used in investment decisions only with other factors including financial risk, investment risk, management strategies and operating and financial outlooks.

Trading ideas and investment strategies discussed herein may give rise to significant risk and are not suitable for all investors. Investors should have experience in relevant markets and the financial resources to absorb any losses arising from applying these ideas or strategies.
BofA Securities does and seeks to do business with issuers covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.
Refer to important disclosures on page 45 to 46. **12499045**

Exhibit 67: BofA Universe Sector/Industry Factor Evaluation
As of 11/30/2022

					Valuation Analysis						Expectation Analysis							
	# of Comp	% Univ BofA	Impl. Return	Reqd Return	DDM Alpha	Eqty. Duration	BofA Adj ßeta	P/E Ratio	Price/ Book	Yield	Surprise	Earnings (Decile) Risk	Torp	Disp	Est. Rev.	PR 5yr Growth	EPS Growth 2022E	2023E
Economic Sectors																		
Energy	69	5.31	15.9	12.5	3.4	18.4	1.30	8.8	2.45	3.4	6	8	9	7	4	29.5	121	4
Materials	67	2.91	10.1	10.6	-0.5	35.7	1.09	14.4	2.91	2.0	5	5	4	5	7	9.1	-1	-11
Industrials	173	8.18	10.4	11.1	-0.7	34.3	1.11	20.7	4.15	1.6	5	4	6	4	5	20.0	36	18
Consumer Discretionary	201	11.42	9.7	11.7	-2	39.9	1.18	27.4	5.15	0.9	5	5	5	6	6	14.0	8	32
Consumer Staples	55	7.02	9.0	8.3	0.7	39.1	0.73	21.9	5.11	2.4	6	5	4	2	5	5.9	2	3
Health Care	281	14.59	10.5	9.0	1.5	36.4	0.83	20.5	4.75	1.4	5	6	5	3	5	8.8	7	0
Financials	157	8.80	11.2	10.5	0.7	31.7	1.08	12.6	1.75	2.6	5	4	5	4	6	6.1	-14	13
Information Technology	205	25.22	10.3	11.2	-0.9	37.8	1.10	22.7	7.58	1.0	7	4	5	4	6	12.8	6	4
Communication Services	53	10.76	10.4	10.5	-0.1	38.6	1.01	15	3.24	0.6	3	4	4	5	7	7.3	-8	11
Utilities	60	2.97	9.8	8.0	1.8	33.6	0.71	18.5	2.14	3.0	4	5	5	2	5	8.5	15	4
Real Estate	95	2.81	9.8	9.5	0.3	32.1	0.89	18.3	2.36	3.6	6	5	6	2	5	8.7	11	1
Capitalization Sectors ($ Million)																		
14 To 1397	284	0.41	12.0	11.2	0.8	24.9	1.17	nm	1.46	1.5	5	7	6	7	6	22.8	53	48
1399 To 3793	283	1.74	11.6	11.6	0	29.0	1.17	20.4	2.02	1.9	6	7	6	6	6	23.2	5	10
3812 To 9018	283	4.10	11.4	11.7	-0.3	31.3	1.19	16.2	2.13	1.6	5	6	5	5	6	14.9	13	11
9021 To 27738	283	11.24	10.7	10.6	0.1	33.2	1.05	15.7	2.59	1.7	5	5	5	4	5	13.5	22	10
27969 To 2360105	283	82.50	10.3	10.5	-0.2	36.6	1.01	18.4	4.36	1.6	5	5	5	4	6	11.2	5	7
Risk Sectors																		
-3.21 To 0.82	246	23.47	10.1	8.0	2.1	35.5	0.70	18.9	3.96	2.3	6	5	5	3	5	8.7	6	4
0.83 To 1.01	253	21.60	10.2	9.9	0.3	36.2	0.94	19.9	4.11	1.7	6	4	5	3	6	10.5	3	10
1.02 To 1.17	238	34.74	10.2	11.1	-0.9	37.3	1.10	17.7	4.40	1.1	5	4	5	4	6	11.4	4	10
1.18 To 1.37	251	10.81	11.8	12.5	-0.7	30.5	1.26	14.1	2.58	1.8	6	6	5	5	6	18.2	10	0
1.38 To 3.40	236	7.80	10.7	15.2	-4.5	37.3	1.60	17.9	3.07	1.1	6	6	6	6	6	15.0	18	12
Uncoded	192	1.59	13.7			26.1		3257	2.93	1.0	5	6	7	7	5	32.7	88	415
DDM Alpha																		
Most Undervalued	160	6.86	16.9	11.0	5.9	18.4	1.08	12.4	2.73	1.8	5	7	8	6	5	52.6	108	18
Undervalued	160	14.84	11.8	9.2	2.6	29.8	0.85	15.3	2.91	2.2	5	6	6	3	5	13.9	12	8
Fair Value	160	15.92	10.1	9.5	0.6	35.2	0.89	19.5	4.15	2.0	6	4	6	3	5	10.0	9	10
Overvalued	160	30.04	9.7	10.8	-1.1	39.4	1.05	19.6	5.91	1.1	6	3	5	3	6	9.0	2	6
Most Overvalued	161	13.99	8.9	12.7	-3.8	44.9	1.29	26.8	5.76	0.7	5	5	5	5	7	9.7	-4	18
Uncoded	615	18.35	13.7	-23.0	36.7	26.1	1.02	16.6	2.47	2.2	6	5	4	5	6	0.8	-3	-1

Source: BofA US Equity and Quant Strategy, FactSet

Exhibit 67: BofA Universe Sector/Industry Factor Evaluation
As of 11/30/2022

					Valuation Analysis						Expectation Analysis							
	# of Comp	% Univ BofA	Impl. Return	Reqd Return	DDM Alpha	Eqty. Duration	BofA Adj ßeta	P/E Ratio	Price/ Book	Yield	Surprise	Earnings (Decile) Risk	Torp	Disp	Est. Rev.	PR 5yr Growth	EPS Growth 2022E	2023E
Economic Sectors																		
Energy	69	5.31	15.9	12.5	3.4	18.4	1.30	8.8	2.45	3.4	6	8	9	7	4	29.5	121	4
Materials	67	2.91	10.1	10.6	-0.5	35.7	1.09	14.4	2.91	2.0	5	5	4	5	7	9.1	-1	-11
Industrials	173	8.18	10.4	11.1	-0.7	34.3	1.11	20.7	4.15	1.6	5	4	6	4	5	20.0	36	18
Consumer Discretionary	201	11.42	9.7	11.7	-2	39.9	1.18	27.4	5.15	0.9	5	5	5	6	6	14.0	8	32
Consumer Staples	55	7.02	9.0	8.3	0.7	39.1	0.73	21.9	5.11	2.4	6	5	4	2	5	5.9	2	3
Health Care	281	14.59	10.5	9.0	1.5	36.4	0.83	20.5	4.75	1.4	5	6	5	3	5	8.8	7	0
Financials	157	8.80	11.2	10.5	0.7	31.7	1.08	12.6	1.75	2.6	5	4	5	4	6	6.1	-14	13
Information Technology	205	25.22	10.3	11.2	-0.9	37.8	1.10	22.7	7.58	1.0	7	4	5	4	6	12.8	6	4
Communication Services	53	10.76	10.4	10.5	-0.1	38.6	1.01	15	3.24	0.6	3	4	4	5	7	7.3	-8	11
Utilities	60	2.97	9.8	8.0	1.8	33.6	0.71	18.5	2.14	3.0	4	5	5	2	5	8.5	15	4
Real Estate	95	2.81	9.8	9.5	0.3	32.1	0.89	18.3	2.36	3.6	6	5	6	2	5	8.7	11	1
Capitalization Sectors ($ Million)																		
14 To 1397	284	0.41	12.0	11.2	0.8	24.9	1.17	nm	1.46	1.5	5	7	6	7	6	22.8	53	48
1399 To 3793	283	1.74	11.6	11.6	0	29.0	1.17	20.4	2.02	1.9	6	7	6	6	6	23.2	5	10
3812 To 9018	283	4.10	11.4	11.7	-0.3	31.3	1.19	16.2	2.13	1.6	5	6	5	5	6	14.9	13	11
9021 To 27738	283	11.24	10.7	10.6	0.1	33.2	1.05	15.7	2.59	1.7	5	5	5	4	5	13.5	22	10
27969 To 2360105	283	82.50	10.3	10.5	-0.2	36.6	1.01	18.4	4.36	1.6	5	5	5	4	6	11.2	5	7
Risk Sectors																		
-3.21 To 0.82	246	23.47	10.1	8.0	2.1	35.5	0.70	18.9	3.96	2.3	6	5	5	3	5	8.7	6	4
0.83 To 1.01	253	21.60	10.2	9.9	0.3	36.2	0.94	19.9	4.11	1.7	6	4	5	3	6	10.5	3	10
1.02 To 1.17	238	34.74	10.2	11.1	-0.9	37.3	1.10	17.7	4.40	1.1	5	4	5	4	6	11.4	4	10
1.18 To 1.37	251	10.81	11.8	12.5	-0.7	30.5	1.26	14.1	2.58	1.8	6	6	5	5	6	18.2	10	0
1.38 To 3.40	236	7.80	10.7	15.2	-4.5	37.3	1.60	17.9	3.07	1.1	6	6	6	6	6	15.0	18	12
Uncoded	192	1.59	13.7			26.1		3257	2.93	1.0	5	6	7	7	5	32.7	88	415
DDM Alpha																		
Most Undervalued	160	6.86	16.9	11.0	5.9	18.4	1.08	12.4	2.73	1.8	5	7	8	6	5	52.6	108	18
Undervalued	160	14.84	11.8	9.2	2.6	29.8	0.85	15.3	2.91	2.2	5	6	6	3	5	13.9	12	8
Fair Value	160	15.92	10.1	9.5	0.6	35.2	0.89	19.5	4.15	2.0	6	4	6	3	5	10.0	9	10
Overvalued	160	30.04	9.7	10.8	-1.1	39.4	1.05	19.6	5.91	1.1	6	3	5	3	6	9.0	2	6
Most Overvalued	161	13.99	8.9	12.7	-3.8	44.9	1.29	26.8	5.76	0.7	5	5	5	5	7	9.7	-4	18
Uncoded	615	18.35	13.7	-23.0	36.7	26.1	1.02	16.6	2.47	2.2	6	5	4	5	6	0.8	-3	-1

Source: BofA US Equity and Quant Strategy, FactSet

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Quantitative Profiles

Regime model hits a fork in the road: hard or soft landing?

Teetering at the brink of Late Cycle

The US Regime Indicator inched lower in its Late Cycle phase, now just nine bp away from triggering a Downturn which is marked by poor and deteriorating macro conditions. Of its eight inputs, four improved last month (the Earnings Revision Ratio, inflation, 10-yr yields, and credit spreads) and four deteriorated (GDP forecasts, ISM PMI, Leading Indicators, Capacity Utilization). We ran scenarios for January (Exhibit 4) and found that if indicators flat-line, we would still hit a Downturn. But if indicators improve, which is suggested by the building narrative around a soft landing, we could avoid a Downturn - but we would need to see a marked improvement (10%+) to drive us back into Mid Cycle. For example, ISM PMI would need to jump to 53 from 48 etc. Since 1990, the Downturn phase occurred seven times, lasting an average of 12mths, ranging between six (9/95 – 3/96) and 19mths (GFC). If we extend the US Regime Indicator back to the 70s and lost a few variables, the average downturn lasted 11mths.

2022: "bird in hand" beat "buy the dream"

Only a handful of factors yielded positive returns last year, with winners focused on "bird in the hand" strategies like high dividend yield, short equity duration, and positive earnings revisions. Value outperformed Growth (Russell 1000 Value beat Growth by 22ppt in 2022), but absolute returns for all Value factors were negative. But one of the biggest pain trades was in expensive stocks (the bottom decile of Value factors) which lagged almost all other factors that we track - investors who stuck with expensive winners suffered most. Long equity duration stocks ("buy the dream" growth stocks that are most susceptible to a rising discount rate) was among the worst five factors in '22.

The reign of King Cash should continue

All cash return strategies we track outperformed in 2022 as cash yields rose by 431bp. We expect this to continue: cash flow based valuation has an outperforming factor during most periods with one exception: the last decade's zero interest rate policy (cash was worthless). We prefer companies generating cash rather than those that need capital to grow. Not only are rates likely to remain higher than they have been in recent past, but we are likely exiting an era of hyper-accommodative monetary policy. Factor positioning suggests that our view is contrarian: long duration stocks are still the most crowded and the most expensive of the factors we follow (Exhibit 15).

Signs point to micro over macro in 2023

2022's market dominated by macro variables, where real rates explained more of stock returns than fundamentals, may be a thing of the past. Returns have shifted to fundamental drivers: reactions to Fed-speak pale in comparison to earnings surprises, valuation dispersion suggest plenty of mean reversion alpha. In 2023, we continue to focus on free cash flow-based valuation factors. This theme is still underweight but has historically provided protection during Downturns. And Risk and Small Size are deeply underweight and extremely inexpensive, but could prove to be good hedges against things going right, given that they tend to bounce back the most in recoveries.

See the Research Library for all data in Excel.

Trading ideas and investment strategies discussed herein may give rise to significant risk and are not suitable for all investors. Investors should have experience in relevant markets and the financial resources to absorb any losses arising from applying these ideas or strategies.
BofA Securities does and seeks to do business with issuers covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.
Refer to important disclosures on page 35 to 37. 12506201

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Exhibit 1: Best and worst performing screens
As of 12/30/2022

Top 5/Bottom 5 screens in 2022

Top 5 screens	Perf.
Dividend Yield (Total Return)	3.7%
High EPS Estimate Dispersion	3.6%
Upward Estimate Revisions	3.4%
High Dividend Growth (Total Ret.)	-1.9%
Price Returns (12-Month)	-2.7%
S&P 500 (Equal weighted)	**-13.0%**

Bottom 5 screens	Perf.
ROE (5-Yr Avg. Adj. by Debt)	-27.5%
ROE (1-Yr Avg. Adj. by Debt)	-25.7%
High Duration	-25.5%
ROA	-23.9%
ROC	-21.3%
S&P 500 (Equal weighted)	**-13.0%**

Source: FactSet, BofA US Equity & Quant Strategy

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Disclaimer: The valuations and screens contained herein are useful in assessing comparative valuations and comparative earnings prospects and are not intended to recommend transactions relating to any specific security. These indicators should be used in investment decisions only with other factors including financial risk, investment risk, management strategies and operating and financial outlooks.

Exhibit 52: BofA Universe Sector/Industry Factor Evaluation
As of 12/30/2022

					Valuation Analysis					Expectation Analysis								
	# of Comp	% Univ BofA	Impl. Return	Reqd Return	DDM Alpha	Eqty. Duration	BofA Adj ßeta	P/E Ratio	Price/ Book	Yield	Surprise	Earnings (Decile) Risk	Torp	Disp	Est. Rev.	PR 5yr Growth	EPS Growth 2023E	EPS Growth 2024E
Economic Sectors																		
Energy	69	5.52	16.2	12.6	3.6	18.1	1.28	8.5	2.36	3.6	6	8	9	6	5	29.5	121	5
Materials	67	2.98	10.3	10.7	-0.4	35.1	1.08	13.7	2.76	2.2	5	5	4	5	7	8.5	-1	-11
Industrials	174	8.46	10.5	11.2	-0.7	33.8	1.10	20.1	3.95	1.7	5	4	6	4	5	20.6	34	18
Consumer Discretionary	202	10.82	10.0	11.6	-1.6	38.0	1.16	23.9	4.38	1.0	5	5	5	6	6	13.1	10	31
Consumer Staples	54	6.78	9.0	8.4	0.6	39.0	0.73	21.2	5.10	2.5	6	5	4	2	5	5.8	3	3
Health Care	282	15.40	10.4	8.9	1.5	36.1	0.83	20.1	4.64	1.5	5	6	5	3	5	8.9	7	-1
Financials	157	8.92	11.4	10.6	0.8	31.1	1.07	11.9	1.65	2.8	5	4	5	4	6	6.1	-14	13
Information Technology	203	24.78	10.5	11.4	-0.9	36.5	1.11	20.8	6.94	1.1	7	4	5	4	6	12.6	5	3
Communication Services	54	10.29	10.5	10.8	-0.3	37.3	1.03	13.4	2.95	0.7	3	4	3	5	7	6.2	-8	9
Utilities	60	3.18	9.7	8.1	1.6	33.6	0.70	18.3	2.11	3.1	5	5	5	2	5	8.1	15	4
Real Estate	95	2.87	10.0	9.6	0.4	30.9	0.89	17.3	2.23	3.8	6	5	6	2	6	8.7	11	1
Capitalization Sectors ($ Million)																		
4 To 1307	284	0.40	12.4	11.4	1	24.3	1.19	nm	1.22	1.8	6	8	6	7	6	22.3	42	43
1316 To 3468	283	1.76	11.9	11.7	0.2	28.1	1.17	22.6	1.94	2.0	5	7	5	6	6	23.6	27	15
3471 To 8372	283	4.18	11.5	11.7	-0.2	30.6	1.18	13.7	1.96	1.7	5	6	6	5	5	15.3	22	5
8420 To 26286	283	11.47	10.8	10.7	0.1	32.5	1.04	15.5	2.51	1.8	6	5	6	4	5	13.3	28	13
26329 To 2071529	284	82.19	10.4	10.6	-0.2	35.5	1.01	17.1	4.06	1.7	5	5	5	4	6	11.0	4	6
Risk Sectors																		
-3.03 To 0.82	249	23.79	10.0	8.1	1.9	35.2	0.70	18.1	3.81	2.5	5	5	5	3	5	8.1	7	3
0.83 To 1.01	246	22.32	10.4	10.1	0.3	35.7	0.94	19.1	3.95	1.7	6	4	5	3	6	10.5	3	7
1.02 To 1.17	260	34.65	10.4	11.3	-0.9	35.8	1.10	16.3	4.01	1.2	5	4	5	5	6	11.6	4	11
1.18 To 1.37	244	11.17	12.1	12.6	-0.5	29.9	1.26	13.3	2.28	1.9	6	6	5	5	6	17.6	8	0
1.38 To 3.36	238	6.72	11.5	15.5	-4	33.7	1.61	15.8	3.03	1.1	5	6	6	7	6	16.4	27	13
Uncoded	180	1.34	12.9			28.0		324	2.44	1.2	6	6	7	7	5	27.6	88	nm
DDM Alpha																		
Most Undervalued	161	6.57	17.4	11.3	6.1	17.6	1.10	11.6	2.53	2.0	5	7	8	6	5	55.2	114	18
Undervalued	161	15.28	12.1	9.3	2.8	29.1	0.85	15.1	2.74	2.2	5	6	6	3	5	14.6	12	9
Fair Value	161	17.50	10.1	9.5	0.6	34.5	0.87	17.9	4.21	2.3	6	5	5	2	5	8.8	8	8
Overvalued	161	25.33	9.9	10.7	-0.8	38.0	1.03	17.7	4.57	1.2	5	4	5	3	6	8.7	1	7
Most Overvalued	162	18.64	9.4	12.5	-3.1	42.4	1.24	23.3	6.94	0.8	6	4	5	5	7	9.6	-2	11
Uncoded	611	16.68	12.8	-22.0	34.8	28.0	1.05	15.2	2.18	2.4	6	5	4	6	6	-0.7	-2	-3

Source: BofA US Equity and Quant Strategy, FactSet

Exhibit 55: BofA Universe Sector/Industry Factor Evaluation (cont'd)
As of 12/30/2022

			Valuation Analysis							Expectation Analysis								
	# of Comp	% Univ BofA	Impl. Return	Reqd Return	DDM Alpha	Eqty. Duration	BofA Adj ßeta	P/E Ratio	Price/ Book	Yield	Surprise	Earnings (Decile) Risk	Torp	Disp	Est. Rev.	PR 5yr Growth	EPS Growth 2023E	EPS Growth 2024E
CONSUMER STAPLES	54	6.78	9.0	8.4	0.6	39.0	0.73	21.2	5.10	2.5	6	5	4	2	5	5.8	3	3
FOOD & STAPLES RETAILING	8	1.30	9.4	9.0	0.4	40.6	0.76	20.1	5.00	1.5	6	4	5	3	5	6.9	2	5
BEVERAGES	11	1.98	8.6	8.6	0	40.7	0.77	26.7	6.84	2.3	7	6	5	1	5	7.1	5	6
FOOD PRODUCTS	22	1.22	9.5	7.9	1.6	36.6	0.70	17.6	2.77	2.4	7	3	5	3	5	5.0	6	-3
TOBACCO	2	0.64	10.4	8.9	1.5	28.9	0.79	13.4		6.1	4	3	4	3	6	2.6	0	0
HOUSEHOLD PRODUCTS	5	1.36	8.6	7.4	1.2	40.0	0.61	25.0	8.23	2.5	7	6	4	1	7	5.1	-2	6
PERSONAL PRODUCTS	6	0.28	8.0	10.6	-2.6	49.0	1.01	32.5	10.44	0.9	6	7	4	6	9	6.2	0	3
HEALTH CARE	282	15.40	10.4	8.9	1.5	36.1	0.83	20.1	4.64	1.5	5	6	5	3	5	8.9	7	-1
HEALTH CARE EQUIP	47	2.58	9.1	9.8	-0.7	43.1	0.93	26.4	4.28	1.1	7	6	4	2	5	8.9	-2	-1
HEALTH CARE PROV	51	3.53	10.8	9.4	1.4	34.4	0.88	18.0	3.45	1.1	5	3	6	2	4	10.7	6	10
HEALTH CARE TECH	12	0.12	9.8	10.1	-0.3	41.6	0.94	nm	2.82	0.0	5	2	5	4	5	14.2	-47	139
BIOTECH	121	2.90	11.0	7.9	3.1	32.8	0.80	27.3	6.08	1.7	6	7	4	6	5	6.0	4	-14
PHARMACEUTICALS	29	4.44	10.1	7.7	2.4	34.9	0.68	14.9	5.46	2.4	4	8	5	4	5	7.2	17	-5
LIFE SCIENCES	22	1.83	13.5	10.5	3	31.0	0.97	25.6	4.66	0.2	3	4	4	2	5	12.9	-3	1
FINANCIALS	157	8.92	11.4	10.6	0.8	31.1	1.07	11.9	1.65	2.8	5	4	5	4	6	6.1	-14	13
BANKS	32	2.94	12.0	11.2	0.8	26.7	1.11	9.4	1.20	3.3	3	4	4	5	5	3.8	-15	11
THRIFTS & MORTGAGE FINANCE	9	0.06	14.1	11.6	2.5	21.4	1.13	6.4	0.81	4.0	3	3	4	4	4	5.0	-2	-14
DIV FINANCIALS	4	0.15	13.9	14.0	-0.1	26.0	1.43	9.8	1.62	2.9	7	9	7	3	4	12.4	-17	32
CONSUMER FINANCE	11	0.54	10.2	11.3	-1.1	36.5	1.18	9.0	1.78	1.9	6	7	3	7	5	0.1	-18	-7
CAPITAL MARKETS	52	2.95	11.8	10.8	1	29.9	1.12	13.9	2.09	2.8	5	4	5	4	6	5.5	-11	13
MORTGAGE REITS	15	0.09	12.3	13.6	-1.3	20.8	1.28	6.4	0.90	13.2	7	6	3	5	4	9.0	-10	-7
INSURANCE	34	2.18	10.5	9.4	1.1	37.0	0.89	17.4	2.32	1.8	7	5	6	3	7	11.0	-14	31
INFO TECH	203	24.78	10.5	11.4	-0.9	36.5	1.11	20.8	6.94	1.1	7	4	5	4	6	12.6	5	3
INTERNET SOFTWARE	1	0.02		13.6		16.0	1.38	21.2		0.0	2		9	9	1	76.6	290	-31
IT SERVICES	45	4.27	10.6	10.9	-0.3	34.6	1.07	21.7	5.56	1.2	5	3	6	3	5	13.1	11	15
SOFTWARE	89	8.70	10.5	10.6	-0.1	37.1	1.01	25.0	8.11	0.8	7	5	6	3	6	16.1	8	11
COMMUNICA. EQUIP	9	0.71	11.5	10.6	0.9	29.9	1.01	13.8	4.54	2.4	5	9	6	2	5	11.9	6	10
COMPUTERS & PERIPH	9	5.85	10.2	11.9	-1.7	39.0	1.17	19.7	24.81	0.8	10	1	5	3	7	10.6	5	-4
ELECTR EQUIP & INSTR	17	0.64	10.1	11.7	-1.6	37.3	1.14	14.1	3.10	1.2	5	5	6	3	5	7.9	17	3
SEMICONDUCTORS	33	4.59	10.5	13.0	-2.5	34.7	1.26	18.2	4.18	1.7	6	5	4	6	7	8.8	-5	-7
COMMUNICATION SERVICES	54	10.29	10.5	10.8	-0.3	37.3	1.03	13.4	2.95	0.7	3	4	3	5	7	6.2	-8	9
DIVERSIFIED TELECOM SVS	5	0.84	12.4	8.2	4.2	25.8	0.65	7.6	1.26	6.4	8	6	2	3	4	0.4	-15	-5
WIRELESS TELECOM SVS	1	0.22	19.2	8.1	11.1	20.6	0.70	37.0	2.49	0.0	1	4	8	6	9	44.9	-1	233
MEDIA	16	0.86	11.8	10.6	1.2	27.4	1.03	10.1	1.80	2.2	7	6	6	5	6	8.4	4	2
ENTERTAINMENT	17	1.20	11.5	11.6	-0.1	34.9	1.15	25.5	2.22	0.1	5	5	5	6	7	13.9	-14	18
INTERACTIVE MEDIA & SVCS	15	7.16	9.7	11.0	-1.3	40.2	1.07	13.8	4.11	0.0	1	3	3	5	8	4.3	-7	11
UTILITIES	60	3.18	9.7	8.1	1.6	33.6	0.70	18.3	2.11	3.1	5	5	5	2	5	8.1	15	4
ELECTRIC UTILITIES	25	1.95	9.8	7.9	1.9	33.8	0.68	17.8	2.10	3.0	4	4	5	1	5	7.4	15	2
GAS UTILITIES	9	0.14	10.4	8.7	1.7	31.1	0.76	15.9	1.69	3.3	7	4	5	2	6	6.4	4	9
MULTI-UTILITIES	15	0.84	9.5	7.9	1.6	33.8	0.68	19.0	2.06	3.4	6	5	4	2	5	5.0	0	6
WATER UTILITIES	2	0.11	8.6	8.3	0.3	41.2	0.73	31.2	3.07	1.9	9	3	5	1	5	7.6	5	6
INDEP POWER PROD & ENERGY TRAD	9	0.14	10.4	10.3	0.1	26.8	0.95	17.6	2.82	2.9	7	7	7	6	6	40.0	nm	24
REAL ESTATE	95	2.87	10.0	9.6	0.4	30.9	0.89	17.3	2.23	3.8	6	5	6	2	6	8.7	11	1
REITS	91	2.84	10.0	9.6	0.4	31.1	0.89	17.0	2.24	3.8	6	5	6	2	6	8.2	12	2
REAL ESTATE MGMT & DEV	4	0.03	13.6	12.7	0.9	16.8	1.28	nm	1.46	1.2		8				57.1	nm	nm
BofA UNIVERSE	1417	100.00	10.5	10.6	-0.1	34.9	1.02	17.5	3.54	1.7						11.6	9	8
S&P 500	503	92.82	10.4	10.6	-0.2	35.4	1.01	17.0	3.59	1.7						10.6	6	6

Source: BofA US Equity and Quant Strategy, FactSet

Quantitative Profiles
"Valuation doesn't matter…"

"…price is the best predictor of price."

We hear this a lot, and the unwavering faith in price momentum investing is likely attributable to a liquidity-fueled decade that saw tremendous serial correlation across price returns. The average portfolio manager (~45 y.o.) has seen a financial crisis during which statistically cheap stocks were traps, followed by a decade during which value factors destroyed alpha almost every year while investing based on past price return turned in hefty alpha. The few value investors left see post-COVID shifts as a sort of come-uppance: Value (proxied by long-short EV/EBITDA) has returned 30 ppt since December 2020, whereas price return produced no alpha. And prior to the GFC, valuation was a far better signal than basing future forecasts on past price returns (cover chart).

Flirting with a downturn, but watch crosscurrents

Our US Regime Indicator dipped its toe into Downturn territory this month; if we get a 2nd month of confirmation, we will officially be out of Late Cycle and in the abyss (Exhibit 5). But January saw market returns more similar to early cycle, as during a downturn, quality, low risk and large caps outperformed most (Exhibit 9). Today's economic crosscurrents suggest things may be a bit different (see Crosscurrents note). One rule that we think still holds: cash-rich companies fare better, especially Free Cash Flow to Enterprise Value, a factor we like for Late Cycle and Downturn phases (Exhibit 8). Value doesn't always lag Growth during Downturns, as it led for 12m into the 70s' Downturn (Exhibit 22). Downturns last about 12m on avg. based on data back to the 70s.

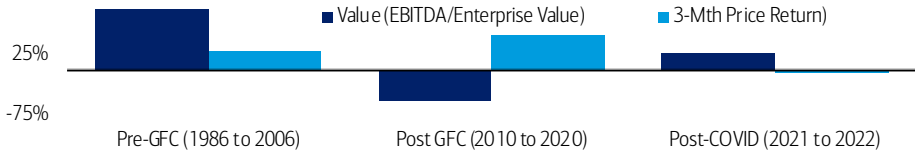
Jan. effect on steroids: junk, small caps & positioning led

And continued to lead into February. Risk factors (+13.3% avg) turned in best returns typical of January (see Jan effect note). Consistent with January (Exhibit 10), Low Quality and Small Size which lagged in 2022 reversed in January. The 50 most shorted stocks in the S&P 500 bounced 14%, a two standard deviation move based on S&P 500 decile returns back to 1993, and short interest explained an unusually high percentage of returns variability in January (ρ=34%) the highest explanatory power in over 20yrs.

Labor intensive stocks lag regardless of wage pressure

The Fed is keenly focused on cooling wage inflation, a potential boon to labor intensive company's margins. But labor intensive companies have been almost constant laggards relative to labor-light counterparts. Since '86, the former lagged the latter 70% of the time, trailing by ~2ppt p.a. (Exhibit 4). When CPI outpaced wage costs, performance was even worse, as underperformance margin was 7.4ppt p.a. (**Data in Research Library.**)

Exhibit 2: Valuation matters, it just hasn't for the last 10 years
EBITDA/EV and 3-mth Price Return performance



Source: US Equity & Quant Strategy, FactSet

BofA GLOBAL RESEARCH

Trading ideas and investment strategies discussed herein may give rise to significant risk and are not suitable for all investors. Investors should have experience in relevant markets and the financial resources to absorb any losses arising from applying these ideas or strategies.
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Refer to important disclosures on page 35 to 37. 12521562

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Exhibit 1: Best and worst performing screens
As of 1/31/2023

Top 5/Bottom 5 screens in January

Top 5 screens	Perf.
High EPS Estimate Dispersion	18.0%
High Beta	15.2%
Low EPS Torpedo	14.4%
Small Size	13.3%
Low Price to Book Value	12.8%
S&P 500 (Equal weighted)	**7.5%**

Bottom 5 screens	Perf.
Price Returns (12-m plus 1-m)	0.8%
Relative Strength (30wk/75wk)	1.3%
Price Returns (12-Month)	1.7%
Price Returns (9-Month)	1.9%
Price Returns (11-m since 1 y ago)	2.4%
S&P 500 (Equal weighted)	**7.5%**

Source: FactSet, BofA US Equity & Quant Strategy

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Disclaimer: The valuations and screens contained herein are useful in assessing comparative valuations and comparative earnings prospects and are not intended to recommend transactions relating to any specific security. These indicators should be used in investment decisions only with other factors including financial risk, investment risk, management strategies and operating and financial outlooks.

Exhibit 51: BofA Universe Sector/Industry Factor Evaluation
As of 1/31/2023

					Valuation Analysis						Expectation Analysis							
	# of Comp	% Univ BofA	Impl. Return	Reqd Return	DDM Alpha	Eqty. Duration	BofA Adj ßeta	P/E Ratio	Price/ Book	Yield	Earnings Surprise	Earnings (Decile) Risk	Torp	Disp	Est. Rev.	PR 5yr Growth	EPS Growth 2023E	EPS Growth 2024E
Economic Sectors																		
Energy	70	5.48	17.4	12.5	4.9	17.5	1.28	9.5	2.41	3.6	7	8	5	7	6	29.9	115	-15
Materials	67	3.17	10.0	10.8	-0.8	35.8	1.11	15.7	2.98	2.0	5	6	4	5	6	7.9	3	-6
Industrials	176	8.43	10.5	11.2	-0.7	33.6	1.11	20.1	4.08	1.7	6	4	6	4	6	16.7	16	12
Consumer Discretionary	204	11.70	9.7	11.8	-2.1	39.8	1.19	27.4	5.03	0.9	5	5	5	6	6	12.4	6	29
Consumer Staples	55	6.19	9.2	8.3	0.9	38.1	0.73	20.8	5.20	2.6	6	5	4	2	5	6.7	2	3
Health Care	277	14.66	10.5	8.8	1.7	35.0	0.81	19.6	4.56	1.5	5	5	5	3	6	8.6	9	-7
Financials	157	9.17	11.0	10.8	0.2	31.3	1.08	13.1	1.63	2.6	7	5	6	4	6	7.2	-6	16
Information Technology	202	25.83	10.1	11.4	-1.3	38.1	1.12	23.3	7.38	1.0	6	4	5	3	7	12.8	8	3
Communication Services	54	9.20	11.0	10.7	0.3	35.1	1.04	15.2	3.23	0.8	3	4	4	5	8	8.3	-6	15
Utilities	60	3.05	9.7	8.0	1.7	33.4	0.70	18.1	2.08	3.1	4	5	5	2	5	8.0	15	3
Real Estate	95	3.12	9.6	9.8	-0.2	33.2	0.92	19.4	2.43	3.4	6	5	5	2	6	8.6	11	3
Capitalization Sectors ($ Million)																		
22 To 1505	284	0.49	12.1	11.5	0.6	26.1	1.23	nm	1.47	1.5	6	7	6	7	6	22.6	54	44
1515 To 3900	283	1.97	11.7	11.9	-0.2	29.1	1.19	23.9	2.22	1.9	5	6	5	6	5	24.5	-2	6
3924 To 9349	283	4.55	11.2	11.8	-0.6	31.6	1.20	15.6	2.22	1.6	5	5	6	5	5	15.2	22	4
9365 To 26540	283	11.84	10.6	10.7	-0.1	33.3	1.06	17.1	2.68	1.7	6	5	6	4	6	13.1	20	4
26543 To 2157659	284	81.16	10.4	10.6	-0.2	35.8	1.02	18.5	4.28	1.6	5	5	5	4	6	11.0	7	3
Risk Sectors																		
-3.01 To 0.83	254	23.16	10.2	8.0	2.2	34.0	0.69	17.5	3.63	2.5	5	5	5	3	5	8.5	12	0
0.84 To 1.02	249	22.02	9.9	9.9	0	37.0	0.94	20.5	4.27	1.6	5	4	5	3	6	10.5	8	9
1.03 To 1.19	260	34.88	10.7	11.3	-0.6	35.5	1.12	18.4	4.29	1.2	6	4	5	5	7	12.3	7	7
1.20 To 1.39	242	10.65	11.6	12.7	-1.1	30.5	1.29	14.9	2.40	1.9	6	6	5	5	6	15.0	14	-7
1.40 To 3.37	251	7.89	10.2	15.5	-5.3	38.9	1.64	19.1	3.49	1.1	5	7	5	7	6	15.0	21	1
Uncoded	161	1.40	12.5			27.5		nm	2.82	0.5	5	5	7	7	5	28.2	66	114
DDM Alpha																		
Most Undervalued	168	8.68	15.6	10.6	5	19.4	1.02	13.5	3.06	2.1	6	8	6	6	6	45.2	98	-3
Undervalued	168	15.41	11.3	9.1	2.2	30.9	0.83	15.9	2.68	2.3	5	5	6	3	5	11.7	7	5
Fair Value	168	16.00	10.5	10.0	0.5	33.5	0.94	17.8	3.96	2.1	6	5	6	3	6	10.6	14	9
Overvalued	168	23.77	9.8	10.7	-0.9	38.1	1.03	19.1	4.77	1.3	5	4	5	3	6	9.1	4	8
Most Overvalued	169	19.03	9.1	12.6	-3.5	44.2	1.28	26.9	7.55	0.7	6	4	5	5	7	9.6	-7	10
Uncoded	576	17.09	12.5	-21.5	34	27.5	1.05	17.9	2.49	2.1	6	5	4	6	6	-0.3	10	-8

Source: BofA US Equity and Quant Strategy, FactSet

Exhibit 54: BofA Universe Sector/Industry Factor Evaluation (cont'd)

As of 1/31/2023

						Valuation Analysis					Expectation Analysis							
	# of Comp	% Univ BofA	Impl. Return	Reqd Return	DDM Alpha	Eqty. Duration	BofA Adj ßeta	P/E Ratio	Price/ Book	Yield	Surprise	Earnings (Decile) Risk	Torp	Disp	Est. Rev.	PR 5yr Growth	EPS Growth 2023E	EPS Growth 2024E
CONSUMER STAPLES	55	6.19	9.2	8.3	0.9	38.1	0.73	20.8	5.20	2.6	6	5	4	2	5	6.7	2	3
FOOD & STAPLES RETAILING	8	1.34	9.1	8.9	0.2	42.4	0.76	21.3	5.74	1.4	6	4	5	3	4	7.1	3	4
BEVERAGES	12	1.67	8.7	8.5	0.2	39.7	0.76	26.3	7.14	2.4	6	6	5	1	5	9.6	6	7
FOOD PRODUCTS	22	1.06	9.6	7.6	2	35.5	0.70	16.6	2.55	2.5	6	3	4	3	4	4.0	2	-2
TOBACCO	2	0.63	10.3	8.8	1.5	29.5	0.80	13.6		6.0	4	3	4	3	7	2.6	0	0
HOUSEHOLD PRODUCTS	5	1.26	9.2	7.3	1.9	37.3	0.61	23.2	8.50	2.6	6	6	4	1	5	6.6	0	5
PERSONAL PRODUCTS	6	0.22	8.0	10.8	-2.8	49.6	1.05	34.5	10.17	0.7	6	7	4	5	9	7.2	4	4
HEALTH CARE	277	14.66	10.5	8.8	1.7	35.0	0.81	19.6	4.56	1.5	5	5	5	3	6	8.6	9	-7
HEALTH CARE EQUIP	45	2.58	9.7	9.9	-0.2	40.6	0.91	27.2	4.17	1.1	6	6	4	3	6	9.9	-3	2
HEALTH CARE PROV	52	3.31	11.0	9.1	1.9	33.6	0.86	16.8	3.26	1.2	4	3	6	2	5	10.4	1	8
HEALTH CARE TECH	12	0.13	9.3	10.0	-0.7	43.2	0.95	nm	3.10	0.0	5	2	5	4	5	14.1	-47	142
BIOTECH	118	2.81	10.5	8.0	2.5	32.4	0.77	27.4	6.13	1.7	6	7	4	6	5	9.2	5	-22
PHARMACEUTICALS	30	4.02	10.5	7.6	2.9	32.9	0.67	13.9	6.34	2.5	4	8	4	3	6	8.2	23	-15
LIFE SCIENCES	20	1.81	10.4	11.3	-0.9	39.8	0.96	26.7	4.78	0.2	3	4	4	2	6	3.4	-6	-1
FINANCIALS	157	9.17	11.0	10.8	0.2	31.3	1.08	13.1	1.63	2.6	7	5	6	4	6	7.2	-6	16
BANKS	32	3.14	11.5	11.1	0.4	28.3	1.12	10.8	1.26	3.0	9	4	6	5	7	3.9	26	27
THRIFTS & MORTGAGE FINANCE	9	0.07	13.5	11.5	2	22.6	1.14	7.4	0.92	3.8	4	3	4	4	5	5.3	-2	-16
DIV FINANCIALS	4	0.16	13.4	13.8	-0.4	25.8	1.43	10.3	1.80	2.6	10	9	7	3	3	14.0	13	-38
CONSUMER FINANCE	11	0.58	12.1	12.3	-0.2	29.7	1.24	11.1	1.88	1.6	5	7	5	5	7	10.1	73	80
CAPITAL MARKETS	52	2.93	10.9	11.2	-0.3	32.1	1.13	15.4	2.07	2.6	7	4	7	4	6	5.2	2	14
MORTGAGE REITS	15	0.11	10.7	14.0	-3.3	26.6	1.32	7.3	1.02	11.5	6	6	3	5	4	7.6	-9	-7
INSURANCE	34	2.18	9.9	9.4	0.5	35.3	0.89	17.1	2.19	1.7	6	5	6	4	5	13.4	-22	38
INFO TECH	202	25.83	10.1	11.4	-1.3	38.1	1.12	23.3	7.38	1.0	6	4	5	3	7	12.8	8	3
INTERNET SOFTWARE	1	0.03		13.9		16.6	1.43	28.2		0.0	1		9	9	1	77.5	295	-29
IT SERVICES	44	4.40	10.3	10.9	-0.6	36.2	1.08	22.7	5.80	1.2	5	3	6	3	5	12.0	8	16
SOFTWARE	89	8.86	10.2	10.5	-0.3	38.5	1.00	27.4	8.41	0.7	4	4	5	3	6	16.6	8	11
COMMUNICA. EQUIP	9	0.70	11.4	10.4	1	30.4	1.00	14.0	4.58	2.4	5	9	6	2	5	11.5	6	9
COMPUTERS & PERIPH	9	5.98	10.0	11.9	-1.9	40.5	1.19	22.4	26.84	0.8	10	2	5	3	8	10.6	4	-5
ELECTR EQUIP & INSTR	17	0.67	9.8	11.7	-1.9	39.6	1.15	15.2	3.28	1.1	5	5	5	3	6	6.5	19	0
SEMICONDUCTORS	33	5.19	9.9	12.8	-2.9	37.4	1.29	22.5	4.80	1.5	6	5	4	6	7	10.3	10	-8
COMMUNICATION SERVICES	54	9.20	11.0	10.7	0.3	35.1	1.04	15.2	3.23	0.8	3	4	4	5	8	8.3	-6	15
DIVERSIFIED TELECOM SVS	5	0.89	10.8	8.6	2.2	32.4	0.67	8.8	1.50	5.9	8	6	3	4	8	-0.5	-5	-52
WIRELESS TELECOM SVS	1	0.23	18.8	8.1	10.7	20.9	0.71	39.5	2.66	0.0	1	4	8	6	2	44.9	-1	233
MEDIA	16	0.82	11.8	10.5	1.3	27.7	1.04	10.9	2.08	2.3	4	6	6	4	7	8.6	-12	2
ENTERTAINMENT	17	1.43	12.7	11.5	1.2	30.5	1.16	29.4	2.72	0.1	4	5	7	6	7	16.7	-14	21
INTERACTIVE MEDIA & SVCS	15	5.83	10.2	10.9	-0.7	38.2	1.07	15.5	4.60	0.0	1	3	3	6	8	6.2	-6	14
UTILITIES	60	3.05	9.7	8.0	1.7	33.4	0.70	18.1	2.08	3.1	4	5	5	2	5	8.0	15	3
ELECTRIC UTILITIES	25	1.82	9.8	7.8	2	33.4	0.68	17.3	2.02	3.1	4	5	5	1	5	6.8	15	1
GAS UTILITIES	9	0.14	10.2	8.7	1.5	31.8	0.78	17.0	1.75	3.0	7	4	6	2	6	6.9	3	5
MULTI-UTILITIES	15	0.84	9.4	7.9	1.5	34.0	0.69	19.3	2.09	3.4	4	5	4	2	5	4.9	1	5
WATER UTILITIES	2	0.11	8.5	8.4	0.1	41.5	0.75	31.5	3.11	1.9	8	3	5	1	5	7.6	5	6
INDEP POWER PROD & ENERGY TRAD	9	0.14	10.2	10.0	0.2	24.8	0.92	17.5	2.86	2.9	3	7	7	6	6	45.1	nm	26
REAL ESTATE	95	3.12	9.6	9.8	-0.2	33.2	0.92	19.4	2.43	3.4	6	5	5	2	6	8.6	11	3
REITS	91	3.08	9.6	9.7	-0.1	33.4	0.92	19.1	2.44	3.5	6	5	5	2	6	8.0	12	4
REAL ESTATE MGMT & DEV	4	0.04	12.9	13.1	-0.2	17.6	1.37	nm	2.02	0.9		8				58.9	nm	nm
BofA UNIVERSE	1417	100.00	10.4	10.7	-0.3	35.2	1.04	19.9	3.73	1.7						11.7	12	4
S&P 500	503	91.13	10.4	10.6	-0.2	35.7	1.02	19.0	3.79	1.6						10.3	9	2

Source: BofA US Equity and Quant Strategy, FactSet

Exhibit ___ (JCN-4)

Capital Asset Pricing Model

CAPITAL ASSET PRICING MODEL -- CURRENT RISK-FREE RATE & VL BETA
K = Rf + β (Rm − Rf)

Company	Ticker	[1] Current 3-month average of 30-year U.S. Treasury bond yield	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
ALLETE, Inc.	ALE	3.75%	0.90	12.27%	8.53%	11.42%
American Electric Power Company, Inc.	AEP	3.75%	0.75	12.27%	8.53%	10.14%
Atmos Energy Corporation	ATO	3.75%	0.85	12.27%	8.53%	11.00%
Avista Corporation	AVA	3.75%	0.90	12.27%	8.53%	11.42%
Black Hills Corporation	BKH	3.75%	0.95	12.27%	8.53%	11.85%
CMS Energy Corporation	CMS	3.75%	0.80	12.27%	8.53%	10.57%
Edison International	EIX	3.75%	0.95	12.27%	8.53%	11.85%
Eversource Energy	ES	3.75%	0.90	12.27%	8.53%	11.42%
Exelon Corporation	EXC	3.75%	NMF	12.27%	8.53%	
Hawaiian Electric Industries, Inc.	HE	3.75%	0.85	12.27%	8.53%	11.00%
NiSource Inc.	NI	3.75%	0.90	12.27%	8.53%	11.42%
Northwest Natural Gas Company	NWN	3.75%	0.80	12.27%	8.53%	10.57%
NorthWestern Corporation	NWE	3.75%	0.90	12.27%	8.53%	11.42%
OGE Energy Corporation	OGE	3.75%	1.00	12.27%	8.53%	12.27%
ONE Gas, Inc.	OGS	3.75%	0.80	12.27%	8.53%	10.57%
PPL Corporation	PPL	3.75%	1.05	12.27%	8.53%	12.70%
Public Service Enterprise Group Inc.	PEG	3.75%	0.90	12.27%	8.53%	11.42%
Spire, Inc.	SR	3.75%	0.85	12.27%	8.53%	11.00%
Xcel Energy Inc.	XEL	3.75%	0.80	12.27%	8.53%	10.57%
Mean			0.88			11.26%
Median			0.90			11.42%

Notes:
[1] Source: Bloomberg Professional
[2] Source: Value Line
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

CAPITAL ASSET PRICING MODEL -- NEAR-TERM PROJECTED RISK-FREE RATE & VL BETA
K = Rf + β (Rm − Rf)

Company	Ticker	[1] Near-term projected 30-year U.S. Treasury bond yield (Q2 2023 - Q2 2024)	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
ALLETE, Inc.	ALE	3.88%	0.90	12.27%	8.39%	11.44%
American Electric Power Company, Inc.	AEP	3.88%	0.75	12.27%	8.39%	10.18%
Atmos Energy Corporation	ATO	3.88%	0.85	12.27%	8.39%	11.02%
Avista Corporation	AVA	3.88%	0.90	12.27%	8.39%	11.44%
Black Hills Corporation	BKH	3.88%	0.95	12.27%	8.39%	11.86%
CMS Energy Corporation	CMS	3.88%	0.80	12.27%	8.39%	10.60%
Edison International	EIX	3.88%	0.95	12.27%	8.39%	11.86%
Eversource Energy	ES	3.88%	0.90	12.27%	8.39%	11.44%
Exelon Corporation	EXC	3.88%	NMF	12.27%	8.39%	
Hawaiian Electric Industries, Inc.	HE	3.88%	0.85	12.27%	8.39%	11.02%
NiSource Inc.	NI	3.88%	0.90	12.27%	8.39%	11.44%
Northwest Natural Gas Company	NWN	3.88%	0.80	12.27%	8.39%	10.60%
NorthWestern Corporation	NWE	3.88%	0.90	12.27%	8.39%	11.44%
OGE Energy Corporation	OGE	3.88%	1.00	12.27%	8.39%	12.27%
ONE Gas, Inc.	OGS	3.88%	0.80	12.27%	8.39%	10.60%
PPL Corporation	PPL	3.88%	1.05	12.27%	8.39%	12.69%
Public Service Enterprise Group Inc.	PEG	3.88%	0.90	12.27%	8.39%	11.44%
Spire, Inc.	SR	3.88%	0.85	12.27%	8.39%	11.02%
Xcel Energy Inc.	XEL	3.88%	0.80	12.27%	8.39%	10.60%
Mean			0.88			11.27%
Median			0.90			11.44%

Notes:
[1] Source: Blue Chip Financial Forecasts, Vol. 42, No. 3, March 1, 2023 at 2
[2] Source: Value Line
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

CAPITAL ASSET PRICING MODEL -- LONG-TERM PROJECTED RISK-FREE RATE & VL BETA
K = Rf + β (Rm − Rf)

Company	Ticker	[1] Projected 30-year U.S. Treasury bond yield (2024 - 2028) [3]	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
ALLETE, Inc.	ALE	3.90%	0.90	12.27%	8.37%	11.44%
American Electric Power Company, Inc.	AEP	3.90%	0.75	12.27%	8.37%	10.18%
Atmos Energy Corporation	ATO	3.90%	0.85	12.27%	8.37%	11.02%
Avista Corporation	AVA	3.90%	0.90	12.27%	8.37%	11.44%
Black Hills Corporation	BKH	3.90%	0.95	12.27%	8.37%	11.86%
CMS Energy Corporation	CMS	3.90%	0.80	12.27%	8.37%	10.60%
Edison International	EIX	3.90%	0.95	12.27%	8.37%	11.86%
Eversource Energy	ES	3.90%	0.90	12.27%	8.37%	11.44%
Exelon Corporation	EXC	3.90%	NMF	12.27%	8.37%	
Hawaiian Electric Industries, Inc.	HE	3.90%	0.85	12.27%	8.37%	11.02%
NiSource Inc.	NI	3.90%	0.90	12.27%	8.37%	11.44%
Northwest Natural Gas Company	NWN	3.90%	0.80	12.27%	8.37%	10.60%
NorthWestern Corporation	NWE	3.90%	0.90	12.27%	8.37%	11.44%
OGE Energy Corporation	OGE	3.90%	1.00	12.27%	8.37%	12.27%
ONE Gas, Inc.	OGS	3.90%	0.80	12.27%	8.37%	10.60%
PPL Corporation	PPL	3.90%	1.05	12.27%	8.37%	12.69%
Public Service Enterprise Group Inc.	PEG	3.90%	0.90	12.27%	8.37%	11.44%
Spire, Inc.	SR	3.90%	0.85	12.27%	8.37%	11.02%
Xcel Energy Inc.	XEL	3.90%	0.80	12.27%	8.37%	10.60%
Mean			0.88			11.27%
Median			0.90			11.44%

Notes:
[1] Source: Blue Chip Financial Forecasts, Vol. 41, No. 12, December 1, 2022 at 14
[2] Source: Value Line
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

CAPITAL ASSET PRICING MODEL -- CURRENT RISK-FREE RATE & BLOOMBERG BETA
K = Rf + β (Rm − Rf)

Company	Ticker	[1] Current 3-month average of 30-year U.S. Treasury bond yield	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
ALLETE, Inc.	ALE	3.75%	0.83	12.27%	8.53%	10.83%
American Electric Power Company, Inc.	AEP	3.75%	0.76	12.27%	8.53%	10.26%
Atmos Energy Corporation	ATO	3.75%	0.76	12.27%	8.53%	10.22%
Avista Corporation	AVA	3.75%	0.75	12.27%	8.53%	10.16%
Black Hills Corporation	BKH	3.75%	0.91	12.27%	8.53%	11.49%
CMS Energy Corporation	CMS	3.75%	0.75	12.27%	8.53%	10.18%
Edison International	EIX	3.75%	0.84	12.27%	8.53%	10.94%
Eversource Energy	ES	3.75%	0.81	12.27%	8.53%	10.65%
Exelon Corporation	EXC	3.75%	0.86	12.27%	8.53%	11.09%
Hawaiian Electric Industries, Inc.	HE	3.75%	0.72	12.27%	8.53%	9.85%
NiSource Inc.	NI	3.75%	0.82	12.27%	8.53%	10.72%
Northwest Natural Gas Company	NWN	3.75%	0.70	12.27%	8.53%	9.75%
NorthWestern Corporation	NWE	3.75%	0.87	12.27%	8.53%	11.16%
OGE Energy Corporation	OGE	3.75%	0.92	12.27%	8.53%	11.60%
ONE Gas, Inc.	OGS	3.75%	0.79	12.27%	8.53%	10.48%
PPL Corporation	PPL	3.75%	0.95	12.27%	8.53%	11.81%
Public Service Enterprise Group Inc.	PEG	3.75%	0.85	12.27%	8.53%	11.02%
Spire, Inc.	SR	3.75%	0.77	12.27%	8.53%	10.34%
Xcel Energy Inc.	XEL	3.75%	0.75	12.27%	8.53%	10.13%
Mean			0.81			10.67%
Median			0.81			10.65%

Notes:
[1] Source: Bloomberg Professional
[2] Source: Bloomberg Professional
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

CAPITAL ASSET PRICING MODEL -- NEAR-TERM PROJECTED RISK-FREE RATE & BLOOMBERG BETA
K = Rf + β (Rm − Rf)

Company	Ticker	[1] Near-term projected 30-year U.S. Treasury bond yield (Q2 2023 - Q2 2024)	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
ALLETE, Inc.	ALE	3.88%	0.83	12.27%	8.39%	10.86%
American Electric Power Company, Inc.	AEP	3.88%	0.76	12.27%	8.39%	10.29%
Atmos Energy Corporation	ATO	3.88%	0.76	12.27%	8.39%	10.25%
Avista Corporation	AVA	3.88%	0.75	12.27%	8.39%	10.20%
Black Hills Corporation	BKH	3.88%	0.91	12.27%	8.39%	11.50%
CMS Energy Corporation	CMS	3.88%	0.75	12.27%	8.39%	10.21%
Edison International	EIX	3.88%	0.84	12.27%	8.39%	10.96%
Eversource Energy	ES	3.88%	0.81	12.27%	8.39%	10.67%
Exelon Corporation	EXC	3.88%	0.86	12.27%	8.39%	11.11%
Hawaiian Electric Industries, Inc.	HE	3.88%	0.72	12.27%	8.39%	9.88%
NiSource Inc.	NI	3.88%	0.82	12.27%	8.39%	10.75%
Northwest Natural Gas Company	NWN	3.88%	0.70	12.27%	8.39%	9.79%
NorthWestern Corporation	NWE	3.88%	0.87	12.27%	8.39%	11.18%
OGE Energy Corporation	OGE	3.88%	0.92	12.27%	8.39%	11.61%
ONE Gas, Inc.	OGS	3.88%	0.79	12.27%	8.39%	10.51%
PPL Corporation	PPL	3.88%	0.95	12.27%	8.39%	11.81%
Public Service Enterprise Group Inc.	PEG	3.88%	0.85	12.27%	8.39%	11.04%
Spire, Inc.	SR	3.88%	0.77	12.27%	8.39%	10.37%
Xcel Energy Inc.	XEL	3.88%	0.75	12.27%	8.39%	10.17%
Mean			0.81			10.69%
Median			0.81			10.67%

Notes:
[1] Source: Blue Chip Financial Forecasts, Vol. 42, No. 3, March 1, 2023 at 2
[2] Source: Bloomberg Professional
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

CAPITAL ASSET PRICING MODEL -- LONG-TERM PROJECTED RISK-FREE RATE & BLOOMBERG BETA
K = Rf + β (Rm − Rf)

Company	Ticker	[1] Projected 30-year U.S. Treasury bond yield (2024 - 2028) [3]	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
ALLETE, Inc.	ALE	3.90%	0.83	12.27%	8.37%	10.86%
American Electric Power Company, Inc.	AEP	3.90%	0.76	12.27%	8.37%	10.30%
Atmos Energy Corporation	ATO	3.90%	0.76	12.27%	8.37%	10.26%
Avista Corporation	AVA	3.90%	0.75	12.27%	8.37%	10.20%
Black Hills Corporation	BKH	3.90%	0.91	12.27%	8.37%	11.50%
CMS Energy Corporation	CMS	3.90%	0.75	12.27%	8.37%	10.21%
Edison International	EIX	3.90%	0.84	12.27%	8.37%	10.97%
Eversource Energy	ES	3.90%	0.81	12.27%	8.37%	10.68%
Exelon Corporation	EXC	3.90%	0.86	12.27%	8.37%	11.11%
Hawaiian Electric Industries, Inc.	HE	3.90%	0.72	12.27%	8.37%	9.89%
NiSource Inc.	NI	3.90%	0.82	12.27%	8.37%	10.75%
Northwest Natural Gas Company	NWN	3.90%	0.70	12.27%	8.37%	9.80%
NorthWestern Corporation	NWE	3.90%	0.87	12.27%	8.37%	11.18%
OGE Energy Corporation	OGE	3.90%	0.92	12.27%	8.37%	11.61%
ONE Gas, Inc.	OGS	3.90%	0.79	12.27%	8.37%	10.52%
PPL Corporation	PPL	3.90%	0.95	12.27%	8.37%	11.82%
Public Service Enterprise Group Inc.	PEG	3.90%	0.85	12.27%	8.37%	11.04%
Spire, Inc.	SR	3.90%	0.77	12.27%	8.37%	10.38%
Xcel Energy Inc.	XEL	3.90%	0.75	12.27%	8.37%	10.17%
Mean			0.81			10.70%
Median			0.81			10.68%

Notes:
[1] Source: Blue Chip Financial Forecasts, Vol. 41, No. 12, December 1, 2022 at 14
[2] Source: Bloomberg Professional
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

ZERO BETA CAPITAL ASSET PRICING MODEL -- CURRENT RISK-FREE RATE & VL BETA
K = Rf + 0.25 (Rm − Rf) + 0.75 (β (Rm − Rf))

Company	Ticker	[1] Current 3-month average of 30-year U.S. Treasury bond yield	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
Atmos Energy Corporation	ATO	3.75%	0.85	12.27%	8.53%	11.32%
ALLETE, Inc.	ALE	3.75%	0.90	12.27%	8.53%	11.64%
American Electric Power Company, Inc.	AEP	3.75%	0.75	12.27%	8.53%	10.68%
Avista Corporation	AVA	3.75%	0.90	12.27%	8.53%	11.64%
Black Hills Corporation	BKH	3.75%	0.95	12.27%	8.53%	11.96%
CMS Energy Corporation	CMS	3.75%	0.80	12.27%	8.53%	11.00%
Edison International	EIX	3.75%	0.95	12.27%	8.53%	11.96%
Eversource Energy	ES	3.75%	0.90	12.27%	8.53%	11.64%
Exelon Corporation	EXC	3.75%	NMF	12.27%	8.53%	
Hawaiian Electric Industries, Inc.	HE	3.75%	0.85	12.27%	8.53%	11.32%
NiSource Inc.	NI	3.75%	0.90	12.27%	8.53%	11.64%
Northwest Natural Gas Company	NWN	3.75%	0.80	12.27%	8.53%	11.00%
NorthWestern Corporation	NWE	3.75%	0.90	12.27%	8.53%	11.64%
OGE Energy Corporation	OGE	3.75%	1.00	12.27%	8.53%	12.27%
ONE Gas, Inc.	OGS	3.75%	0.80	12.27%	8.53%	11.00%
PPL Corporation	PPL	3.75%	1.05	12.27%	8.53%	12.59%
Public Service Enterprise Group Inc.	PEG	3.75%	0.90	12.27%	8.53%	11.64%
Spire, Inc.	SR	3.75%	0.85	12.27%	8.53%	11.32%
Xcel Energy Inc.	XEL	3.75%	0.80	12.27%	8.53%	11.00%
Mean			0.88			11.51%
Median			0.90			11.64%

Notes:
[1] Source: Bloomberg Professional
[2] Source: Value Line
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

ZERO BETA CAPITAL ASSET PRICING MODEL -- NEAR-TERM PROJECTED RISK-FREE RATE & VL BETA
K = Rf + 0.25 (Rm − Rf) + 0.75 (β (Rm − Rf))

Company	Ticker	[1] Near-term projected 30-year U.S. Treasury bond yield (Q3 2020 - Q3 2021)	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
Atmos Energy Corporation	ATO	3.88%	0.85	12.27%	8.39%	11.33%
ALLETE, Inc.	ALE	3.88%	0.90	12.27%	8.39%	11.65%
American Electric Power Company, Inc.	AEP	3.88%	0.75	12.27%	8.39%	10.70%
Avista Corporation	AVA	3.88%	0.90	12.27%	8.39%	11.65%
Black Hills Corporation	BKH	3.88%	0.95	12.27%	8.39%	11.96%
CMS Energy Corporation	CMS	3.88%	0.80	12.27%	8.39%	11.02%
Edison International	EIX	3.88%	0.95	12.27%	8.39%	11.96%
Eversource Energy	ES	3.88%	0.90	12.27%	8.39%	11.65%
Exelon Corporation	EXC	3.88%	NMF	12.27%	8.39%	
Hawaiian Electric Industries, Inc.	HE	3.88%	0.85	12.27%	8.39%	11.33%
NiSource Inc.	NI	3.88%	0.90	12.27%	8.39%	11.65%
Northwest Natural Gas Company	NWN	3.88%	0.80	12.27%	8.39%	11.02%
NorthWestern Corporation	NWE	3.88%	0.90	12.27%	8.39%	11.65%
OGE Energy Corporation	OGE	3.88%	1.00	12.27%	8.39%	12.27%
ONE Gas, Inc.	OGS	3.88%	0.80	12.27%	8.39%	11.02%
PPL Corporation	PPL	3.88%	1.05	12.27%	8.39%	12.59%
Public Service Enterprise Group Inc.	PEG	3.88%	0.90	12.27%	8.39%	11.65%
Spire, Inc.	SR	3.88%	0.85	12.27%	8.39%	11.33%
Xcel Energy Inc.	XEL	3.88%	0.80	12.27%	8.39%	11.02%
Mean			0.88			11.52%
Median			0.90			11.65%

Notes:
[1] Source: Blue Chip Financial Forecasts, Vol. 42, No. 3, March 1, 2023 at 2
[2] Source: Value Line
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

ZERO BETA CAPITAL ASSET PRICING MODEL -- LONG-TERM PROJECTED RISK-FREE RATE & VL BETA
K = Rf + 0.25 (Rm − Rf) + 0.75 (β (Rm − Rf))

Company	Ticker	[1] Projected 30-year U.S. Treasury bond yield (2022 - 2026)	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
Atmos Energy Corporation	ATO	3.90%	0.85	12.27%	8.37%	11.33%
ALLETE, Inc.	ALE	3.90%	0.90	12.27%	8.37%	11.65%
American Electric Power Company, Inc.	AEP	3.90%	0.75	12.27%	8.37%	10.70%
Avista Corporation	AVA	3.90%	0.90	12.27%	8.37%	11.65%
Black Hills Corporation	BKH	3.90%	0.95	12.27%	8.37%	11.96%
CMS Energy Corporation	CMS	3.90%	0.80	12.27%	8.37%	11.02%
Edison International	EIX	3.90%	0.95	12.27%	8.37%	11.96%
Eversource Energy	ES	3.90%	0.90	12.27%	8.37%	11.65%
Exelon Corporation	EXC	3.90%	NMF	12.27%	8.37%	
Hawaiian Electric Industries, Inc.	HE	3.90%	0.85	12.27%	8.37%	11.33%
NiSource Inc.	NI	3.90%	0.90	12.27%	8.37%	11.65%
Northwest Natural Gas Company	NWN	3.90%	0.80	12.27%	8.37%	11.02%
NorthWestern Corporation	NWE	3.90%	0.90	12.27%	8.37%	11.65%
OGE Energy Corporation	OGE	3.90%	1.00	12.27%	8.37%	12.27%
ONE Gas, Inc.	OGS	3.90%	0.80	12.27%	8.37%	11.02%
PPL Corporation	PPL	3.90%	1.05	12.27%	8.37%	12.59%
Public Service Enterprise Group Inc.	PEG	3.90%	0.90	12.27%	8.37%	11.65%
Spire, Inc.	SR	3.90%	0.85	12.27%	8.37%	11.33%
Xcel Energy Inc.	XEL	3.90%	0.80	12.27%	8.37%	11.02%
Mean			0.88			11.52%
Median			0.90			11.65%

Notes:
[1] Source: Blue Chip Financial Forecasts, Vol. 41, No. 12, December 1, 2022 at 14
[2] Source: Value Line
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

ZERO BETA CAPITAL ASSET PRICING MODEL -- CURRENT RISK-FREE RATE & BLOOMBERG BETA
K = Rf + 0.25 (Rm − Rf) + 0.75 (β (Rm − Rf))

Company	Ticker	[1] Current 3-month average of 30-year U.S. Treasury bond yield	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
Atmos Energy Corporation	ATO	3.75%	0.76	12.27%	8.53%	10.74%
ALLETE, Inc.	ALE	3.75%	0.83	12.27%	8.53%	11.19%
American Electric Power Company, Inc.	AEP	3.75%	0.76	12.27%	8.53%	10.76%
Avista Corporation	AVA	3.75%	0.75	12.27%	8.53%	10.69%
Black Hills Corporation	BKH	3.75%	0.91	12.27%	8.53%	11.68%
CMS Energy Corporation	CMS	3.75%	0.75	12.27%	8.53%	10.70%
Edison International	EIX	3.75%	0.84	12.27%	8.53%	11.28%
Eversource Energy	ES	3.75%	0.81	12.27%	8.53%	11.06%
Exelon Corporation	EXC	3.75%	0.86	12.27%	8.53%	11.38%
Hawaiian Electric Industries, Inc.	HE	3.75%	0.72	12.27%	8.53%	10.45%
NiSource Inc.	NI	3.75%	0.82	12.27%	8.53%	11.11%
Northwest Natural Gas Company	NWN	3.75%	0.70	12.27%	8.53%	10.38%
NorthWestern Corporation	NWE	3.75%	0.87	12.27%	8.53%	11.44%
OGE Energy Corporation	OGE	3.75%	0.92	12.27%	8.53%	11.77%
ONE Gas, Inc.	OGS	3.75%	0.79	12.27%	8.53%	10.93%
PPL Corporation	PPL	3.75%	0.95	12.27%	8.53%	11.92%
Public Service Enterprise Group Inc.	PEG	3.75%	0.85	12.27%	8.53%	11.33%
Spire, Inc.	SR	3.75%	0.77	12.27%	8.53%	10.83%
Xcel Energy Inc.	XEL	3.75%	0.75	12.27%	8.53%	10.67%
Mean			0.81			11.07%
Median			0.81			11.06%

Notes:
[1] Source: Bloomberg Professional
[2] Source: Bloomberg Professional
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

ZERO BETA CAPITAL ASSET PRICING MODEL -- NEAR-TERM PROJECTED RISK-FREE RATE & BLOOMBERG BETA
K = Rf + 0.25 (Rm − Rf) + 0.75 (β (Rm − Rf))

Company	Ticker	[1] Near-term projected 30-year U.S. Treasury bond yield (Q3 2020 - Q3 2021)	[2] Beta (β)	[3] Market Return (Rm)	[4] Market Risk Premium (Rm − Rf)	[5] ROE (K)
Atmos Energy Corporation	ATO	3.88%	0.76	12.27%	8.39%	10.76%
ALLETE, Inc.	ALE	3.88%	0.83	12.27%	8.39%	11.21%
American Electric Power Company, Inc.	AEP	3.88%	0.76	12.27%	8.39%	10.79%
Avista Corporation	AVA	3.88%	0.75	12.27%	8.39%	10.72%
Black Hills Corporation	BKH	3.88%	0.91	12.27%	8.39%	11.69%
CMS Energy Corporation	CMS	3.88%	0.75	12.27%	8.39%	10.72%
Edison International	EIX	3.88%	0.84	12.27%	8.39%	11.29%
Eversource Energy	ES	3.88%	0.81	12.27%	8.39%	11.07%
Exelon Corporation	EXC	3.88%	0.86	12.27%	8.39%	11.40%
Hawaiian Electric Industries, Inc.	HE	3.88%	0.72	12.27%	8.39%	10.48%
NiSource Inc.	NI	3.88%	0.82	12.27%	8.39%	11.13%
Northwest Natural Gas Company	NWN	3.88%	0.70	12.27%	8.39%	10.41%
NorthWestern Corporation	NWE	3.88%	0.87	12.27%	8.39%	11.46%
OGE Energy Corporation	OGE	3.88%	0.92	12.27%	8.39%	11.77%
ONE Gas, Inc.	OGS	3.88%	0.79	12.27%	8.39%	10.95%
PPL Corporation	PPL	3.88%	0.95	12.27%	8.39%	11.93%
Public Service Enterprise Group Inc.	PEG	3.88%	0.85	12.27%	8.39%	11.35%
Spire, Inc.	SR	3.88%	0.77	12.27%	8.39%	10.85%
Xcel Energy Inc.	XEL	3.88%	0.75	12.27%	8.39%	10.70%
Mean			0.81			11.09%
Median			0.81			11.07%

Notes:
[1] Source: Blue Chip Financial Forecasts, Vol. 42, No. 3, March 1, 2023 at 2
[2] Source: Bloomberg Professional
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

ZERO BETA CAPITAL ASSET PRICING MODEL -- LONG-TERM PROJECTED RISK-FREE RATE & BLOOMBERG BETA
K = Rf + 0.25 (Rm − Rf) + 0.75 (β (Rm − Rf))

		[1]	[2]	[3]	[4]	[5]
Company	Ticker	Projected 30-year U.S. Treasury bond yield (2022 - 2026)	Beta (β)	Market Return (Rm)	Market Risk Premium (Rm − Rf)	ROE (K)
Atmos Energy Corporation	ATO	3.90%	0.76	12.27%	8.37%	10.76%
ALLETE, Inc.	ALE	3.90%	0.83	12.27%	8.37%	11.21%
American Electric Power Company, Inc.	AEP	3.90%	0.76	12.27%	8.37%	10.79%
Avista Corporation	AVA	3.90%	0.75	12.27%	8.37%	10.72%
Black Hills Corporation	BKH	3.90%	0.91	12.27%	8.37%	11.69%
CMS Energy Corporation	CMS	3.90%	0.75	12.27%	8.37%	10.73%
Edison International	EIX	3.90%	0.84	12.27%	8.37%	11.29%
Eversource Energy	ES	3.90%	0.81	12.27%	8.37%	11.08%
Exelon Corporation	EXC	3.90%	0.86	12.27%	8.37%	11.40%
Hawaiian Electric Industries, Inc.	HE	3.90%	0.72	12.27%	8.37%	10.49%
NiSource Inc.	NI	3.90%	0.82	12.27%	8.37%	11.13%
Northwest Natural Gas Company	NWN	3.90%	0.70	12.27%	8.37%	10.42%
NorthWestern Corporation	NWE	3.90%	0.87	12.27%	8.37%	11.46%
OGE Energy Corporation	OGE	3.90%	0.92	12.27%	8.37%	11.77%
ONE Gas, Inc.	OGS	3.90%	0.79	12.27%	8.37%	10.96%
PPL Corporation	PPL	3.90%	0.95	12.27%	8.37%	11.93%
Public Service Enterprise Group Inc.	PEG	3.90%	0.85	12.27%	8.37%	11.35%
Spire, Inc.	SR	3.90%	0.77	12.27%	8.37%	10.85%
Xcel Energy Inc.	XEL	3.90%	0.75	12.27%	8.37%	10.70%
Mean			0.81			11.09%
Median			0.81			11.08%

Notes:
[1] Source: Blue Chip Financial Forecasts, Vol. 41, No. 12, December 1, 2022 at 14
[2] Source: Bloomberg Professional
[3] Source: Exhibit __ (JCN-5)
[4] Equals [3] - [1]
[5] Equals [1] + [2] x [4]

Exhibit ___ (JCN-5)

Market Risk Premium Derived from Analysts' L/T Growth Estimates

MARKET RISK PREMIUM - DERIVED FROM ANALYSTS' LONG-TERM GROWTH ESTIMATES

[1] Estimated Weighted Average Dividend Yield	1.66%
[2] Estimated Weighted Average Long-Term Growth Rate	12.32%
[3] S&P 500 Estimated Required Market Return	14.08%

STANDARD AND POOR'S 500 INDEX

Name	Ticker	[4] Weight in Index	[5] Current Dividend Yield	[6] Cap-Weighted Dividend Yield	[7] Bloomberg Long-Term Growth Est.	[8] Value Line Long-Term Growth Est.	[9] Cap-Weighted Long-Term Growth Est.
LyondellBasell Industries NV	LYB	0.09%	5.07%	0.00%	13.50%	3.00%	0.01%
American Express Co	AXP	0.35%	1.45%	0.01%	10.16%	10.00%	0.04%
Verizon Communications Inc	VZ	0.46%	6.71%	0.03%	1.80%	2.50%	0.01%
Broadcom Inc	AVGO	0.76%	2.87%	0.02%	13.21%	30.00%	0.16%
Boeing Co/The	BA	0.36%	n/a	n/a	119.56%		0.43%
Caterpillar Inc	CAT	0.34%	2.10%	0.01%	11.43%	10.50%	0.04%
JPMorgan Chase & Co	JPM	1.09%	3.07%	0.03%	0.33%	5.00%	0.03%
Chevron Corp	CVX	0.88%	3.70%	0.03%	6.18%	45.00%	0.23%
Coca-Cola Co/The	KO	0.76%	2.97%	0.02%	6.66%	8.00%	0.06%
AbbVie Inc	ABBV	0.80%	3.71%	0.03%	3.25%	2.00%	0.02%
Walt Disney Co/The	DIS	0.52%	n/a	n/a	22.81%	86.00%	0.28%
FleetCor Technologies Inc	FLT	0.04%	n/a	n/a	13.03%	10.50%	0.01%
Extra Space Storage Inc	EXR	0.06%	3.98%	0.00%	3.30%	6.50%	0.00%
Exxon Mobil Corp	XOM	1.27%	3.32%	0.04%	7.58%		0.10%
Phillips 66	PSX	0.13%	4.14%	0.01%	13.25%		0.02%
General Electric Co	GE	0.30%	0.33%	0.00%	20.03%	21.00%	0.06%
HP Inc	HPQ	0.08%	3.58%	0.00%	1.00%	12.50%	0.01%
Home Depot Inc/The	HD	0.85%	2.83%	0.02%	10.64%	9.00%	0.08%
Monolithic Power Systems Inc	MPWR	0.07%	0.80%	0.00%	27.93%	21.00%	0.02%
International Business Machines Corp	IBM	0.34%	5.03%	0.02%	4.84%	3.00%	0.01%
Johnson & Johnson	JNJ	1.14%	2.92%	0.03%	4.49%	8.00%	0.07%
McDonald's Corp	MCD	0.58%	2.17%	0.01%	7.55%	9.00%	0.05%
Merck & Co Inc	MRK	0.77%	2.74%	0.02%	11.01%	8.50%	0.07%
3M Co	MMM	0.16%	5.71%	0.01%	3.60%	7.50%	0.01%
American Water Works Co Inc	AWK	0.08%	1.79%	0.00%	7.40%	3.00%	0.00%
Bank of America Corp	BAC	0.65%	3.08%	0.02%	1.95%	8.50%	0.03%
Pfizer Inc	PFE	0.65%	4.02%	0.03%	-7.20%	2.00%	-0.02%
Procter & Gamble Co/The	PG	0.99%	2.46%	0.02%	5.29%	5.50%	0.05%
AT&T Inc	T	0.39%	5.77%	0.02%	3.36%	1.00%	0.01%
Travelers Cos Inc/The	TRV	0.11%	2.17%	0.00%	15.71%	7.50%	0.01%
Raytheon Technologies Corp	RTX	0.41%	2.25%	0.01%	7.52%	14.00%	0.04%
Analog Devices Inc	ADI	0.28%	1.74%	0.00%	9.53%	11.50%	0.03%
Walmart Inc	WMT	1.13%	1.55%	0.02%	6.60%	7.50%	0.08%
Cisco Systems Inc	CSCO	0.61%	2.98%	0.02%	6.80%	8.50%	0.05%
Intel Corp	INTC	0.38%	1.53%	0.01%	2.65%		0.01%
General Motors Co	GM	0.15%	0.98%	0.00%	0.77%	8.50%	0.01%
Microsoft Corp	MSFT	6.09%	0.94%	0.06%	14.55%	15.00%	0.90%
Dollar General Corp	DG	0.13%	1.12%	0.00%	8.24%	10.00%	0.01%
Cigna Group/The	CI	0.22%	1.93%	0.00%	11.32%	10.00%	0.02%
Kinder Morgan Inc	KMI	0.11%	6.34%	0.01%	3.70%	18.50%	0.01%
Citigroup Inc	C	0.26%	4.35%	0.01%	-4.76%	3.50%	0.00%
American International Group Inc	AIG	0.10%	2.54%	0.00%	2.05%	6.50%	0.00%
Altria Group Inc	MO	0.23%	8.43%	0.02%	1.35%	6.00%	0.01%
HCA Healthcare Inc	HCA	0.21%	0.91%	0.00%	7.81%	12.50%	0.02%
International Paper Co	IP	0.04%	5.13%	0.00%	4.75%	9.50%	0.00%
Hewlett Packard Enterprise Co	HPE	0.06%	3.01%	0.00%	5.55%	7.50%	0.00%
Abbott Laboratories	ABT	0.50%	2.01%	0.01%	1.73%	6.50%	0.02%
Aflac Inc	AFL	0.11%	2.60%	0.00%	n/a	8.00%	0.01%
Air Products and Chemicals Inc	APD	0.18%	2.44%	0.00%	10.84%	11.50%	0.02%
Royal Caribbean Cruises Ltd	RCL	n/a	n/a	n/a	n/a		n/a
Hess Corp	HES	0.11%	1.32%	0.00%	5.50%		0.01%
Archer-Daniels-Midland Co	ADM	0.12%	2.26%	0.00%	-6.02%	13.00%	0.00%
Automatic Data Processing Inc	ADP	0.26%	2.25%	0.01%	13.35%	10.00%	0.03%
Verisk Analytics Inc	VRSK	0.08%	0.71%	0.00%	12.73%	13.00%	0.01%
AutoZone Inc	AZO	0.13%	n/a	n/a	12.62%	14.50%	0.02%
Avery Dennison Corp	AVY	0.04%	1.68%	0.00%	7.25%	9.50%	0.00%
Enphase Energy Inc	ENPH	0.08%	n/a	n/a	45.85%	24.50%	0.03%
MSCI Inc	MSCI	0.13%	0.99%	0.00%	13.40%	12.50%	0.02%
Ball Corp	BALL	0.05%	1.45%	0.00%	11.27%	21.50%	0.01%
Ceridian HCM Holding Inc	CDAY	0.03%	n/a	n/a	79.90%		0.03%
Carrier Global Corp	CARR	0.11%	1.62%	0.00%	7.56%		0.01%
Bank of New York Mellon Corp/The	BK	0.10%	3.26%	0.00%	10.00%	6.00%	0.01%
Otis Worldwide Corp	OTIS	0.10%	1.37%	0.00%	8.95%		0.01%
Baxter International Inc	BAX	0.06%	2.86%	0.00%	0.48%	7.00%	0.00%
Becton Dickinson & Co	BDX	0.20%	1.47%	0.00%	7.07%	4.50%	0.01%
Berkshire Hathaway Inc	BRK/B	n/a	n/a	n/a	n/a		n/a
Best Buy Co Inc	BBY	0.05%	4.70%	0.00%	3.80%	4.00%	0.00%

MARKET RISK PREMIUM - DERIVED FROM ANALYSTS' LONG-TERM GROWTH ESTIMATES

[1] Estimated Weighted Average Dividend Yield	1.66%
[2] Estimated Weighted Average Long-Term Growth Rate	12.32%
[3] S&P 500 Estimated Required Market Return	14.08%

STANDARD AND POOR'S 500 INDEX

Name	Ticker	[4] Weight in Index	[5] Current Dividend Yield	[6] Cap-Weighted Dividend Yield	[7] Bloomberg Long-Term Growth Est.	[8] Value Line Long-Term Growth Est.	[9] Cap-Weighted Long-Term Growth Est.
Boston Scientific Corp	BSX	0.20%	n/a	n/a	11.17%	15.50%	0.03%
Bristol-Myers Squibb Co	BMY	0.41%	3.29%	0.01%	3.71%		0.02%
Brown-Forman Corp	BF/B	0.06%	1.28%	0.00%	8.76%		0.00%
Coterra Energy Inc	CTRA	0.05%	9.29%	0.00%	26.15%		0.01%
Campbell Soup Co	CPB	0.05%	2.69%	0.00%	5.02%	5.00%	0.00%
Hilton Worldwide Holdings Inc	HLT	0.11%	0.43%	0.00%	42.95%		0.05%
Carnival Corp	CCL	n/a	n/a	n/a	n/a		n/a
Qorvo Inc	QRVO	0.03%	n/a	n/a	-0.77%	14.50%	0.00%
UDR Inc	UDR	0.04%	4.09%	0.00%	8.99%	17.00%	0.00%
Clorox Co/The	CLX	0.06%	2.98%	0.00%	3.76%	7.00%	0.00%
Paycom Software Inc	PAYC	0.05%	n/a	n/a	22.75%	21.00%	0.01%
CMS Energy Corp	CMS	0.05%	3.18%	0.00%	6.97%	6.50%	0.00%
Newell Brands Inc	NWL	n/a	7.40%	n/a	n/a		n/a
Colgate-Palmolive Co	CL	0.18%	2.55%	0.00%	6.42%	6.00%	0.01%
EPAM Systems Inc	EPAM	0.05%	n/a	n/a	16.17%	20.50%	0.01%
Comerica Inc	CMA	0.02%	6.54%	0.00%	1.94%	8.50%	0.00%
Conagra Brands Inc	CAG	0.05%	3.51%	0.00%	7.28%	3.50%	0.00%
Consolidated Edison Inc	ED	0.10%	3.39%	0.00%	4.83%	4.50%	0.00%
Corning Inc	GLW	0.08%	3.17%	0.00%	10.06%	17.50%	0.01%
Cummins Inc	CMI	0.10%	2.63%	0.00%	8.60%	8.50%	0.01%
Caesars Entertainment Inc	CZR	n/a	n/a	n/a	n/a		n/a
Danaher Corp	DHR	0.52%	0.43%	0.00%	7.46%	16.00%	0.06%
Target Corp	TGT	0.22%	2.61%	0.01%	7.50%	12.00%	0.02%
Deere & Co	DE	0.35%	1.21%	0.00%	15.97%	12.50%	0.05%
Dominion Energy Inc	D	0.13%	4.78%	0.01%	4.70%	4.00%	0.01%
Dover Corp	DOV	0.06%	1.33%	0.00%	11.75%	9.00%	0.01%
Alliant Energy Corp	LNT	0.04%	3.39%	0.00%	6.19%	6.00%	0.00%
Steel Dynamics Inc	STLD	0.06%	1.50%	0.00%	-14.81%	2.00%	0.00%
Duke Energy Corp	DUK	0.21%	4.17%	0.01%	4.20%	5.00%	0.01%
Regency Centers Corp	REG	0.03%	4.25%	0.00%	3.91%	10.50%	0.00%
Eaton Corp PLC	ETN	0.19%	2.01%	0.00%	12.50%	12.00%	0.02%
Ecolab Inc	ECL	0.13%	1.28%	0.00%	15.33%	6.00%	0.01%
PerkinElmer Inc	PKI	0.05%	0.21%	0.00%	-0.60%	4.00%	0.00%
Emerson Electric Co	EMR	0.14%	2.39%	0.00%	10.00%	6.50%	0.01%
EOG Resources Inc	EOG	0.19%	2.88%	0.01%	4.42%	26.00%	0.03%
Aon PLC	AON	0.18%	0.71%	0.00%	10.93%	7.50%	0.02%
Entergy Corp	ETR	0.06%	3.97%	0.00%	6.33%	0.50%	0.00%
Equifax Inc	EFX	0.07%	0.77%	0.00%	13.82%	7.00%	0.01%
EQT Corp	EQT	0.03%	1.88%	0.00%	30.17%		0.01%
IQVIA Holdings Inc	IQV	0.10%	n/a	n/a	14.36%	14.50%	0.02%
Gartner Inc	IT	0.07%	n/a	n/a	6.25%	17.50%	0.01%
FedEx Corp	FDX	0.16%	2.01%	0.00%	6.00%	9.00%	0.01%
FMC Corp	FMC	0.04%	1.90%	0.00%	7.40%	10.50%	0.00%
Brown & Brown Inc	BRO	0.05%	0.80%	0.00%	7.00%	8.00%	0.00%
Ford Motor Co	F	0.14%	4.76%	0.01%	1.66%	27.50%	0.02%
NextEra Energy Inc	NEE	0.43%	2.43%	0.01%	10.80%	10.00%	0.05%
Franklin Resources Inc	BEN	0.04%	4.45%	0.00%	-6.26%	3.50%	0.00%
Garmin Ltd	GRMN	0.05%	2.89%	0.00%	7.75%	5.00%	0.00%
Freeport-McMoRan Inc	FCX	0.17%	1.47%	0.00%	-13.38%	18.50%	0.00%
Dexcom Inc	DXCM	0.13%	n/a	n/a	24.56%		0.03%
General Dynamics Corp	GD	0.18%	2.31%	0.00%	10.20%	9.50%	0.02%
General Mills Inc	GIS	0.14%	2.53%	0.00%	8.00%	4.00%	0.01%
Genuine Parts Co	GPC	0.07%	2.27%	0.00%	10.77%	10.50%	0.01%
Atmos Energy Corp	ATO	0.05%	2.63%	0.00%	8.00%	7.00%	0.00%
WW Grainger Inc	GWW	0.10%	1.00%	0.00%	9.45%	9.00%	0.01%
Halliburton Co	HAL	0.08%	2.02%	0.00%	27.20%	32.50%	0.02%
L3Harris Technologies Inc	LHX	0.11%	2.32%	0.00%	1.79%	17.00%	0.01%
Healthpeak Properties Inc	PEAK	0.03%	5.46%	0.00%	-7.28%	14.50%	0.00%
Insulet Corp	PODD	0.06%	n/a	n/a	80.41%		0.05%
Catalent Inc	CTLT	0.03%	n/a	n/a	9.86%	21.00%	0.01%
Fortive Corp	FTV	0.07%	0.41%	0.00%	8.92%	12.00%	0.01%
Hershey Co/The	HSY	0.11%	1.63%	0.00%	8.30%	9.00%	0.01%
Synchrony Financial	SYF	0.04%	3.16%	0.00%	27.80%	9.50%	0.01%
Hormel Foods Corp	HRL	0.06%	2.76%	0.00%	5.63%	7.50%	0.00%
Arthur J Gallagher & Co	AJG	0.12%	1.15%	0.00%	12.58%	18.50%	0.02%
Mondelez International Inc	MDLZ	0.27%	2.21%	0.01%	12.26%	7.50%	0.03%
CenterPoint Energy Inc	CNP	0.05%	2.58%	0.00%	6.05%	6.50%	0.00%

MARKET RISK PREMIUM - DERIVED FROM ANALYSTS' LONG-TERM GROWTH ESTIMATES

[1] Estimated Weighted Average Dividend Yield	1.66%
[2] Estimated Weighted Average Long-Term Growth Rate	12.32%
[3] S&P 500 Estimated Required Market Return	14.08%

STANDARD AND POOR'S 500 INDEX

Name	Ticker	[4] Weight in Index	[5] Current Dividend Yield	[6] Cap-Weighted Dividend Yield	[7] Bloomberg Long-Term Growth Est.	[8] Value Line Long-Term Growth Est.	[9] Cap-Weighted Long-Term Growth Est.
Humana Inc	HUM	0.17%	0.73%	0.00%	12.00%	12.50%	0.02%
Willis Towers Watson PLC	WTW	0.07%	1.45%	0.00%	13.94%	8.50%	0.01%
Illinois Tool Works Inc	ITW	0.21%	2.15%	0.00%	6.58%	11.00%	0.02%
CDW Corp/DE	CDW	0.07%	1.21%	0.00%	13.10%	8.50%	0.01%
Trane Technologies PLC	TT	0.12%	1.63%	0.00%	10.62%		0.01%
Interpublic Group of Cos Inc/The	IPG	0.04%	3.33%	0.00%	5.71%	10.00%	0.00%
International Flavors & Fragrances Inc	IFF	0.07%	3.52%	0.00%	4.21%	6.00%	0.00%
Generac Holdings Inc	GNRC	0.02%	n/a	n/a	10.00%	19.00%	0.00%
NXP Semiconductors NV	NXPI	0.14%	2.18%	0.00%	13.50%	11.00%	0.02%
Kellogg Co	K	0.07%	3.52%	0.00%	5.90%	3.50%	0.00%
Broadridge Financial Solutions Inc	BR	0.05%	1.98%	0.00%	n/a	8.50%	0.00%
Kimberly-Clark Corp	KMB	0.13%	3.52%	0.00%	8.74%	7.00%	0.01%
Kimco Realty Corp	KIM	0.03%	4.71%	0.00%	5.58%	11.00%	0.00%
Oracle Corp	ORCL	0.71%	1.72%	0.01%	12.05%	10.00%	0.08%
Kroger Co/The	KR	0.10%	2.11%	0.00%	10.22%	6.50%	0.01%
Lennar Corp	LEN	0.08%	1.43%	0.00%	-1.75%	8.50%	0.00%
Eli Lilly & Co	LLY	0.93%	1.32%	0.01%	22.91%	11.50%	0.16%
Bath & Body Works Inc	BBWI	0.02%	2.19%	0.00%	12.46%	26.50%	0.00%
Charter Communications Inc	CHTR	0.15%	n/a	n/a	27.86%	15.50%	0.03%
Lincoln National Corp	LNC	0.01%	8.01%	0.00%	81.14%	30.50%	0.01%
Loews Corp	L	0.04%	0.43%	0.00%	n/a	18.50%	0.01%
Lowe's Cos Inc	LOW	0.34%	2.10%	0.01%	7.46%	11.00%	0.03%
IDEX Corp	IEX	0.05%	1.04%	0.00%	15.00%	11.00%	0.01%
Marsh & McLennan Cos Inc	MMC	0.23%	1.42%	0.00%	9.73%	11.00%	0.02%
Masco Corp	MAS	0.03%	2.29%	0.00%	6.45%	8.00%	0.00%
S&P Global Inc	SPGI	0.32%	1.04%	0.00%	5.90%	6.50%	0.02%
Medtronic PLC	MDT	0.30%	3.37%	0.01%	3.39%	7.50%	0.02%
Viatris Inc	VTRS	0.03%	4.99%	0.00%	-1.60%		0.00%
CVS Health Corp	CVS	0.27%	3.26%	0.01%	6.89%	6.00%	0.02%
DuPont de Nemours Inc	DD	0.09%	2.01%	0.00%	10.05%	10.00%	0.01%
Micron Technology Inc	MU	0.19%	0.76%	0.00%	-11.00%	9.50%	0.00%
Motorola Solutions Inc	MSI	0.14%	1.23%	0.00%	11.70%	10.50%	0.02%
Cboe Global Markets Inc	CBOE	0.04%	1.49%	0.00%	n/a	10.00%	0.00%
Laboratory Corp of America Holdings	LH	0.06%	1.26%	0.00%	-4.05%	1.50%	0.00%
Newmont Corp	NEM	0.11%	3.26%	0.00%	11.65%	8.00%	0.01%
NIKE Inc	NKE	0.43%	1.11%	0.00%	8.55%	24.00%	0.07%
NiSource Inc	NI	0.03%	3.58%	0.00%	7.90%	9.50%	0.00%
Norfolk Southern Corp	NSC	0.14%	2.55%	0.00%	4.61%	10.00%	0.01%
Principal Financial Group Inc	PFG	0.05%	3.44%	0.00%	8.26%	6.50%	0.00%
Eversource Energy	ES	0.08%	3.45%	0.00%	6.00%	6.50%	0.00%
Northrop Grumman Corp	NOC	0.20%	1.50%	0.00%	3.23%	9.50%	0.01%
Wells Fargo & Co	WFC	0.40%	3.21%	0.01%	11.96%	12.00%	0.05%
Nucor Corp	NUE	0.11%	1.32%	0.00%	-11.99%	9.50%	0.00%
Occidental Petroleum Corp	OXY	0.16%	1.15%	0.00%	69.00%		0.11%
Omnicom Group Inc	OMC	0.05%	2.97%	0.00%	3.80%	6.50%	0.00%
ONEOK Inc	OKE	0.08%	6.01%	0.00%	8.35%	11.50%	0.01%
Raymond James Financial Inc	RJF	0.06%	1.80%	0.00%	n/a	15.00%	0.01%
PG&E Corp	PCG	0.09%	n/a	n/a	2.80%	7.50%	0.00%
Parker-Hannifin Corp	PH	0.12%	1.58%	0.00%	11.53%	15.50%	0.02%
Rollins Inc	ROL	0.05%	1.39%	0.00%	11.80%	10.50%	0.01%
PPL Corp	PPL	0.06%	3.45%	0.00%	5.75%	3.50%	0.00%
ConocoPhillips	COP	0.34%	0.60%	0.00%	11.55%	20.00%	0.05%
PulteGroup Inc	PHM	0.04%	1.10%	0.00%	n/a	7.00%	0.00%
Pinnacle West Capital Corp	PNW	0.03%	4.37%	0.00%	-1.99%	0.50%	0.00%
PNC Financial Services Group Inc/The	PNC	0.14%	4.72%	0.01%	11.36%	12.00%	0.02%
PPG Industries Inc	PPG	0.09%	1.86%	0.00%	9.89%	4.00%	0.01%
Progressive Corp/The	PGR	0.24%	0.28%	0.00%	33.16%	6.50%	0.05%
Public Service Enterprise Group Inc	PEG	0.09%	3.65%	0.00%	5.13%	4.50%	0.00%
Robert Half International Inc	RHI	0.02%	2.38%	0.00%	4.98%	9.50%	0.00%
Edison International	EIX	0.08%	4.18%	0.00%	4.50%	16.00%	0.01%
Schlumberger NV	SLB	0.20%	2.04%	0.00%	27.90%	28.50%	0.06%
Charles Schwab Corp/The	SCHW	0.27%	1.91%	0.01%	9.57%	9.00%	0.02%
Sherwin-Williams Co/The	SHW	0.16%	1.08%	0.00%	7.20%	7.00%	0.01%
West Pharmaceutical Services Inc	WST	0.07%	0.22%	0.00%	9.86%	17.00%	0.01%
J M Smucker Co/The	SJM	0.05%	2.59%	0.00%	5.45%	4.00%	0.00%
Snap-on Inc	SNA	0.04%	2.62%	0.00%	5.39%	4.50%	0.00%
AMETEK Inc	AME	0.09%	0.69%	0.00%	10.91%	10.00%	0.01%

MARKET RISK PREMIUM - DERIVED FROM ANALYSTS' LONG-TERM GROWTH ESTIMATES

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STANDARD AND POOR'S 500 INDEX

Name	Ticker	[4] Weight in Index	[5] Current Dividend Yield	[6] Cap-Weighted Dividend Yield	[7] Bloomberg Long-Term Growth Est.	[8] Value Line Long-Term Growth Est.	[9] Cap-Weighted Long-Term Growth Est.
Southern Co/The	SO	0.21%	3.91%	0.01%	4.40%	6.50%	0.01%
Truist Financial Corp	TFC	0.13%	6.10%	0.01%	7.61%	5.50%	0.01%
Southwest Airlines Co	LUV	0.05%	2.21%	0.00%	42.42%		0.02%
W R Berkley Corp	WRB	0.05%	0.64%	0.00%	12.00%	17.50%	0.01%
Stanley Black & Decker Inc	SWK	0.03%	3.97%	0.00%	-14.00%	6.00%	0.00%
Public Storage	PSA	0.15%	3.97%	0.01%	3.90%	7.50%	0.01%
Arista Networks Inc	ANET	0.15%	n/a	n/a	18.17%	10.00%	0.02%
Sysco Corp	SYY	0.11%	2.54%	0.00%	46.00%	21.50%	0.04%
Corteva Inc	CTVA	0.12%	0.99%	0.00%	15.20%	15.50%	0.02%
Texas Instruments Inc	TXN	0.48%	2.67%	0.01%	7.50%	4.50%	0.03%
Textron Inc	TXT	0.04%	0.11%	0.00%	11.18%	10.50%	0.00%
Thermo Fisher Scientific Inc	TMO	0.63%	0.24%	0.00%	5.00%	11.00%	0.05%
TJX Cos Inc/The	TJX	0.26%	1.70%	0.00%	10.85%	17.00%	0.04%
Globe Life Inc	GL	0.03%	0.82%	0.00%	n/a	8.50%	0.00%
Johnson Controls International plc	JCI	0.12%	2.39%	0.00%	15.92%	12.50%	0.02%
Ulta Beauty Inc	ULTA	0.08%	n/a	n/a	8.03%	16.50%	0.01%
Union Pacific Corp	UNP	0.35%	2.58%	0.01%	7.30%	9.50%	0.03%
Keysight Technologies Inc	KEYS	0.08%	n/a	n/a	6.39%	13.00%	0.01%
UnitedHealth Group Inc	UNH	1.25%	1.40%	0.02%	12.10%	12.00%	0.15%
Marathon Oil Corp	MRO	0.04%	1.67%	0.00%	9.00%		0.00%
Bio-Rad Laboratories Inc	BIO	0.03%	n/a	n/a	n/a	11.50%	0.00%
Ventas Inc	VTR	0.05%	4.15%	0.00%	4.22%	23.50%	0.01%
VF Corp	VFC	0.03%	5.24%	0.00%	-10.30%	9.00%	0.00%
Vulcan Materials Co	VMC	0.06%	1.00%	0.00%	11.67%	9.00%	0.01%
Weyerhaeuser Co	WY	0.06%	2.52%	0.00%	-9.40%	5.00%	0.00%
Whirlpool Corp	WHR	0.02%	5.30%	0.00%	0.88%	6.00%	0.00%
Williams Cos Inc/The	WMB	0.10%	5.99%	0.01%	8.90%	11.00%	0.01%
Constellation Energy Corp	CEG	0.07%	1.44%	0.00%	-93.74%		-0.07%
WEC Energy Group Inc	WEC	0.08%	3.29%	0.00%	6.64%	6.00%	0.01%
Adobe Inc	ADBE	0.50%	n/a	n/a	14.10%	13.00%	0.07%
AES Corp/The	AES	0.05%	2.76%	0.00%	7.90%	14.00%	0.01%
Amgen Inc	AMGN	0.37%	3.52%	0.01%	8.03%	5.50%	0.02%
Apple Inc	AAPL	7.40%	0.56%	0.04%	10.20%	10.50%	0.77%
Autodesk Inc	ADSK	0.13%	n/a	n/a	20.14%	14.00%	0.02%
Cintas Corp	CTAS	0.13%	0.99%	0.00%	10.32%	14.00%	0.02%
Comcast Corp	CMCSA	0.45%	3.06%	0.01%	11.34%	8.50%	0.04%
Molson Coors Beverage Co	TAP	0.03%	3.17%	0.00%	4.94%	49.50%	0.01%
KLA Corp	KLAC	0.16%	1.30%	0.00%	2.22%	20.00%	0.02%
Marriott International Inc/MD	MAR	0.15%	0.96%	0.00%	18.50%	17.50%	0.03%
McCormick & Co Inc/MD	MKC	0.06%	1.87%	0.00%	5.27%	4.50%	0.00%
PACCAR Inc	PCAR	0.11%	1.37%	0.00%	6.35%	5.00%	0.01%
Costco Wholesale Corp	COST	0.62%	0.72%	0.00%	9.73%	10.50%	0.06%
First Republic Bank/CA	FRC	0.01%	n/a	n/a	-10.20%	11.50%	0.00%
Stryker Corp	SYK	0.31%	1.05%	0.00%	9.18%	6.50%	0.02%
Tyson Foods Inc	TSN	0.05%	3.24%	0.00%	-3.02%	6.00%	0.00%
Lamb Weston Holdings Inc	LW	0.04%	1.07%	0.00%	n/a	11.50%	0.01%
Applied Materials Inc	AMAT	0.29%	1.04%	0.00%	2.10%	10.50%	0.02%
American Airlines Group Inc	AAL	0.03%	n/a	n/a	61.20%		0.02%
Cardinal Health Inc	CAH	0.06%	2.63%	0.00%	11.57%	5.00%	0.00%
Cincinnati Financial Corp	CINF	0.05%	2.68%	0.00%	17.66%	9.00%	0.01%
Paramount Global	PARA	0.04%	4.30%	0.00%	16.18%	4.50%	0.00%
DR Horton Inc	DHI	0.10%	1.02%	0.00%	-6.19%	1.00%	0.00%
Electronic Arts Inc	EA	0.09%	0.63%	0.00%	9.71%	13.00%	0.01%
Fair Isaac Corp	FICO	0.05%	n/a	n/a	n/a	16.00%	0.01%
Expeditors International of Washington Inc	EXPD	0.05%	1.22%	0.00%	-9.80%	10.00%	0.00%
Fastenal Co	FAST	0.09%	2.60%	0.00%	8.50%	6.50%	0.01%
M&T Bank Corp	MTB	0.06%	4.35%	0.00%	7.53%	9.00%	0.00%
Xcel Energy Inc	XEL	0.11%	3.08%	0.00%	6.88%	6.00%	0.01%
Fiserv Inc	FISV	0.20%	n/a	n/a	11.32%	11.00%	0.02%
Fifth Third Bancorp	FITB	0.05%	4.95%	0.00%	14.15%	10.00%	0.01%
Gilead Sciences Inc	GILD	0.29%	3.62%	0.01%	1.06%	12.00%	0.02%
Hasbro Inc	HAS	0.02%	5.22%	0.00%	5.11%	7.50%	0.00%
Huntington Bancshares Inc/OH	HBAN	0.05%	5.54%	0.00%	2.31%	12.50%	0.00%
Welltower Inc	WELL	0.10%	3.40%	0.00%	21.40%	12.00%	0.02%
Biogen Inc	BIIB	0.11%	n/a	n/a	5.28%	-10.50%	0.00%
Northern Trust Corp	NTRS	0.05%	3.40%	0.00%	n/a	8.00%	0.00%
Packaging Corp of America	PKG	0.04%	3.60%	0.00%	1.47%	11.00%	0.00%

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STANDARD AND POOR'S 500 INDEX

Name	Ticker	[4] Weight in Index	[5] Current Dividend Yield	[6] Cap-Weighted Dividend Yield	[7] Bloomberg Long-Term Growth Est.	[8] Value Line Long-Term Growth Est.	[9] Cap-Weighted Long-Term Growth Est.
Paychex Inc	PAYX	0.12%	2.76%	0.00%	7.00%	10.50%	0.01%
QUALCOMM Inc	QCOM	0.40%	2.35%	0.01%	6.90%	9.50%	0.03%
Roper Technologies Inc	ROP	0.13%	0.62%	0.00%	7.60%	3.50%	0.01%
Ross Stores Inc	ROST	0.10%	1.26%	0.00%	9.60%	14.00%	0.01%
IDEXX Laboratories Inc	IDXX	0.12%	n/a	n/a	14.09%	11.50%	0.02%
Starbucks Corp	SBUX	0.34%	2.04%	0.01%	19.04%	16.00%	0.06%
KeyCorp	KEY	0.03%	6.55%	0.00%	5.10%	7.50%	0.00%
Fox Corp	FOXA	0.03%	1.47%	0.00%	8.76%	12.00%	0.00%
Fox Corp	FOX	0.02%	1.60%	0.00%	8.76%		0.00%
State Street Corp	STT	0.07%	3.33%	0.00%	10.42%	8.50%	0.01%
Norwegian Cruise Line Holdings Ltd	NCLH	n/a	n/a	n/a	n/a		n/a
US Bancorp	USB	0.16%	5.33%	0.01%	11.17%	7.00%	0.01%
A O Smith Corp	AOS	0.02%	1.74%	0.00%	10.00%	11.50%	0.00%
Gen Digital Inc	GEN	0.03%	2.91%	0.00%	n/a	10.50%	0.00%
T Rowe Price Group Inc	TROW	0.07%	4.32%	0.00%	-7.19%	4.50%	0.00%
Waste Management Inc	WM	0.19%	1.72%	0.00%	n/a	6.50%	0.01%
Constellation Brands Inc	STZ	0.12%	1.42%	0.00%	7.73%	6.00%	0.01%
DENTSPLY SIRONA Inc	XRAY	0.02%	1.43%	0.00%	8.87%	12.00%	0.00%
Zions Bancorp NA	ZION	0.01%	5.48%	0.00%	3.60%	6.50%	0.00%
Alaska Air Group Inc	ALK	0.02%	n/a	n/a	19.47%		0.00%
Invesco Ltd	IVZ	0.02%	4.57%	0.00%	-0.70%	10.00%	0.00%
Intuit Inc	INTU	0.35%	0.70%	0.00%	17.48%	16.50%	0.06%
Morgan Stanley	MS	0.42%	3.53%	0.01%	-0.43%	8.50%	0.02%
Microchip Technology Inc	MCHP	0.13%	1.71%	0.00%	15.89%	10.00%	0.02%
Chubb Ltd	CB	0.23%	1.71%	0.00%	13.50%	14.50%	0.03%
Hologic Inc	HOLX	0.06%	n/a	n/a	-15.00%	25.00%	0.00%
Citizens Financial Group Inc	CFG	0.04%	5.53%	0.00%	2.62%	8.00%	0.00%
O'Reilly Automotive Inc	ORLY	0.15%	n/a	n/a	10.44%	13.00%	0.02%
Allstate Corp/The	ALL	0.08%	3.21%	0.00%	31.13%	2.50%	0.01%
Equity Residential	EQR	0.06%	4.42%	0.00%	5.50%	-5.00%	0.00%
BorgWarner Inc	BWA	0.03%	1.38%	0.00%	13.12%	9.50%	0.00%
Keurig Dr Pepper Inc	KDP	0.14%	2.27%	0.00%	6.35%	11.50%	0.01%
Organon & Co	OGN	0.02%	4.76%	0.00%	5.61%		0.00%
Host Hotels & Resorts Inc	HST	0.03%	2.91%	0.00%	14.80%	51.00%	0.01%
Incyte Corp	INCY	0.05%	n/a	n/a	45.45%	27.00%	0.02%
Simon Property Group Inc	SPG	0.10%	6.43%	0.01%	2.47%	3.50%	0.00%
Eastman Chemical Co	EMN	0.03%	3.75%	0.00%	6.98%	7.00%	0.00%
AvalonBay Communities Inc	AVB	0.07%	3.93%	0.00%	14.64%	7.00%	0.01%
Prudential Financial Inc	PRU	0.09%	6.04%	0.01%	5.26%	3.00%	0.00%
United Parcel Service Inc	UPS	0.40%	3.34%	0.01%	0.92%	7.50%	0.02%
Walgreens Boots Alliance Inc	WBA	0.08%	5.55%	0.00%	1.58%	3.00%	0.00%
STERIS PLC	STE	0.05%	0.98%	0.00%	n/a	10.00%	0.01%
McKesson Corp	MCK	0.14%	0.61%	0.00%	8.72%	10.00%	0.01%
Lockheed Martin Corp	LMT	0.34%	2.54%	0.01%	8.14%	7.00%	0.03%
AmerisourceBergen Corp	ABC	0.09%	1.21%	0.00%	6.37%	8.50%	0.01%
Capital One Financial Corp	COF	0.10%	2.50%	0.00%	2.44%		0.00%
Waters Corp	WAT	0.05%	n/a	n/a	9.20%	6.00%	0.00%
Nordson Corp	NDSN	0.04%	1.17%	0.00%	23.00%	12.00%	0.01%
Dollar Tree Inc	DLTR	0.09%	n/a	n/a	11.65%	12.00%	0.01%
Darden Restaurants Inc	DRI	0.05%	3.12%	0.00%	10.08%	17.50%	0.01%
Evergy Inc	EVRG	0.04%	4.01%	0.00%	5.22%	7.50%	0.00%
Match Group Inc	MTCH	0.03%	n/a	n/a	36.73%	21.00%	0.01%
Domino's Pizza Inc	DPZ	0.03%	1.47%	0.00%	11.00%	13.00%	0.00%
NVR Inc	NVR	0.05%	n/a	n/a	n/a	5.50%	0.00%
NetApp Inc	NTAP	0.04%	3.13%	0.00%	8.79%	8.50%	0.00%
DXC Technology Co	DXC	0.02%	n/a	n/a	7.80%	12.00%	0.00%
Old Dominion Freight Line Inc	ODFL	0.11%	0.47%	0.00%	8.37%	12.50%	0.01%
DaVita Inc	DVA	0.02%	n/a	n/a	3.37%	7.50%	0.00%
Hartford Financial Services Group Inc/The	HIG	0.06%	2.44%	0.00%	7.00%	6.50%	0.00%
Iron Mountain Inc	IRM	0.04%	4.68%	0.00%	4.00%	10.00%	0.00%
Estee Lauder Cos Inc/The	EL	0.16%	1.07%	0.00%	5.88%	14.00%	0.02%
Cadence Design Systems Inc	CDNS	0.16%	n/a	n/a	14.50%	12.00%	0.02%
Tyler Technologies Inc	TYL	0.04%	n/a	n/a	n/a	12.00%	0.01%
Universal Health Services Inc	UHS	0.02%	0.63%	0.00%	8.73%	5.50%	0.00%
Skyworks Solutions Inc	SWKS	0.05%	2.10%	0.00%	8.48%	9.00%	0.00%
Quest Diagnostics Inc	DGX	0.04%	2.01%	0.00%	-0.92%	5.00%	0.00%
Activision Blizzard Inc	ATVI	0.19%	0.55%	0.00%	7.73%	11.50%	0.02%

MARKET RISK PREMIUM - DERIVED FROM ANALYSTS' LONG-TERM GROWTH ESTIMATES

[1] Estimated Weighted Average Dividend Yield	1.66%
[2] Estimated Weighted Average Long-Term Growth Rate	12.32%
[3] S&P 500 Estimated Required Market Return	14.08%

STANDARD AND POOR'S 500 INDEX

Name	Ticker	[4] Weight in Index	[5] Current Dividend Yield	[6] Cap-Weighted Dividend Yield	[7] Bloomberg Long-Term Growth Est.	[8] Value Line Long-Term Growth Est.	[9] Cap-Weighted Long-Term Growth Est.
Rockwell Automation Inc	ROK	0.10%	1.61%	0.00%	11.54%	9.50%	0.01%
Kraft Heinz Co/The	KHC	0.13%	4.14%	0.01%	3.14%	6.50%	0.01%
American Tower Corp	AMT	0.27%	3.05%	0.01%	17.95%	6.00%	0.03%
Regeneron Pharmaceuticals Inc	REGN	0.25%	n/a	n/a	0.00%	5.00%	0.01%
Amazon.com Inc	AMZN	3.00%	n/a	n/a	2.50%	26.50%	0.44%
Jack Henry & Associates Inc	JKHY	0.03%	1.38%	0.00%	9.00%	8.50%	0.00%
Ralph Lauren Corp	RL	0.01%	2.57%	0.00%	9.29%	12.00%	0.00%
Boston Properties Inc	BXP	0.02%	7.24%	0.00%	-5.66%	-1.00%	0.00%
Amphenol Corp	APH	0.14%	1.03%	0.00%	5.76%	12.50%	0.01%
Howmet Aerospace Inc	HWM	0.05%	0.38%	0.00%	20.00%	14.00%	0.01%
Pioneer Natural Resources Co	PXD	0.14%	10.93%	0.01%	5.97%	21.00%	0.02%
Valero Energy Corp	VLO	0.15%	2.92%	0.00%	18.16%	29.50%	0.03%
Synopsys Inc	SNPS	0.17%	n/a	n/a	16.66%	12.50%	0.02%
Etsy Inc	ETSY	0.04%	n/a	n/a	10.57%	24.50%	0.01%
CH Robinson Worldwide Inc	CHRW	0.03%	2.46%	0.00%	5.00%	8.50%	0.00%
Accenture PLC	ACN	0.54%	1.57%	0.01%	11.47%	12.50%	0.06%
TransDigm Group Inc	TDG	0.11%	n/a	n/a	17.97%	20.00%	0.02%
Yum! Brands Inc	YUM	0.10%	1.83%	0.00%	11.38%	10.50%	0.01%
Prologis Inc	PLD	0.33%	2.79%	0.01%	8.40%	2.50%	0.02%
FirstEnergy Corp	FE	0.07%	3.89%	0.00%	4.63%	3.00%	0.00%
VeriSign Inc	VRSN	0.06%	n/a	n/a	n/a	11.00%	0.01%
Quanta Services Inc	PWR	0.07%	0.19%	0.00%	11.50%	15.50%	0.01%
Henry Schein Inc	HSIC	0.03%	n/a	n/a	5.24%	6.00%	0.00%
Ameren Corp	AEE	0.06%	2.92%	0.00%	n/a	6.50%	0.00%
ANSYS Inc	ANSS	0.08%	n/a	n/a	9.15%	8.50%	0.01%
FactSet Research Systems Inc	FDS	0.05%	0.86%	0.00%	10.00%	10.50%	0.00%
NVIDIA Corp	NVDA	1.95%	0.06%	0.00%	23.36%	23.00%	0.45%
Sealed Air Corp	SEE	0.02%	1.74%	0.00%	5.18%	9.00%	0.00%
Cognizant Technology Solutions Corp	CTSH	0.09%	1.90%	0.00%	11.35%	8.00%	0.01%
Intuitive Surgical Inc	ISRG	0.25%	n/a	n/a	14.90%	10.00%	0.03%
Take-Two Interactive Software Inc	TTWO	0.06%	n/a	n/a	1.80%	3.00%	0.00%
Republic Services Inc	RSG	0.12%	1.46%	0.00%	8.84%	12.50%	0.01%
eBay Inc	EBAY	0.07%	2.25%	0.00%	6.26%	12.50%	0.01%
Goldman Sachs Group Inc/The	GS	0.31%	3.06%	0.01%	11.68%	5.00%	0.03%
SBA Communications Corp	SBAC	0.08%	1.30%	0.00%	8.00%	35.50%	0.02%
Sempra Energy	SRE	0.13%	3.15%	0.00%	4.94%	7.50%	0.01%
Moody's Corp	MCO	0.16%	1.01%	0.00%	n/a	4.00%	0.01%
ON Semiconductor Corp	ON	0.10%	n/a	n/a	3.56%	18.50%	0.01%
Booking Holdings Inc	BKNG	0.28%	n/a	n/a	20.63%	22.00%	0.06%
F5 Inc	FFIV	0.02%	n/a	n/a	13.89%	10.00%	0.00%
Akamai Technologies Inc	AKAM	0.03%	n/a	n/a	11.30%	5.50%	0.00%
Charles River Laboratories International Inc	CRL	0.03%	n/a	n/a	12.00%	12.00%	0.00%
MarketAxess Holdings Inc	MKTX	0.04%	0.74%	0.00%	14.30%	10.00%	0.01%
Devon Energy Corp	DVN	0.09%	7.03%	0.01%	16.34%	27.50%	0.02%
Bio-Techne Corp	TECH	0.03%	0.43%	0.00%	15.50%	13.00%	0.00%
Alphabet Inc	GOOGL	1.75%	n/a	n/a	16.15%		0.28%
Teleflex Inc	TFX	0.03%	0.54%	0.00%	7.18%	10.00%	0.00%
Bunge Ltd	BG	0.04%	2.62%	0.00%	-5.54%	2.50%	0.00%
Netflix Inc	NFLX	0.44%	n/a	n/a	25.32%	14.50%	0.09%
Allegion plc	ALLE	0.03%	1.69%	0.00%	10.70%	11.00%	0.00%
Agilent Technologies Inc	A	0.12%	0.65%	0.00%	11.50%	12.00%	0.01%
Warner Bros Discovery Inc	WBD	0.10%	n/a	n/a	93.00%		0.10%
Elevance Health Inc	ELV	0.31%	1.29%	0.00%	9.11%	12.50%	0.03%
Trimble Inc	TRMB	0.04%	n/a	n/a	10.00%	7.00%	0.00%
CME Group Inc	CME	0.20%	2.30%	0.00%	5.36%	8.50%	0.01%
Juniper Networks Inc	JNPR	0.03%	2.56%	0.00%	7.46%	11.00%	0.00%
BlackRock Inc	BLK	0.29%	2.99%	0.01%	7.46%	8.50%	0.02%
DTE Energy Co	DTE	0.06%	3.48%	0.00%	5.50%	4.50%	0.00%
Nasdaq Inc	NDAQ	0.08%	1.46%	0.00%	8.00%	8.50%	0.01%
Celanese Corp	CE	0.03%	2.57%	0.00%	6.73%	7.50%	0.00%
Philip Morris International Inc	PM	0.43%	5.22%	0.02%	7.43%	5.00%	0.03%
Salesforce Inc	CRM	0.57%	n/a	n/a	21.45%	19.50%	0.12%
Ingersoll Rand Inc	IR	0.07%	0.14%	0.00%	10.00%		0.01%
Huntington Ingalls Industries Inc	HII	0.02%	2.40%	0.00%	40.00%	10.00%	0.01%
MetLife Inc	MET	0.13%	3.45%	0.00%	8.63%	7.50%	0.01%
Tapestry Inc	TPR	0.03%	2.78%	0.00%	12.10%	13.50%	0.00%
CSX Corp	CSX	0.17%	1.47%	0.00%	4.67%	10.50%	0.01%

MARKET RISK PREMIUM - DERIVED FROM ANALYSTS' LONG-TERM GROWTH ESTIMATES

[1] Estimated Weighted Average Dividend Yield	1.66%
[2] Estimated Weighted Average Long-Term Growth Rate	12.32%
[3] S&P 500 Estimated Required Market Return	14.08%

STANDARD AND POOR'S 500 INDEX

Name	Ticker	[4] Weight in Index	[5] Current Dividend Yield	[6] Cap-Weighted Dividend Yield	[7] Bloomberg Long-Term Growth Est.	[8] Value Line Long-Term Growth Est.	[9] Cap-Weighted Long-Term Growth Est.
Edwards Lifesciences Corp	EW	0.14%	n/a	n/a	7.88%	11.00%	0.01%
Ameriprise Financial Inc	AMP	0.09%	1.63%	0.00%	8.40%	13.50%	0.01%
Zebra Technologies Corp	ZBRA	0.05%	n/a	n/a	n/a	11.50%	0.01%
Zimmer Biomet Holdings Inc	ZBH	0.08%	0.74%	0.00%	1.81%	4.50%	0.00%
CBRE Group Inc	CBRE	0.06%	n/a	n/a	n/a	8.50%	0.01%
Camden Property Trust	CPT	0.03%	3.82%	0.00%	8.87%	-4.00%	0.00%
Mastercard Inc	MA	0.97%	0.63%	0.01%	22.48%	18.50%	0.20%
CarMax Inc	KMX	0.03%	n/a	n/a	8.81%	-3.00%	0.00%
Intercontinental Exchange Inc	ICE	0.17%	1.61%	0.00%	13.30%	7.00%	0.02%
Fidelity National Information Services Inc	FIS	0.09%	3.83%	0.00%	5.82%	52.00%	0.03%
Chipotle Mexican Grill Inc	CMG	0.13%	n/a	n/a	23.60%	20.00%	0.03%
Wynn Resorts Ltd	WYNN	0.04%	n/a	n/a	167.88%	27.00%	0.04%
Live Nation Entertainment Inc	LYV	n/a	n/a	n/a	n/a		n/a
Assurant Inc	AIZ	0.02%	2.33%	0.00%	11.91%	15.50%	0.00%
NRG Energy Inc	NRG	0.02%	4.40%	0.00%	n/a	-2.50%	0.00%
Regions Financial Corp	RF	0.05%	4.31%	0.00%	-0.07%	11.50%	0.00%
Monster Beverage Corp	MNST	0.16%	n/a	n/a	22.12%	10.50%	0.03%
Mosaic Co/The	MOS	0.04%	1.74%	0.00%	17.75%	7.50%	0.01%
Baker Hughes Co	BKR	0.08%	2.63%	0.00%	17.00%		0.01%
Expedia Group Inc	EXPE	0.04%	n/a	n/a	12.95%		0.01%
CF Industries Holdings Inc	CF	0.04%	2.21%	0.00%	-14.20%	11.00%	0.00%
Leidos Holdings Inc	LDOS	0.04%	1.56%	0.00%	7.89%	8.00%	0.00%
APA Corp	APA	0.03%	2.77%	0.00%	3.96%		0.00%
Alphabet Inc	GOOG	1.76%	n/a	n/a	16.15%	18.50%	0.30%
First Solar Inc	FSLR	0.07%	n/a	n/a	33.94%	24.50%	0.02%
TE Connectivity Ltd	TEL	0.12%	1.80%	0.00%	2.75%	10.50%	0.01%
Cooper Cos Inc/The	COO	0.05%	0.02%	0.00%	9.50%	12.00%	0.01%
Discover Financial Services	DFS	0.07%	2.43%	0.00%	-0.18%	8.50%	0.00%
Linde PLC	LIN	0.49%	1.43%	0.01%	11.05%	10.00%	0.05%
Visa Inc	V	1.04%	0.80%	0.01%	16.53%	13.50%	0.16%
Mid-America Apartment Communities Inc	MAA	0.05%	3.71%	0.00%	n/a	-12.50%	-0.01%
Xylem Inc/NY	XYL	0.05%	1.26%	0.00%	13.20%	9.00%	0.01%
Marathon Petroleum Corp	MPC	0.17%	2.23%	0.00%	4.90%		0.01%
Tractor Supply Co	TSCO	0.07%	1.75%	0.00%	10.12%	13.50%	0.01%
Advanced Micro Devices Inc	AMD	0.45%	n/a	n/a	23.96%	25.50%	0.11%
ResMed Inc	RMD	0.09%	0.80%	0.00%	15.95%	8.50%	0.01%
Mettler-Toledo International Inc	MTD	0.10%	n/a	n/a	12.48%	13.50%	0.01%
Jacobs Solutions Inc	J	0.04%	0.89%	0.00%	10.91%	12.00%	0.00%
Copart Inc	CPRT	0.10%	n/a	n/a	n/a	7.00%	0.01%
VICI Properties Inc	VICI	0.09%	4.78%	0.00%	6.78%	7.00%	0.01%
Albemarle Corp	ALB	0.07%	0.72%	0.00%	41.50%	21.50%	0.02%
Fortinet Inc	FTNT	0.15%	n/a	n/a	17.20%	21.50%	0.03%
Moderna Inc	MRNA	0.17%	n/a	n/a	-59.18%	-2.50%	-0.05%
Essex Property Trust Inc	ESS	0.04%	4.42%	0.00%	9.86%	-3.00%	0.00%
CoStar Group Inc	CSGP	0.08%	n/a	n/a	19.10%	13.00%	0.01%
Realty Income Corp	O	0.12%	4.83%	0.01%	0.59%	5.50%	0.00%
Westrock Co	WRK	0.02%	3.61%	0.00%	-0.85%	10.00%	0.00%
Westinghouse Air Brake Technologies Corp	WAB	0.05%	0.67%	0.00%	9.25%	9.50%	0.00%
Pool Corp	POOL	0.04%	1.17%	0.00%	1.71%	14.00%	0.00%
Western Digital Corp	WDC	0.03%	n/a	n/a	-8.94%	4.00%	0.00%
PepsiCo Inc	PEP	0.71%	2.52%	0.02%	7.69%	6.50%	0.05%
Diamondback Energy Inc	FANG	0.07%	8.73%	0.01%	21.94%		0.02%
ServiceNow Inc	NOW	0.27%	n/a	n/a	27.50%	45.50%	0.10%
Church & Dwight Co Inc	CHD	0.06%	1.23%	0.00%	5.73%	6.00%	0.00%
Federal Realty Investment Trust	FRT	0.02%	4.37%	0.00%	7.29%	2.50%	0.00%
MGM Resorts International	MGM	0.05%	n/a	n/a	-13.00%	25.00%	0.00%
American Electric Power Co Inc	AEP	0.13%	3.65%	0.00%	6.07%	6.00%	0.01%
SolarEdge Technologies Inc	SEDG	0.05%	n/a	n/a	31.30%	27.00%	0.01%
Invitation Homes Inc	INVH	0.05%	3.33%	0.00%	6.65%		0.00%
PTC Inc	PTC	0.04%	n/a	n/a	14.93%	29.00%	0.01%
JB Hunt Transport Services Inc	JBHT	0.05%	0.96%	0.00%	12.75%	10.00%	0.01%
Lam Research Corp	LRCX	0.20%	1.30%	0.00%	1.74%	14.00%	0.02%
Mohawk Industries Inc	MHK	0.02%	n/a	n/a	-3.73%	10.00%	0.00%
GE HealthCare Technologies Inc	GEHC	n/a	n/a	n/a	n/a		n/a
Pentair PLC	PNR	0.03%	1.59%	0.00%	7.48%	12.00%	0.00%
Vertex Pharmaceuticals Inc	VRTX	0.23%	n/a	n/a	6.57%	13.50%	0.02%
Amcor PLC	AMCR	0.05%	4.31%	0.00%	5.82%	14.50%	0.00%

MARKET RISK PREMIUM - DERIVED FROM ANALYSTS' LONG-TERM GROWTH ESTIMATES

[1] Estimated Weighted Average Dividend Yield	1.66%
[2] Estimated Weighted Average Long-Term Growth Rate	12.32%
[3] S&P 500 Estimated Required Market Return	14.08%

STANDARD AND POOR'S 500 INDEX

Name	Ticker	[4] Weight in Index	[5] Current Dividend Yield	[6] Cap-Weighted Dividend Yield	[7] Bloomberg Long-Term Growth Est.	[8] Value Line Long-Term Growth Est.	[9] Cap-Weighted Long-Term Growth Est.
Meta Platforms Inc	META	1.34%	n/a	n/a	12.61%	11.00%	0.16%
T-Mobile US Inc	TMUS	0.50%	n/a	n/a	57.04%	16.00%	0.18%
United Rentals Inc	URI	0.08%	1.50%	0.00%	12.82%	18.00%	0.01%
Honeywell International Inc	HON	0.36%	2.16%	0.01%	8.75%	12.00%	0.04%
Alexandria Real Estate Equities Inc	ARE	0.06%	3.85%	0.00%	-6.54%	11.00%	0.00%
Delta Air Lines Inc	DAL	0.06%	n/a	n/a	29.81%		0.02%
Seagate Technology Holdings PLC	STX	0.04%	4.23%	0.00%	-3.90%	12.00%	0.00%
United Airlines Holdings Inc	UAL	0.04%	n/a	n/a	41.53%		0.02%
News Corp	NWS	0.01%	1.15%	0.00%	1.20%		0.00%
Centene Corp	CNC	0.10%	n/a	n/a	8.63%	9.00%	0.01%
Martin Marietta Materials Inc	MLM	0.06%	0.74%	0.00%	13.04%	4.50%	0.01%
Teradyne Inc	TER	0.05%	0.41%	0.00%	13.61%	19.00%	0.01%
PayPal Holdings Inc	PYPL	0.24%	n/a	n/a	13.46%	12.00%	0.03%
Tesla Inc	TSLA	1.86%	n/a	n/a	27.50%	21.50%	0.46%
Arch Capital Group Ltd	ACGL	0.07%	n/a	n/a	13.00%	21.50%	0.01%
DISH Network Corp	DISH	0.01%	n/a	n/a	-11.53%	-4.00%	0.00%
Dow Inc	DOW	0.11%	5.11%	0.01%	-0.27%	8.50%	0.00%
Everest Re Group Ltd	RE	0.04%	1.84%	0.00%	31.67%	9.50%	0.01%
Teledyne Technologies Inc	TDY	0.06%	n/a	n/a	6.78%	9.50%	0.00%
News Corp	NWSA	0.02%	1.16%	0.00%	1.20%		0.00%
Exelon Corp	EXC	0.12%	3.44%	0.00%	-6.60%		-0.01%
Global Payments Inc	GPN	0.08%	0.95%	0.00%	10.39%	17.00%	0.01%
Crown Castle Inc	CCI	0.16%	4.68%	0.01%	7.40%	13.50%	0.02%
Aptiv PLC	APTV	0.09%	n/a	n/a	11.63%	30.00%	0.02%
Advance Auto Parts Inc	AAP	0.02%	4.93%	0.00%	3.29%	12.00%	0.00%
Align Technology Inc	ALGN	0.07%	n/a	n/a	15.96%	17.00%	0.01%
Illumina Inc	ILMN	0.10%	n/a	n/a	19.49%	6.50%	0.01%
Targa Resources Corp	TRGP	0.05%	1.92%	0.00%	9.85%		0.00%
LKQ Corp	LKQ	0.04%	1.94%	0.00%	2.30%	13.00%	0.00%
Zoetis Inc	ZTS	0.22%	0.90%	0.00%	10.24%	9.00%	0.02%
Equinix Inc	EQIX	0.19%	1.89%	0.00%	11.96%	15.00%	0.03%
Digital Realty Trust Inc	DLR	0.08%	4.96%	0.00%	24.80%	-1.00%	0.01%
Molina Healthcare Inc	MOH	0.04%	n/a	n/a	19.58%	12.50%	0.01%
Las Vegas Sands Corp	LVS	n/a	n/a	n/a	n/a		n/a

Notes:
[1] Equals Sum ([6])
[2] Equals Sum ([9])
[3] Equals ([1] x (1 + (0.5 x [2]))) + [2]
[4] Equals weight in S&P 500 based on market capitalization
[5] Source: Bloomberg Professional
[6] Equals [4] x [5]
[7] Source: Bloomberg Professional
[8] Source: Value Line
[9] Equals [4] x (([7] + [8]) / 2)

MARKET RISK PREMIUM - DERIVED FROM MERRILL LYNCH ESTIMATES

Merrill Lynch Cost of Market	Implied	Required
Dec-22	10.30%	10.50%
Jan-23	10.40%	10.60%
Feb-23	10.40%	10.60%

Merrill Lynch Expected Market Return [1]: 10.47%

Sources:
[1] *Merrill Lynch cost of market figure is average of Implied and Required Returns for the 3 months ending February 2023*

Exhibit ___ (JCN-6)

Comparison of KEDNY/KEDLI and Proxy Group Companies Risk Assessment

COMPARISON OF KEDNY/KEDLI AND PROXY GROUP COMPANIES
RISK ASSESSMENT

Company	Ticker	Jurisdiction/Service	[1] Test Year	[2] Fuel/Purchased Power Cost Recovery Mechanism	[3] Revenue Decoupling	[4] Capital Cost Recovery Mechanism
ALLETE, Inc.	ALE	Minnesota - Electric	Fully Forecast	Yes	No	No
American Electric Power Company, Inc.	AEP	Arkansas - Electric	Partially Forecast	Yes	Partial	Yes
	AEP	Indiana - Electric	Fully Forecast	Yes	Partial	Yes
	AEP	Kentucky - Electric	Fully Forecast	Yes	Partial	No
	AEP	Louisiana - Electric	Historical	Yes	Partial	No
	AEP	Michigan - Electric	Fully Forecast	Yes	No	No
	AEP	Ohio - Electric	Partially Forecast	Yes	Partial	Yes
	AEP	Oklahoma - Electric	Historical	Yes	Partial	Yes
	AEP	Tennessee - Electric	Fully Forecast	Yes	No	No
	AEP	Texas - Electric	Historical	Yes	No	Yes
	AEP	Virginia - Electric	Historical	Yes	No	Yes
	AEP	West Virginia - Electric	Historical	Yes	No	No
Atmos Energy Corporation	ATO	Kansas - Gas	Historical	Yes	Partial	Yes
	ATO	Kentucky - Gas	Fully Forecast	Yes	Partial	Yes
	ATO	Louisiana - Gas	Historical	Yes	Partial	Yes
	ATO	Mississippi - Gas	Partially Forecast	Yes	Partial	Yes
	ATO	Tennessee - Gas	Fully Forecast	Yes	Partial	No
	ATO	Texas - Gas	Historical	Yes	Partial	Yes
Avista Corporation	AVA	Alaska - Electric	Historical	Yes	No	No
	AVA	Idaho - Electric	Historical	Yes	Full	No
	AVA	Idaho - Gas	Historical	Yes	Full	No
	AVA	Oregon - Gas	Fully Forecast	Yes	Full	No
	AVA	Washington - Electric	Historical	Yes	Partial	No
	AVA	Washington - Gas	Historical	Yes	Partial	No
Black Hills Corp	BKH	Arkansas - Gas	Partially Forecast	Yes	Full	Yes
	BKH	Colorado - Electric	Historical	Yes	No	Yes
	BKH	Colorado - Gas	Historical	Yes	No	No
	BKH	Iowa - Gas	Historical	Yes	No	Yes
	BKH	Kansas - Gas	Historical	Yes	Partial	Yes
	BKH	Nebraska - Gas	Historical	Yes	No	Yes
	BKH	South Dakota - Electric	Historical	Yes	Partial	No
	BKH	Wyoming - Electric	Historical	Yes	Partial	No
	BKH	Wyoming - Gas	Historical	Yes	Partial	Yes
CMS Energy Corporation	CMS	Michigan - Electric	Fully Forecast	Yes	No	No
	CMS	Michigan - Gas	Fully Forecast	Yes	Partial	Yes
Edison International	EIX	California - Electric	Fully Forecast	Yes	Full	No
Eversource Energy	ES	Connecticut - Electric	Fully Forecast	N/A	Full	Yes
		Connecticut - Gas	Fully Forecast	Yes	No	Yes
		Massachusetts - Electric	Historical	N/A	Full	Yes
		Massachusetts - Gas	Historical	Yes	Full	Yes
		New Hampshire - Electric	Historical	Yes	Partial	Yes
Exelon Corporation	EXC	Delaware - Electric	Partially Forecast	Yes	No	Yes
	EXC	Delaware - Gas	Partially Forecast	Yes	No	Yes
	EXC	District of Columbia - Electric	Partially Forecast	Yes	Partial	Yes
	EXC	Illinois - Electric	Historical	Yes	No	Yes
	EXC	Maryland - Electric	Partially Forecast	Yes	Full	No
	EXC	Maryland - Gas	Partially Forecast	Yes	Full	Yes
	EXC	New Jersey - Electric	Partially Forecast	Yes	No	Yes
	EXC	Pennsylvania - Electric	Fully Forecast	Yes	No	Yes
	EXC	Pennsylvania - Gas	Fully Forecast	Yes	No	Yes
Hawaiian Electric Industries, Inc.	HE	Hawaii - Electric	Fully Forecast	Yes	Full	Yes

COMPARISON OF KEDNY/KEDLI AND PROXY GROUP COMPANIES
RISK ASSESSMENT

Company	Ticker	Jurisdiction/Service	[1] Test Year	[2] Fuel/Purchased Power Cost Recovery Mechanism	[3] Revenue Decoupling	[4] Capital Cost Recovery Mechanism
NiSource Inc.	NI	Indiana - Electric	Historical	Yes	Partial	Yes
		Indiana - Gas	Historical	Yes	No	Yes
		Kentucky - Gas	Fully Forecast	Yes	Partial	Yes
		Maryland - Gas	Partially Forecast	Yes	Partial	Yes
		Massachusetts - Gas	Historical	Yes	Full	Yes
		Ohio - Gas	Partially Forecast	N/A	No	Yes
		Pennsylvania - Gas	Fully Forecast	Yes	Partial	Yes
		Virginia - Gas	Historical	Yes	Partial	Yes
Northwest Natural Gas Company	NWN	Oregon - Gas	Fully Forecast	Yes	Partial	No
	NWN	Washington - Gas	Historical	Yes	No	No
NorthWestern Corporation	NWE	Montana - Electric	Historical	Yes	No	No
	NWE	Montana - Gas	Historical	Yes	No	No
	NWE	Nebraska - Gas	Historical	Yes	No	No
	NWE	South Dakota - Electric	Historical	Yes	No	No
	NWE	South Dakota - Gas	Historical	Yes	No	No
OGE Energy Corporation	OGE	Arkansas - Electric	Partially Forecast	Yes	Partial	Yes
	OGE	Oklahoma - Electric	Historical	Yes	Partial	Yes
ONE Gas, Inc.	OGS	Kansas - Gas	Historical	Yes	Partial	Yes
	OGS	Oklahoma - Gas	Historical	Yes	Partial	No
	OGS	Texas - Gas	Historical	Yes	Partial	Yes
PPL Corporation	PPL	Kentucky - Electric	Fully Forecast	Yes	Partial	No
	PPL	Kentucky - Gas	Fully Forecast	Yes	Partial	Yes
	PPL	Pennsylvania - Electric	Fully Forecast	Yes	No	Yes
	PPL	Virginia - Electric	Historical	Yes	No	No
Public Service Enterprise Group	PEG	New Jersey - Electric	Partially Forecast	Yes	No	Yes
	PEG	New Jersey - Gas	Partially Forecast	Yes	Partial	Yes
Spire, Inc.	SR	Alabama - Gas	Fully Forecast	Yes	Partial	No
	SR	Missouri East - Gas	Historical	Yes	Partial	Yes
	SR	Missouri West - Gas	Historical	Yes	No	Yes
Xcel Energy Inc.	XEL	Colorado - Electric	Historical	Yes	No	Yes
	XEL	Colorado - Gas	Historical	Yes	Partial	Yes
	XEL	Minnesota - Electric	Fully Forecast	Yes	Partial	No
	XEL	Minnesota - Gas	Fully Forecast	Yes	No	Yes
	XEL	New Mexico - Electric	Historical	Yes	No	No
	XEL	North Dakota - Electric	Fully Forecast	Yes	No	Yes
	XEL	North Dakota - Gas	Fully Forecast	Yes	No	No
	XEL	South Dakota - Electric	Historical	Yes	Partial	Yes
	XEL	Texas - Electric	Historical	Yes	No	Yes
	XEL	Wisconsin - Electric	Fully Forecast	Yes	No	No
	XEL	Wisconsin - Gas	Fully Forecast	Yes	No	No

Proxy Group Average		Fully Forecast	29	Yes	88	Full	12	Yes	56
		Partially Forecast	15	No	0	Partial	40	No	35
		Historical	47	N/A	3	No	39		

KEDNY	New York - Gas	Fully Forecast	Yes	Yes	Yes
KEDLI	New York - Gas	Fully Forecast	Yes	Yes	Yes

Notes
[1] Source: "Alternative Regulation for Evolving Utility Challenges," Prepared by Pacific Economics Group Research for Edison Electric Institute, Table 6, November 2015; S&P RRA Research; Company Investor Presentations.
[2] - [4] Source: "Adjustment Clauses: A State-by-state Overview," Regulatory Research Associates, November 12, 2019, Operating subsidiaries not covered in this report were excluded from this exhibit.

Exhibit ___ (JCN-7)

Capital Expenditures – Comparison of KEDNY/KEDLI and Proxy Group

2021 - 2023 CAPITAL EXPENDITURES AS A PERCENT OF 2021 NET PLANT
($ Millions)

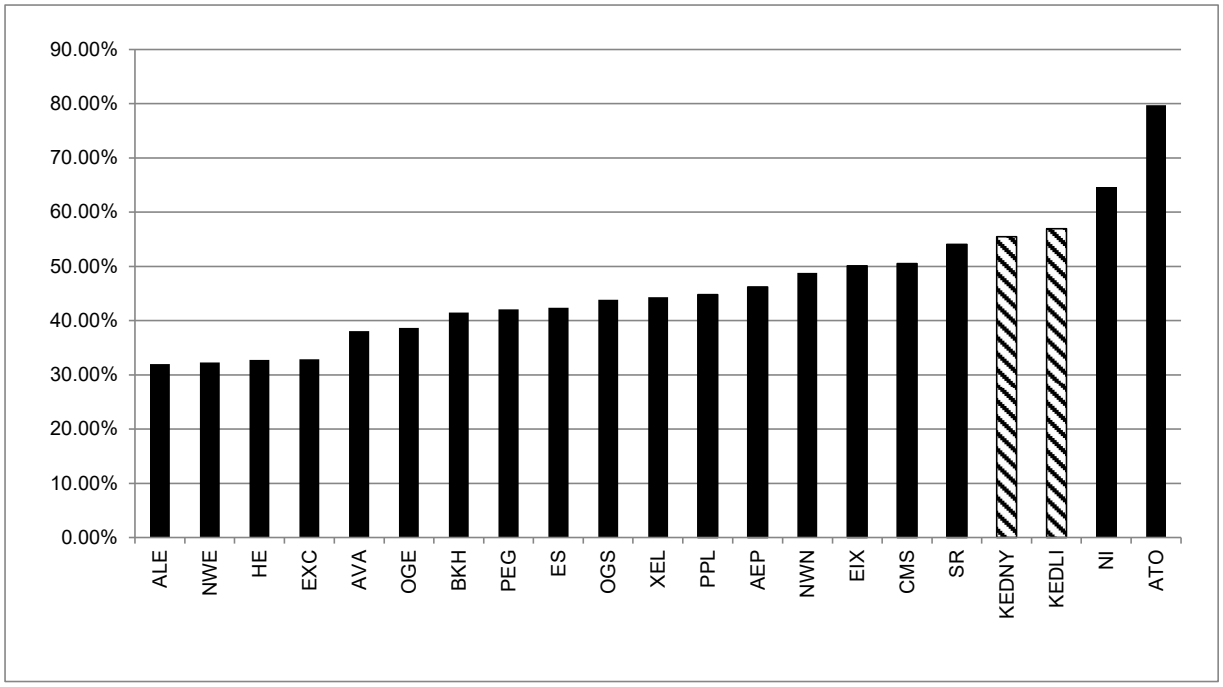
		[1]		[2]	[3]	[4]	[4]	[5]
		2021	2023	2024	2025	2026	2027	2024-27 Cap. Ex. / 2021 Net Plant
ALLETE, Inc.	ALE							
Capital Spending per Share				$5.95	$6.60	$7.25	$7.25	
Common Shares Outstanding				59.00	60.00	61.00	61.00	
Capital Expenditures				$351.1	$396.0	$442.3	$442.3	31.99%
Net Plant		$5,100.2						
American Electric Power Company, Inc.	AEP							
Capital Spending per Share				$14.15	$14.08	$14.00	$14.00	
Common Shares Outstanding				530.00	540.00	550.00	550.00	
Capital Expenditures				$7,499.5	$7,600.5	$7,700.0	$7,700.0	46.21%
Net Plant		$66,001.0						
Atmos Energy Corporation	ATO							
Capital Spending per Share				$18.55	$18.43	$18.30	$18.30	
Common Shares Outstanding				152.00	161.00	170.00	170.00	
Capital Expenditures				$2,819.6	$2,966.4	$3,111.0	$3,111.0	79.71%
Net Plant		$15,064.0						
Avista Corporation	AVA							
Capital Spending per Share			$6.40	$6.20	$6.00	$6.00	$6.00	
Common Shares Outstanding			77.00	80.00	83.00	83.00	83.00	
Capital Expenditures				$496.0	$498.0	$498.0	$498.0	38.08%
Net Plant		$5,225.5						
Black Hills Corporation	BKH							
Capital Spending per Share			$9.30	$9.40	$9.50	$9.50	$9.50	
Common Shares Outstanding			67.50	69.25	71.00	71.00	71.00	
Capital Expenditures				$651.0	$674.5	$674.5	$674.5	41.47%
Net Plant		$6,449.2						
CMS Energy Corporation	CMS							
Capital Spending per Share				$9.00	$9.38	$9.75	$9.75	
Common Shares Outstanding				295.00	297.50	300.00	300.00	
Capital Expenditures			$0.0	$2,655.0	$2,789.1	$2,925.0	$2,925.0	50.53%
Net Plant		$22,352.0						
Edison International	EIX							
Capital Spending per Share			$15.10	$15.80	$16.50	$16.50	$16.50	
Common Shares Outstanding			382.00	386.00	390.00	390.00	390.00	
Capital Expenditures			$5,768.2	$6,098.8	$6,435.0	$6,435.0	$6,435.0	50.11%
Net Plant		$50,700.0						
Eversource Energy	ES							
Capital Spending per Share				$10.10	$9.93	$9.75	$9.75	
Common Shares Outstanding				355.00	357.50	360.00	360.00	
Capital Expenditures				$3,585.5	$3,548.2	$3,510.0	$3,510.0	42.40%
Net Plant		$33,378.0						
Exelon Corporation	EXC							
Capital Spending per Share				$6.80	$6.90	$7.00	$7.00	
Common Shares Outstanding				1,000.00	1,000.00	1,000.00	1,000.00	
Capital Expenditures				$6,800.0	$6,900.0	$7,000.0	$7,000.0	32.89%
Net Plant		$84,219.0						
Hawaiian Electric Industries, Inc.	HE							
Capital Spending per Share			$3.35	$3.68	$4.00	$4.00	$4.00	
Common Shares Outstanding			110.50	111.75	113.00	113.00	113.00	
Capital Expenditures			$370.2	$410.7	$452.0	$452.0	$452.0	32.76%
Net Plant		$5,392.1						
NiSource Inc.	NI							
Capital Spending per Share				$6.55	$6.90	$7.25	$7.25	
Common Shares Outstanding				410.00	412.50	415.00	415.00	
Capital Expenditures				$2,685.5	$2,846.3	$3,008.8	$3,008.8	64.59%
Net Plant		$17,882.0						
Northwest Natural Gas Company	NWN							
Capital Spending per Share				$9.25	$9.38	$9.50	$9.50	
Common Shares Outstanding				36.00	37.00	38.00	38.00	
Capital Expenditures				$333.0	$346.9	$361.0	$361.0	48.82%
Net Plant		$2,871.4						

2021 - 2023 CAPITAL EXPENDITURES AS A PERCENT OF 2021 NET PLANT
($ Millions)

		[1]		[2]	[3]	[4]	[4]	[5]
		2021	2023	2024	2025	2026	2027	2024-27 Cap. Ex. / 2021 Net Plant
NorthWestern Corporation	NWE							
Capital Spending per Share			$9.10	$7.80	$6.50	$6.50	$6.50	
Common Shares Outstanding			62.00	62.00	62.00	62.00	62.00	
Capital Expenditures			$564.2	$483.6	$403.0	$403.0	$403.0	32.26%
Net Plant		$5,247.2						
OGE Energy Corporation	OGE							
Capital Spending per Share				$4.75	$4.75	$4.75	$4.75	
Common Shares Outstanding				200.20	200.20	200.20	200.20	
Capital Expenditures			$0.0	$951.0	$951.0	$951.0	$951.0	38.68%
Net Plant		$9,832.9						
ONE Gas, Inc.	OGS							
Capital Spending per Share				$9.90	$10.08	$10.25	$10.25	
Common Shares Outstanding				55.00	56.00	57.00	57.00	
Capital Expenditures				$544.5	$564.2	$584.3	$584.3	43.87%
Net Plant		$5,190.8						
PPL Corporation	PPL							
Capital Spending per Share				$3.65	$3.83	$4.00	$4.00	
Common Shares Outstanding				737.00	737.50	738.00	738.00	
Capital Expenditures				$2,690.1	$2,820.9	$2,952.0	$2,952.0	44.82%
Net Plant		$25,470.0						
Public Service Enterprise Group Inc.	PEG							
Capital Spending per Share				$7.20	$7.23	$7.25	$7.25	
Common Shares Outstanding				500.00	500.00	500.00	500.00	
Capital Expenditures				$3,600.0	$3,612.5	$3,625.0	$3,625.0	42.08%
Net Plant		$34,366.0						
Spire, Inc.	SR							
Capital Spending per Share				$13.60	$12.80	$12.00	$12.00	
Common Shares Outstanding				53.00	54.00	55.00	55.00	
Capital Expenditures				$720.8	$691.2	$660.0	$660.0	54.04%
Net Plant		$5,055.7						
Xcel Energy Inc.	XEL							
Capital Spending per Share			$9.00	$9.00	$9.00	$9.00	$9.00	
Common Shares Outstanding			550.00	555.50	561.00	561.00	561.00	
Capital Expenditures			$4,950.0	$4,999.5	$5,049.0	$5,049.0	$5,049.0	44.32%
Net Plant		$45,457.0						
Brooklyn Union Gas Company	KEDNY							
		2021		FY24	FY25	FY26	FY27	
Capital Expenditures [7]				$855	$1,013	$1,016	$996	55.51%
Net Plant [8]		$6,989.5						
KeySpan Gas East Corporation	KEDLI							
		2021		FY24	FY25	FY26	FY27	
Capital Expenditures [7]				$521	$671	$722	$722	56.98%
Net Plant [8]		$4,626.2						

Notes:
[1] Source: Value Line
[2] Source: Value Line
[3] Source: Value Line
[4] Source: Value Line
[5] Source: Value Line
[6] Equals Sum ([3], [4], [5]) / [1]
[7] Source: Company Data
[8] Source: S&P Capital IQ Pro

2021 - 2023 CAPITAL EXPENDITURES AS A PERCENT OF 2021 NET PLANT
($ Millions)



			2024-27 Cap. Ex. / 2021 Net Plant
1	ALLETE, Inc.	ALE	31.99%
2	NorthWestern Corporation	NWE	32.26%
3	Hawaiian Electric Industries, Inc.	HE	32.76%
4	Exelon Corporation	EXC	32.89%
5	Avista Corporation	AVA	38.08%
6	OGE Energy Corporation	OGE	38.68%
7	Black Hills Corporation	BKH	41.47%
8	Public Service Enterprise Group Inc.	PEG	42.08%
9	Eversource Energy	ES	42.40%
10	ONE Gas, Inc.	OGS	43.87%
11	Xcel Energy Inc.	XEL	44.32%
12	PPL Corporation	PPL	44.82%
13	American Electric Power Company, Inc.	AEP	46.21%
14	Northwest Natural Gas Company	NWN	48.82%
15	Edison International	EIX	50.11%
16	CMS Energy Corporation	CMS	50.53%
17	Spire, Inc.	SR	54.04%
18	Brooklyn Union Gas Company	KEDNY	55.51%
19	KeySpan Gas East Corporation	KEDLI	56.98%
20	NiSource Inc.	NI	64.59%
21	Atmos Energy Corporation	ATO	79.71%
	Median of Proxy Group (excluding KEDNY and KEDLI)		43.87%
	Mean of Proxy Group (excluding KEDNY and KEDLI)		45.24%

Exhibit ___ (JCN-8)

Capital Structure – Proxy Group Analysis

CAPITAL STRUCTURE
PROXY GROUP ANALYSIS
Common Equity Ratio [1]

Proxy Group Company	Ticker	2021	2020	2019	2019	Average
ALLETE, Inc.	ALE	56.13%	58.15%	59.55%	61.27%	58.78%
American Electric Power Company, Inc.	AEP	47.38%	47.74%	47.89%	47.97%	47.74%
Atmos Energy Corporation	ATO	59.79%	58.19%	58.29%	63.02%	59.82%
Avista Corporation	AVA	50.31%	50.53%	50.76%	49.82%	50.36%
Black Hills Corporation	BKH	68.52%	66.94%	64.87%	58.16%	64.62%
CMS Energy Corporation	CMS	52.10%	50.99%	51.26%	50.05%	51.10%
Edison International	EIX	43.17%	46.13%	46.59%	42.51%	44.60%
Eversource Energy	ES	54.59%	55.40%	53.48%	53.64%	54.28%
Exelon Corporation	EXC	52.98%	52.81%	52.64%	52.80%	52.81%
Hawaiian Electric Industries, Inc.	HE	56.94%	56.55%	55.81%	56.98%	56.57%
NiSource Inc.	NI	55.23%	54.04%	53.87%	54.31%	54.36%
Northwest Natural Gas Company	NWN	49.53%	49.02%	51.34%	50.19%	50.02%
NorthWestern Corporation	NWE	47.74%	47.11%	47.54%	47.79%	47.55%
OGE Energy Corp.	OGE	52.88%	52.47%	54.52%	52.55%	53.11%
ONE Gas, Inc.	OGS	60.34%	59.06%	62.31%	60.97%	60.67%
PPL Corporation	PPL	55.97%	54.68%	53.76%	54.67%	54.77%
Public Service Enterprise Group Inc.	PEG	55.05%	54.39%	54.50%	54.00%	54.48%
Spire, Inc.	SR	55.11%	58.46%	60.92%	59.30%	58.45%
Xcel Energy Inc.	XEL	54.31%	54.71%	54.14%	54.29%	54.36%
MEAN		54.11%	54.07%	54.42%	53.91%	54.13%
LOW		43.17%	46.13%	46.59%	42.51%	44.60%
HIGH		68.52%	66.94%	64.87%	63.02%	64.62%

CAPITAL STRUCTURE
PROXY GROUP ANALYSIS
Common Equity Ratio - Utility Operating Companies

Utility Operating Company	Ticker	2021	2020	2019	2019	Average
ALLETE (Minnesota Power)	ALE	56.08%	58.12%	59.59%	61.39%	58.79%
Superior Water, Light and Power Company	ALE	58.27%	59.27%	58.08%	56.86%	58.12%
AEP Texas Inc.	AEP	42.80%	42.41%	43.77%	45.37%	43.59%
Appalachian Power Company	AEP	47.97%	46.80%	48.25%	48.98%	48.00%
Indiana Michigan Power Company	AEP	47.02%	48.31%	46.40%	44.31%	46.51%
Kentucky Power Company	AEP	43.46%	44.53%	46.47%	44.88%	44.83%
Kingsport Power Company	AEP	52.60%	51.89%	52.86%	48.66%	51.50%
Ohio Power Company	AEP	48.21%	51.86%	53.68%	56.20%	52.49%
Public Service Company of Oklahoma	AEP	53.64%	51.90%	48.66%	48.05%	50.56%
Southwestern Electric Power Company	AEP	48.23%	50.18%	48.17%	46.37%	48.24%
Wheeling Power Company	AEP	53.45%	53.69%	53.32%	54.46%	53.73%
Atmos Energy Corporation	ATO	59.79%	58.19%	58.29%	63.02%	59.82%
Avista Corporation	AVA	50.31%	50.53%	50.76%	49.82%	50.36%
Black Hills Colorado Electric, Inc.	BKH	73.57%	72.97%	72.50%	52.88%	67.98%
Black Hills Energy Arkansas, Inc.	BKH	99.02%	98.90%	98.78%	98.63%	98.83%
Black Hills Power, Inc.	BKH	59.97%	58.08%	56.66%	53.97%	57.17%
Cheyenne Light, Fuel and Power Company	BKH	56.02%	54.30%	48.13%	53.51%	52.99%
Consumers Energy Company	CMS	52.10%	50.99%	51.26%	50.05%	51.10%
Southern California Edison Company	EIX	43.17%	46.13%	46.59%	42.51%	44.60%
Connecticut Light and Power Company	ES	54.78%	55.32%	54.31%	55.10%	54.88%
NSTAR Electric Company	ES	55.08%	54.92%	54.94%	55.31%	55.06%
NSTAR Gas Company	ES	55.51%	55.49%	55.46%	53.59%	55.01%
Public Service Company of New Hampshire	ES	48.99%	48.54%	47.63%	47.79%	48.24%
Yankee Gas Services Company	ES	61.67%	61.87%	62.64%	66.82%	63.25%
Bay State Gas Company	ES	52.15%	68.48%	20.75%	-1.02%	35.09%
Baltimore Gas and Electric Company	EXC	52.80%	52.49%	51.91%	52.67%	52.47%
Commonwealth Edison Company	EXC	54.88%	54.80%	54.61%	54.73%	54.76%
PECO Energy Company	EXC	53.39%	52.98%	53.15%	53.22%	53.18%
Delmarva Power & Light Company	EXC	49.29%	49.56%	49.56%	49.40%	49.45%
Atlantic City Electric Company	EXC	49.76%	49.50%	48.82%	48.59%	49.17%
Potomac Electric Power Company	EXC	49.85%	49.73%	49.68%	49.52%	49.69%
Hawaiian Electric Company, Inc.	HE	56.94%	56.55%	55.81%	56.98%	56.57%
Columbia Gas of Kentucky, Incorporated	NI	53.65%	54.34%	53.82%	54.18%	54.00%
Columbia Gas of Maryland, Incorporated	NI	55.15%	54.84%	52.21%	56.49%	54.67%
Columbia Gas of Ohio, Inc.	NI	50.67%	50.30%	52.79%	54.16%	51.98%
Columbia Gas of Pennsylvania, Inc.	NI	55.95%	55.58%	55.48%	55.56%	55.65%
Columbia Gas of Virginia, Incorporated	NI		43.39%	42.02%	42.14%	42.51%
Northern Indiana Public Service Company	NI	57.97%	57.34%	55.69%	55.57%	56.64%
NorthWestern Corporation	NWE	47.74%	47.11%	47.54%	47.79%	47.55%
Northwest Natural Gas Company	NWN	49.53%	49.02%	51.34%	50.19%	50.02%
Oklahoma Gas and Electric Company	OGE	52.88%	52.47%	54.52%	52.55%	53.11%
Kansas Gas Service Company, Inc.	OGS	60.66%	59.39%	62.66%	61.19%	60.97%
Oklahoma Natural Gas Company	OGS	60.15%	58.84%	62.03%	60.76%	60.45%
Texas Gas Service Company, Inc.	OGS	60.27%	58.99%	62.29%	60.98%	60.63%
Kentucky Utilities Company	PPL	55.43%	54.32%	52.62%	54.51%	54.22%
Louisville Gas and Electric Company	PPL	56.73%	55.50%	53.65%	55.41%	55.32%
PPL Electric Utilities Corporation	PPL	55.93%	54.51%	54.54%	54.41%	54.85%
Public Service Electric and Gas Company	PEG	55.05%	54.39%	54.50%	54.00%	54.48%
Spire Alabama Inc.	SR	57.81%	63.30%	67.94%		63.02%
Spire Gulf Inc.	SR	48.88%	45.30%	36.70%	44.06%	43.73%
Spire Mississippi Inc.	SR	96.78%	96.04%	95.51%	95.04%	95.84%
Spire Missouri Inc.	SR	53.81%	56.47%	58.70%	59.94%	57.23%
Public Service Company of Colorado	XEL	56.32%	56.70%	56.20%	56.16%	56.34%
Northern States Power Company - MN	XEL	52.55%	53.07%	51.99%	52.55%	52.54%
Northern States Power Company - WI	XEL	52.37%	53.68%	54.00%	53.43%	53.37%
Southwestern Public Service Company	XEL	54.19%	54.12%	54.08%	54.09%	54.12%

Notes:
[1] Natural Gas and Electric Operating Subsidiaries with data listed as N/A have been excluded from the analysis.
Source: S&P Capital IQ Pro

CAPITAL STRUCTURE
PROXY GROUP ANALYSIS
Long-Term Debt Ratio [1]

Proxy Group Company	Ticker	2021	2020	2019	2019	Average
ALLETE, Inc.	ALE	43.87%	41.85%	40.45%	38.73%	41.22%
American Electric Power Company, Inc.	AEP	51.82%	51.40%	51.11%	50.84%	51.29%
Atmos Energy Corporation	ATO	40.06%	41.60%	41.47%	36.66%	39.95%
Avista Corporation	AVA	49.60%	49.40%	49.11%	50.04%	49.54%
Black Hills Corporation	BKH	31.08%	32.60%	34.61%	41.33%	34.91%
CMS Energy Corporation	CMS	47.56%	48.67%	48.35%	49.51%	48.52%
Edison International	EIX	51.67%	47.83%	45.79%	48.12%	48.35%
Eversource Energy	ES	44.69%	43.80%	45.52%	45.28%	44.82%
Exelon Corporation	EXC	46.42%	46.47%	46.44%	46.24%	46.39%
Hawaiian Electric Industries, Inc.	HE	42.20%	41.22%	40.83%	41.30%	41.39%
NiSource Inc.	NI	44.15%	45.26%	45.28%	44.75%	44.86%
Northwest Natural Gas Company	NWN	50.39%	50.79%	48.35%	49.46%	49.75%
NorthWestern Corporation	NWE	52.08%	52.76%	52.36%	52.03%	52.31%
OGE Energy Corp.	OGE	46.19%	46.46%	44.34%	46.23%	45.80%
ONE Gas, Inc.	OGS	38.42%	39.30%	36.16%	37.32%	37.80%
PPL Corporation	PPL	43.70%	44.94%	45.84%	44.88%	44.84%
Public Service Enterprise Group Inc.	PEG	44.73%	45.33%	45.10%	45.55%	45.18%
Spire, Inc.	SR	44.27%	40.78%	38.11%	39.94%	40.77%
Xcel Energy Inc.	XEL	45.50%	45.09%	45.58%	45.37%	45.39%
MEAN		45.18%	45.03%	44.46%	44.93%	44.90%
LOW		31.08%	32.60%	34.61%	36.66%	34.91%
HIGH		52.08%	52.76%	52.36%	52.03%	52.31%

CAPITAL STRUCTURE
PROXY GROUP ANALYSIS
Long-Term Debt Ratio - Utility Operating Companies

Utility Operating Company	Ticker	2021	2020	2019	2019	Average
ALLETE (Minnesota Power)	ALE	43.92%	41.88%	40.41%	38.61%	41.21%
Superior Water, Light and Power Company	ALE	41.70%	40.73%	41.92%	43.14%	41.87%
AEP Texas Inc.	AEP	57.19%	57.58%	56.23%	54.62%	56.41%
Appalachian Power Company	AEP	51.27%	52.36%	50.76%	49.94%	51.08%
Indiana Michigan Power Company	AEP	52.22%	50.96%	52.88%	54.98%	52.76%
Kentucky Power Company	AEP	54.93%	53.81%	51.69%	53.28%	53.43%
Kingsport Power Company	AEP	44.48%	45.25%	43.91%	47.16%	45.20%
Ohio Power Company	AEP	50.66%	47.08%	44.81%	41.03%	45.90%
Public Service Company of Oklahoma	AEP	45.04%	46.26%	49.26%	49.69%	47.56%
Southwestern Electric Power Company	AEP	50.81%	48.65%	50.54%	52.34%	50.59%
Wheeling Power Company	AEP	45.52%	45.55%	46.32%	45.24%	45.66%
Atmos Energy Corporation	ATO	40.06%	41.60%	41.47%	36.66%	39.95%
Avista Corporation	AVA	49.60%	49.40%	49.11%	50.04%	49.54%
Black Hills Colorado Electric, Inc.	BKH	25.94%	26.46%	26.84%	46.62%	31.47%
Black Hills Energy Arkansas, Inc.	BKH	0.00%	0.00%	0.00%	0.00%	0.00%
Black Hills Power, Inc.	BKH	39.82%	41.66%	43.06%	45.78%	42.58%
Cheyenne Light, Fuel and Power Company	BKH	43.76%	45.45%	51.49%	46.00%	46.68%
Consumers Energy Company	CMS	47.56%	48.67%	48.35%	49.51%	48.52%
Southern California Edison Company	EIX	51.67%	47.83%	45.79%	48.12%	48.35%
Connecticut Light and Power Company	ES	43.87%	43.22%	43.85%	42.98%	43.48%
NSTAR Electric Company	ES	44.40%	44.50%	44.39%	43.93%	44.30%
NSTAR Gas Company	ES	44.44%	44.43%	44.42%	46.26%	44.88%
Public Service Company of New Hampshire	ES	50.79%	51.22%	52.08%	51.93%	51.50%
Yankee Gas Services Company	ES	38.21%	37.97%	37.08%	32.86%	36.53%
Bay State Gas Company	ES	47.66%	31.28%	78.62%	100.32%	64.47%
Baltimore Gas and Electric Company	EXC	46.08%	46.13%	46.40%	45.45%	46.02%
Commonwealth Edison Company	EXC	44.68%	44.78%	44.79%	44.67%	44.73%
PECO Energy Company	EXC	46.10%	46.33%	45.97%	45.84%	46.06%
Delmarva Power & Light Company	EXC	49.93%	49.47%	49.32%	49.44%	49.54%
Atlantic City Electric Company	EXC	49.68%	49.70%	50.22%	50.28%	49.97%
Potomac Electric Power Company	EXC	49.66%	49.49%	49.35%	49.51%	49.50%
Hawaiian Electric Company, Inc.	HE	42.20%	41.22%	40.83%	41.30%	41.39%
Columbia Gas of Kentucky, Incorporated	NI	45.94%	45.04%	45.42%	45.01%	45.35%
Columbia Gas of Maryland, Incorporated	NI	44.65%	44.96%	47.46%	43.14%	45.05%
Columbia Gas of Ohio, Inc.	NI	49.09%	49.40%	46.81%	45.40%	47.68%
Columbia Gas of Pennsylvania, Inc.	NI	43.88%	44.24%	44.32%	44.23%	44.17%
Columbia Gas of Virginia, Incorporated	NI		55.91%	56.78%	56.53%	56.41%
Northern Indiana Public Service Company	NI	40.97%	41.50%	43.00%	43.00%	42.12%
NorthWestern Corporation	NWE	52.08%	52.76%	52.36%	52.03%	52.31%
Northwest Natural Gas Company	NWN	50.39%	50.79%	48.35%	49.46%	49.75%
Oklahoma Gas and Electric Company	OGE	46.19%	46.46%	44.34%	46.23%	45.80%
Kansas Gas Service Company, Inc.	OGS	38.18%	39.05%	35.94%	37.18%	37.59%
Oklahoma Natural Gas Company	OGS	38.48%	39.48%	36.28%	37.33%	37.89%
Texas Gas Service Company, Inc.	OGS	38.56%	39.34%	36.22%	37.45%	37.90%
Kentucky Utilities Company	PPL	44.03%	45.13%	46.82%	44.87%	45.21%
Louisville Gas and Electric Company	PPL	42.60%	43.81%	45.65%	43.89%	43.99%
PPL Electric Utilities Corporation	PPL	44.02%	45.38%	45.31%	45.39%	45.03%
Public Service Electric and Gas Company	PEG	44.73%	45.33%	45.10%	45.55%	45.18%
Spire Alabama Inc.	SR	40.99%	35.30%	30.46%		35.58%
Spire Gulf Inc.	SR	49.90%	53.39%	62.00%	53.18%	54.62%
Spire Mississippi Inc.	SR	0.00%	0.00%	0.00%	0.00%	0.00%
Spire Missouri Inc.	SR	45.91%	43.16%	40.71%	39.43%	42.30%
Public Service Company of Colorado	XEL	43.46%	43.10%	43.59%	43.57%	43.43%
Northern States Power Company - MN	XEL	47.25%	46.71%	47.61%	46.95%	47.13%
Northern States Power Company - WI	XEL	47.32%	46.01%	45.58%	46.25%	46.29%
Southwestern Public Service Company	XEL	45.73%	45.79%	45.81%	45.77%	45.77%

Notes:
[1] Natural Gas and Electric Operating Subsidiaries with data listed as N/A have been excluded from the analysis.
Source: S&P Capital IQ Pro

CAPITAL STRUCTURE
PROXY GROUP ANALYSIS
Preferred Equity Ratio [1]

Proxy Group Company	Ticker	2021	2020	2019	2019	Average
ALLETE, Inc.	ALE	0.00%	0.00%	0.00%	0.00%	0.00%
American Electric Power Company, Inc.	AEP	0.00%	0.00%	0.00%	0.00%	0.00%
Atmos Energy Corporation	ATO	0.00%	0.00%	0.00%	0.00%	0.00%
Avista Corporation	AVA	0.00%	0.00%	0.00%	0.00%	0.00%
Black Hills Corporation	BKH	0.00%	0.00%	0.00%	0.00%	0.00%
CMS Energy Corporation	CMS	0.21%	0.22%	0.25%	0.27%	0.24%
Edison International	EIX	4.70%	5.37%	6.71%	8.27%	6.26%
Eversource Energy	ES	0.60%	0.65%	0.73%	0.81%	0.70%
Exelon Corporation	EXC	0.00%	0.00%	0.00%	0.00%	0.00%
Hawaiian Electric Industries, Inc.	HE	0.86%	0.91%	0.93%	1.00%	0.93%
NiSource Inc.	NI	0.00%	0.00%	0.00%	0.00%	0.00%
Northwest Natural Gas Company	NWN	0.00%	0.00%	0.00%	0.00%	0.00%
NorthWestern Corporation	NWE	0.00%	0.00%	0.00%	0.00%	0.00%
OGE Energy Corp.	OGE	0.00%	0.00%	0.00%	0.00%	0.00%
ONE Gas, Inc.	OGS	0.00%	0.00%	0.00%	0.00%	0.00%
PPL Corporation	PPL	0.00%	0.00%	0.00%	0.00%	0.00%
Public Service Enterprise Group Inc.	PEG	0.00%	0.00%	0.00%	0.00%	0.00%
Spire, Inc.	SR	0.00%	0.00%	0.00%	0.00%	0.00%
Xcel Energy Inc.	XEL	0.00%	0.00%	0.00%	0.00%	0.00%
MEAN		0.34%	0.38%	0.45%	0.54%	0.43%
LOW		0.00%	0.00%	0.00%	0.00%	0.00%
HIGH		4.70%	5.37%	6.71%	8.27%	6.26%

CAPITAL STRUCTURE
PROXY GROUP ANALYSIS
Preferred Equity Ratio - Utility Operating Companies

Utility Operating Company	Ticker	2021	2020	2019	2019	Average
ALLETE (Minnesota Power)	ALE	0.00%	0.00%	0.00%	0.00%	0.00%
Superior Water, Light and Power Company	ALE	0.00%	0.00%	0.00%	0.00%	0.00%
AEP Texas Inc.	AEP	0.00%	0.00%	0.00%	0.00%	0.00%
Appalachian Power Company	AEP	0.00%	0.00%	0.00%	0.00%	0.00%
Indiana Michigan Power Company	AEP	0.00%	0.00%	0.00%	0.00%	0.00%
Kentucky Power Company	AEP	0.00%	0.00%	0.00%	0.00%	0.00%
Kingsport Power Company	AEP	0.00%	0.00%	0.00%	0.00%	0.00%
Ohio Power Company	AEP	0.00%	0.00%	0.00%	0.00%	0.00%
Public Service Company of Oklahoma	AEP	0.00%	0.00%	0.00%	0.00%	0.00%
Southwestern Electric Power Company	AEP	0.00%	0.00%	0.00%	0.00%	0.00%
Wheeling Power Company	AEP	0.00%	0.00%	0.00%	0.00%	0.00%
Atmos Energy Corporation	ATO	0.00%	0.00%	0.00%	0.00%	0.00%
Avista Corporation	AVA	0.00%	0.00%	0.00%	0.00%	0.00%
Black Hills Colorado Electric, Inc.	BKH	0.00%	0.00%	0.00%	0.00%	0.00%
Black Hills Energy Arkansas, Inc.	BKH	0.00%	0.00%	0.00%	0.00%	0.00%
Black Hills Power, Inc.	BKH	0.00%	0.00%	0.00%	0.00%	0.00%
Cheyenne Light, Fuel and Power Company	BKH	0.00%	0.00%	0.00%	0.00%	0.00%
Consumers Energy Company	CMS	0.21%	0.22%	0.25%	0.27%	0.24%
Southern California Edison Company	EIX	4.70%	5.37%	6.71%	8.27%	6.26%
Connecticut Light and Power Company	ES	1.20%	1.27%	1.44%	1.52%	1.36%
NSTAR Electric Company	ES	0.48%	0.52%	0.57%	0.64%	0.55%
NSTAR Gas Company	ES	0.00%	0.00%	0.00%	0.00%	0.00%
Public Service Company of New Hampshire	ES	0.00%	0.00%	0.00%	0.00%	0.00%
Yankee Gas Services Company	ES	0.00%	0.00%	0.00%	0.00%	0.00%
Bay State Gas Company	ES	0.00%	0.00%	0.00%	0.00%	0.00%
Baltimore Gas and Electric Company	EXC	0.00%	0.00%	0.00%	0.00%	0.00%
Commonwealth Edison Company	EXC	0.00%	0.00%	0.00%	0.00%	0.00%
PECO Energy Company	EXC	0.00%	0.00%	0.00%	0.00%	0.00%
Delmarva Power & Light Company	EXC	0.00%	0.00%	0.00%	0.00%	0.00%
Atlantic City Electric Company	EXC	0.00%	0.00%	0.00%	0.00%	0.00%
Potomac Electric Power Company	EXC	0.00%	0.00%	0.00%	0.00%	0.00%
Hawaiian Electric Company, Inc.	HE	0.86%	0.91%	0.93%	1.00%	0.93%
Columbia Gas of Kentucky, Incorporated	NI	0.00%	0.00%	0.00%	0.00%	0.00%
Columbia Gas of Maryland, Incorporated	NI	0.00%	0.00%	0.00%	0.00%	0.00%
Columbia Gas of Ohio, Inc.	NI	0.00%	0.00%	0.00%	0.00%	0.00%
Columbia Gas of Pennsylvania, Inc.	NI	0.00%	0.00%	0.00%	0.00%	0.00%
Columbia Gas of Virginia, Incorporated	NI		0.00%	0.00%	0.00%	0.00%
Northern Indiana Public Service Company	NI	0.00%	0.00%	0.00%	0.00%	0.00%
NorthWestern Corporation	NWE	0.00%	0.00%	0.00%	0.00%	0.00%
Northwest Natural Gas Company	NWN	0.00%	0.00%	0.00%	0.00%	0.00%
Oklahoma Gas and Electric Company	OGE	0.00%	0.00%	0.00%	0.00%	0.00%
Kansas Gas Service Company, Inc.	OGS	0.00%	0.00%	0.00%	0.00%	0.00%
Oklahoma Natural Gas Company	OGS	0.00%	0.00%	0.00%	0.00%	0.00%
Texas Gas Service Company, Inc.	OGS	0.00%	0.00%	0.00%	0.00%	0.00%
Kentucky Utilities Company	PPL	0.00%	0.00%	0.00%	0.00%	0.00%
Louisville Gas and Electric Company	PPL	0.00%	0.00%	0.00%	0.00%	0.00%
PPL Electric Utilities Corporation	PPL	0.00%	0.00%	0.00%	0.00%	0.00%
Public Service Electric and Gas Company	PEG	0.00%	0.00%	0.00%	0.00%	0.00%
Spire Alabama Inc.	SR	0.00%	0.00%	0.00%		0.00%
Spire Gulf Inc.	SR	0.00%	0.00%	0.00%	0.00%	0.00%
Spire Mississippi Inc.	SR	0.00%	0.00%	0.00%	0.00%	0.00%
Spire Missouri Inc.	SR	0.00%	0.00%	0.00%	0.00%	0.00%
Public Service Company of Colorado	XEL	0.00%	0.00%	0.00%	0.00%	0.00%
Northern States Power Company - MN	XEL	0.00%	0.00%	0.00%	0.00%	0.00%
Northern States Power Company - WI	XEL	0.00%	0.00%	0.00%	0.00%	0.00%
Southwestern Public Service Company	XEL	0.00%	0.00%	0.00%	0.00%	0.00%

Notes:
[1] Natural Gas and Electric Operating Subsidiaries with data listed as N/A have been excluded from the analysis.
Source: S&P Capital IQ Pro

CAPITAL STRUCTURE
PROXY GROUP ANALYSIS
Customer Deposit Ratio [1]

Proxy Group Company	Ticker	2021	2020	2019	2019	Average
ALLETE, Inc.	ALE	0.00%	0.00%	0.00%	0.00%	0.00%
American Electric Power Company, Inc.	AEP	0.80%	0.86%	0.99%	1.20%	0.96%
Atmos Energy Corporation	ATO	0.15%	0.20%	0.24%	0.31%	0.23%
Avista Corporation	AVA	0.09%	0.08%	0.12%	0.13%	0.11%
Black Hills Corporation	BKH	0.40%	0.46%	0.53%	0.50%	0.47%
CMS Energy Corporation	CMS	0.13%	0.11%	0.14%	0.17%	0.14%
Edison International	EIX	0.47%	0.67%	0.90%	1.10%	0.79%
Eversource Energy	ES	0.12%	0.15%	0.26%	0.28%	0.20%
Exelon Corporation	EXC	0.60%	0.72%	0.92%	0.96%	0.80%
Hawaiian Electric Industries, Inc.	HE	0.00%	1.32%	2.43%	0.73%	1.12%
NiSource Inc.	NI	0.62%	0.70%	0.86%	0.94%	0.78%
Northwest Natural Gas Company	NWN	0.08%	0.19%	0.30%	0.35%	0.23%
NorthWestern Corporation	NWE	0.17%	0.14%	0.10%	0.18%	0.15%
OGE Energy Corp.	OGE	0.93%	1.07%	1.14%	1.22%	1.09%
ONE Gas, Inc.	OGS	1.24%	1.64%	1.53%	1.71%	1.53%
PPL Corporation	PPL	0.33%	0.37%	0.40%	0.44%	0.38%
Public Service Enterprise Group Inc.	PEG	0.22%	0.28%	0.41%	0.46%	0.34%
Spire, Inc.	SR	0.62%	0.76%	0.97%	0.76%	0.78%
Xcel Energy Inc.	XEL	0.19%	0.20%	0.28%	0.34%	0.25%
MEAN		0.38%	0.52%	0.66%	0.62%	0.54%
LOW		0.00%	0.00%	0.00%	0.00%	0.00%
HIGH		1.24%	1.64%	2.43%	1.71%	1.53%

CAPITAL STRUCTURE
PROXY GROUP ANALYSIS
Customer Deposit Ratio - Utility Operating Companies

Utility Operating Company	Ticker	2021	2020	2019	2019	Average
ALLETE (Minnesota Power)	ALE	0.00%	0.00%	0.00%	0.00%	0.00%
Superior Water, Light and Power Company	ALE	0.03%	0.00%	0.00%	0.00%	0.01%
AEP Texas Inc.	AEP	0.00%	0.00%	0.00%	0.00%	0.00%
Appalachian Power Company	AEP	0.76%	0.84%	0.99%	1.08%	0.92%
Indiana Michigan Power Company	AEP	0.76%	0.73%	0.72%	0.72%	0.73%
Kentucky Power Company	AEP	1.61%	1.66%	1.84%	1.85%	1.74%
Kingsport Power Company	AEP	2.92%	2.86%	3.23%	4.18%	3.30%
Ohio Power Company	AEP	1.13%	1.06%	1.51%	2.77%	1.62%
Public Service Company of Oklahoma	AEP	1.31%	1.84%	2.09%	2.26%	1.87%
Southwestern Electric Power Company	AEP	0.96%	1.17%	1.28%	1.29%	1.18%
Wheeling Power Company	AEP	1.03%	0.76%	0.36%	0.30%	0.61%
Atmos Energy Corporation	ATO	0.15%	0.20%	0.24%	0.31%	0.23%
Avista Corporation	AVA	0.09%	0.08%	0.12%	0.13%	0.11%
Black Hills Colorado Electric, Inc.	BKH	0.49%	0.57%	0.66%	0.50%	0.55%
Black Hills Energy Arkansas, Inc.	BKH	0.98%	1.10%	1.22%	1.37%	1.17%
Black Hills Power, Inc.	BKH	0.21%	0.26%	0.28%	0.25%	0.25%
Cheyenne Light, Fuel and Power Company	BKH	0.22%	0.25%	0.38%	0.49%	0.33%
Consumers Energy Company	CMS	0.13%	0.11%	0.14%	0.17%	0.14%
Southern California Edison Company	EIX	0.47%	0.67%	0.90%	1.10%	0.79%
Connecticut Light and Power Company	ES	0.14%	0.19%	0.40%	0.39%	0.28%
NSTAR Electric Company	ES	0.05%	0.06%	0.10%	0.12%	0.08%
NSTAR Gas Company	ES	0.06%	0.08%	0.12%	0.15%	0.10%
Public Service Company of New Hampshire	ES	0.22%	0.24%	0.29%	0.28%	0.26%
Yankee Gas Services Company	ES	0.11%	0.16%	0.28%	0.33%	0.22%
Bay State Gas Company	ES	0.19%	0.25%	0.63%	0.70%	0.44%
Baltimore Gas and Electric Company	EXC	1.12%	1.37%	1.69%	1.88%	1.52%
Commonwealth Edison Company	EXC	0.44%	0.42%	0.60%	0.60%	0.51%
PECO Energy Company	EXC	0.50%	0.69%	0.88%	0.94%	0.75%
Delmarva Power & Light Company	EXC	0.78%	0.96%	1.12%	1.16%	1.01%
Atlantic City Electric Company	EXC	0.56%	0.80%	0.96%	1.13%	0.86%
Potomac Electric Power Company	EXC	0.50%	0.79%	0.97%	0.97%	0.81%
Hawaiian Electric Company, Inc.	HE	0.00%	1.32%	2.43%	0.73%	1.12%
Columbia Gas of Kentucky, Incorporated	NI	0.40%	0.61%	0.76%	0.81%	0.65%
Columbia Gas of Maryland, Incorporated	NI	0.20%	0.20%	0.33%	0.36%	0.27%
Columbia Gas of Ohio, Inc.	NI	0.25%	0.30%	0.40%	0.44%	0.35%
Columbia Gas of Pennsylvania, Inc.	NI	0.17%	0.17%	0.20%	0.21%	0.19%
Columbia Gas of Virginia, Incorporated	NI		0.70%	1.20%	1.33%	1.08%
Northern Indiana Public Service Company	NI	1.06%	1.16%	1.32%	1.43%	1.24%
NorthWestern Corporation	NWE	0.17%	0.14%	0.10%	0.18%	0.15%
Northwest Natural Gas Company	NWN	0.08%	0.19%	0.30%	0.35%	0.23%
Oklahoma Gas and Electric Company	OGE	0.93%	1.07%	1.14%	1.22%	1.09%
Kansas Gas Service Company, Inc.	OGS	1.16%	1.56%	1.40%	1.63%	1.44%
Oklahoma Natural Gas Company	OGS	1.38%	1.68%	1.69%	1.91%	1.66%
Texas Gas Service Company, Inc.	OGS	1.17%	1.66%	1.49%	1.57%	1.47%
Kentucky Utilities Company	PPL	0.54%	0.55%	0.56%	0.61%	0.57%
Louisville Gas and Electric Company	PPL	0.67%	0.68%	0.70%	0.70%	0.69%
PPL Electric Utilities Corporation	PPL	0.04%	0.11%	0.15%	0.20%	0.12%
Public Service Electric and Gas Company	PEG	0.22%	0.28%	0.41%	0.46%	0.34%
Spire Alabama Inc.	SR	1.20%	1.39%	1.60%		1.40%
Spire Gulf Inc.	SR	1.22%	1.31%	1.30%	2.76%	1.65%
Spire Mississippi Inc.	SR	3.22%	3.96%	4.49%	4.96%	4.16%
Spire Missouri Inc.	SR	0.27%	0.37%	0.59%	0.63%	0.46%
Public Service Company of Colorado	XEL	0.22%	0.20%	0.22%	0.27%	0.23%
Northern States Power Company - MN	XEL	0.20%	0.23%	0.40%	0.51%	0.33%
Northern States Power Company - WI	XEL	0.30%	0.31%	0.42%	0.32%	0.34%
Southwestern Public Service Company	XEL	0.08%	0.09%	0.11%	0.15%	0.11%

Notes:
[1] Natural Gas and Electric Operating Subsidiaries with data listed as N/A have been excluded from the analysis.
Source: S&P Capital IQ Pro